Advanced Mathematics A

Lecture Notes, Volume I

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知之者不如好之者，好之者不如乐之者 —《论语 · 雍也》

Preface

These lecture notes were prepared for the course *Advanced Mathematics A* at Peking University. They are intended primarily for students taking the course, but they may also serve as a companion for review and independent study. The aim is to present the main ideas of calculus as a coherent mathematical subject, with careful definitions, meaningful examples, and enough detail to support work beyond what can be covered in lectures.

Relationship to the Peking University Textbook

The notes stand in continuity with the third edition of Peking University's *Advanced Mathematics* (《高等数学（第三版）上册》，李忠周建莹编著) textbook. They inherit its central subject matter, its commitment to a substantial calculus course, and many of the standard concepts and methods that generations of students have learned from it. The textbook remains an essential course resource and should be read alongside these notes.

At the same time, this text is not a line-by-line rewriting of the textbook. Some topics are reorganized to make their logical dependencies more visible; some proofs are expanded to clarify their central ideas; and additional examples, counterexamples, questions, and method summaries are included when they may improve understanding. Certain optional extensions are also added to connect elementary calculus with ideas that students will encounter later. The choice and emphasis of material therefore reflect the needs of this particular course as well as the author's own teaching experience.

A Work in Progress

Although considerable care has been taken in the preparation of these notes, their mathematical accuracy, exposition, and pedagogical design remain open to improvement. Errors, unclear arguments, infelicitous examples, and uneven choices of emphasis may still be present. Readers are warmly invited to send the author corrections, suggestions, and comments. Such feedback is valuable not only for improving later versions of the notes but also for improving the course itself.

Use and Circulation

The author acknowledges the ElegantLaTeX template (https://github.com/ElegantLaTeX/), upon which the typesetting of these lecture notes is based.

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Introduction

Course Objectives

This course is designed to provide students with a solid foundation in the fundamental concepts, theories, and methods of calculus, while developing the ability to understand and use mathematical ideas rigorously and effectively. Students should learn not only how to carry out calculations, but also why the definitions, theorems, and methods of calculus take the forms that they do.

Throughout the course, students are expected to understand the principal ideas and concepts, master essential methods and techniques, recognize the logical structure underlying the theory, and apply mathematical knowledge to a range of problems. Computational fluency is important, but it should be supported by conceptual understanding: a correct calculation is most useful when one also knows its hypotheses, its meaning, and its limitations.

Through systematic study and practice, students should gradually develop mathematical intuition, rigorous logical reasoning, and the ability to think abstractly. They should also strengthen their capacity to formulate, analyze, and solve problems using appropriate mathematical language and tools. Ultimately, the course aims to build the mathematical foundation and analytical habits needed for further study in mathematics, science, engineering, and related disciplines.

Advice for Beginning Students

University mathematics asks for a more active kind of learning than simply following a worked calculation. The following habits will make the course more productive.

- **Be an independent learner.** Take responsibility for your own learning and do not rely solely on what is presented in class. Learn to identify what you do not yet understand, consult appropriate resources, and return to a difficult point more than once.

- **Read the textbook and the notes actively.** The textbook is an essential course resource. Assigned readings and materials remain part of the course whether or not every detail is discussed in lectures. When reading, pause to reconstruct arguments, check examples, and write down questions.

- **Do not treat lectures as an examination checklist.** Lectures and examinations are guided by the same syllabus and learning objectives, but they need not contain exactly the same material. Some ideas discussed in class may not appear on an examination, and a problem may require course material that was not worked out explicitly in class.

- **Prepare from the whole course, not from predictions.** When reviewing, pay attention to the syllabus, textbook, lecture notes, assigned work, and stated learning objectives. Trying to infer an examination solely from a list of lecture examples gives a narrow and unreliable picture of the subject.

- **Study consistently throughout the semester.** Mathematical understanding is cumulative. Regular reading, review, problem solving, and independent reflection are more effective than a short period of intensive preparation immediately before an assessment.

- **Interrogate the basic concepts.** Ask how a definition arises, what problem it resolves, why each condition is present, and what the concept is really designed to capture. Being able to repeat a definition is not the same as understanding it.

- **Learn from examples and counterexamples.** A good example shows how a concept works; a counterexample shows where an apparently plausible statement fails. Use both positive and negative examples to test your understanding and to distinguish hypotheses from conclusions.
- **Practise with purpose.** Technique requires training, and solving exercises is indispensable. Nevertheless, indiscriminately accumulating difficult, unusual, or highly specialized problems is not the goal. After solving a problem, ask which idea made the solution possible, what alternatives were available, and how the method might apply elsewhere. Mathematical ideas and habits of thought are more fundamental than the number of problems completed.
- **Write mathematics precisely.** A solution should communicate its assumptions, reasoning, and conclusion. Practise stating definitions accurately, citing hypotheses when a theorem is used, and explaining the decisive step rather than presenting only a chain of formulas.

A Common Path to Mathematics

It is normal for serious mathematics to feel difficult. A definition may seem artificial on first reading, and a proof may remain opaque until one has examined several examples or attempted to reconstruct it independently. Temporary confusion is not evidence that one is incapable of learning the subject; it is often the stage at which deeper understanding begins.

Do not measure progress only by speed. Learn to remain with a question, to break it into smaller parts, to test simple cases, and to seek help after a genuine attempt. Over time, ideas that initially appeared isolated will begin to support one another. The goal is not to avoid ever becoming stuck, but to develop the patience, judgment, and mathematical tools needed to move forward.

Calculus rewards sustained attention. If you read thoughtfully, practise regularly, and keep asking why the theory works, you will gain more than a collection of techniques: you will acquire a language and a way of thinking that will remain useful far beyond this course.

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Chapter 1 Functions and Limits

Calculus studies change, but change can be discussed precisely only after its basic objects and limiting processes have been fixed. We therefore begin with the real number system, pass to functions as rules relating variable quantities, and then develop the limit of a sequence as our first rigorous limiting notion. Function limits lead to continuity, whose local definition yields global conclusions on closed bounded intervals.

1.1 Real Numbers and Basic Inequalities

知识点概要

- *Review rational and irrational numbers and understand why calculus needs the full real number system.*
- *Interpret completeness as the property that prevents gaps in the number line.*
- *Use intervals and neighbourhoods to describe location and distance on the real line.*
- *Use the Archimedean property and density to choose integers and rational approximations.*
- *Control errors using absolute values, AM–GM, Bernoulli's inequality, and square-root estimates.*

Calculus repeatedly asks what happens after an approximation is refined without end. For example, the decimal approximations

$$1.4, \quad 1.41, \quad 1.414, \quad 1.4142, \quad \ldots$$

approach the length of the diagonal of a unit square. If our number system contained only rational numbers, that limiting length would be missing. The real numbers provide a number system in which such limiting processes can be carried out without falling into a gap.

1.1.1 From rational numbers to real numbers

Throughout these notes, the natural numbers (自然数) are

$$\mathbb{N} = \{0, 1, 2, 3, \ldots\},$$

and the positive integers are denoted by $\mathbb{N}_{>0} = \{1, 2, 3, \ldots\}$. We also use $\mathbb{Z}$ for the integers (整数) and $\mathbb{R}$ for the real numbers (实数).

Definition 1.1

A real number x is rational (有理数) *if*

$$x = \frac{p}{q}$$

for some $p \in \mathbb{Z}$ and $q \in \mathbb{Z} \setminus \{0\}$. The set of all rational numbers is $\mathbb{Q}$. A real number that is not rational is irrational (无理数)*; the set of irrational numbers is $\mathbb{R} \setminus \mathbb{Q}$.* ♣

A rational number has a terminating or eventually periodic decimal expansion, and conversely every terminating or eventually periodic decimal represents a rational number. Irrational numbers have nonperiodic infinite decimal expansions. This decimal description is useful for intuition, but the decisive distinction is whether the number can be represented as a ratio of integers.

Proposition 1.1

If p is a prime number, then $\sqrt{p}$ is irrational.

> ***Proof Strategy.*** We take *Euclid's lemma* (欧几里得引理) as known prerequisite knowledge: if p is prime and $a, b \in \mathbb{Z}$, then
> $$p \mid ab \implies p \mid a \text{ or } p \mid b.$$
> Assume that $\sqrt{p} = m/n$ is a fraction in lowest terms. After squaring, apply the lemma twice to force p to divide both m and n, contradicting the lowest-terms assumption.

Proof Suppose, to the contrary, that

$$\sqrt{p} = \frac{m}{n},$$

where $m, n \in \mathbb{N}_{>0}$ and m, n have no common factor greater than 1. Then

$$m^2 = pn^2.$$

Thus $p \mid m^2$. By Euclid's lemma applied to $m^2 = m \cdot m$, we have $p \mid m$, so $m = pk$ for some $k \in \mathbb{N}_{>0}$. Substitution gives

$$p^2k^2 = pn^2,$$

hence $n^2 = pk^2$. By Euclid's lemma again, applied to $n^2 = n \cdot n$, we obtain $p \mid n$. This contradicts the assumption that m and n have no common factor greater than 1. Consequently, $\sqrt{p}$ is irrational. □

In particular, $\sqrt{2}$ and $\sqrt{3}$ are irrational. The essential step is the contradiction with coprimality, not the particular prime chosen.

Example 1.1 A sum of two irrational numbers may require a new argument Prove that $\sqrt{2} + \sqrt{3}$ is irrational.

> ***Idea.*** The sum of two irrational numbers need not be irrational, so their irrationality alone is insufficient. Assume the sum is rational, square it, and isolate $\sqrt{6}$.

Proof Suppose that $r = \sqrt{2} + \sqrt{3}$ is rational. Then

$$r^2 = 5 + 2\sqrt{6},$$

so

$$\sqrt{6} = \frac{r^2 - 5}{2}$$

would also be rational. But $\sqrt{6}$ is irrational by the same lowest-terms argument: if $\sqrt{6} = m/n$ in lowest terms, then $m^2 = 6n^2$ forces both 2 and 3 to divide m, hence $6 \mid m$; substitution then forces $6 \mid n$, a contradiction. Therefore $\sqrt{2} + \sqrt{3}$ is irrational. □

The rational numbers are closed under addition, subtraction, multiplication, and division by a nonzero number. The real numbers have the same algebraic closure and, in addition, carry an order compatible with these operations. What distinguishes $\mathbb{R}$ from $\mathbb{Q}$ for calculus is a further property: completeness.

1.1.2 Order and completeness

The real numbers form an *ordered field* (有序域). In particular, the usual algebraic laws hold, exactly one of $a < b$, $a = b$, and $a > b$ holds for any $a, b \in \mathbb{R}$, and inequalities are compatible with addition and multiplication by positive numbers.

Completeness is best understood through bounded sets.

Definition 1.2

Let $E \subseteq \mathbb{R}$ be nonempty. A number $M \in \mathbb{R}$ is an upper bound(上界) *of E if*

$$x \leq M$$

for every $x \in E$. The set E is bounded above(有上界) *if it has an upper bound. A number $s \in \mathbb{R}$ is the* supremum(上确界) *or* least upper bound *of E if*

(i) s is an upper bound of E;

(ii) every upper bound M of E satisfies $s \leq M$.

It is denoted by $s = \sup E$. A lower bound(下界) *is a number m satisfying $m \leq x$ for every $x \in E$. The set is* bounded below(有下界) *if such an m exists. Its* infimum(下确界)*, denoted by* $\inf E$*, is its greatest lower bound. A set is* bounded(有界) *if it is bounded both above and below.* ♣

Example 1.2 A supremum need not be attained The set $E = (0, 1)$ has $\inf E = 0$ and $\sup E = 1$, but neither boundary belongs to E. In contrast, $F = (0, 1]$ has the same infimum and supremum, and 1 is also its *maximum*(最大值). A maximum is an upper bound that belongs to the set; a supremum need not belong to it. The corresponding distinction holds between an infimum and a *minimum*(最小值).

Proposition 1.2

Let E be nonempty and bounded above. A number s equals $\sup E$ *if and only if it is an upper bound and, for every $\varepsilon > 0$, some $x \in E$ satisfies*

$$s - \varepsilon < x \leq s.$$

♠

Proof If $s = \sup E$, then $s - \varepsilon$ cannot be an upper bound, so such an x exists. Conversely, suppose the stated approximation property holds. Any $M < s$ fails to be an upper bound: take $\varepsilon = s - M$ to obtain $x > M$. Thus no upper bound is smaller than s, which is the definition of $\sup E$. □

Axiom 1.1 (Least-Upper-Bound Property)

Every nonempty subset of $\mathbb{R}$ that is bounded above has a supremum in $\mathbb{R}$. ♡

We regard the least-upper-bound property(确界原理)as a fundamental property of $\mathbb{R}$, not as a theorem to be proved from algebra alone. It is one precise formulation of the *completeness*(完备性)of $\mathbb{R}$. It rules out gaps: whenever a nonempty set has points only up to some finite height, there is a real number marking its least possible upper boundary. Other completeness principles, such as the convergence of bounded monotone sequences and the Cauchy criterion, are equivalent to it. We shall derive the monotone sequence theorem in Section 1.3.

> ***Comment.*** The rational numbers do not satisfy this property when they are considered by themselves. For example,
>
> $$E = \{q \in \mathbb{Q} \mid q^2 < 2\}$$
>
> is nonempty and bounded above in $\mathbb{Q}$, but it has no rational least upper bound. Its missing boundary is $\sqrt{2}$.

Completeness also implies the Archimedean property(阿基米德性质): the positive integers eventually exceed every prescribed real number.

Proposition 1.3 (Archimedean Property)

For every $x \in \mathbb{R}$, there exists $n \in \mathbb{N}_{>0}$ such that $n > x$. Equivalently, for every $\varepsilon > 0$, there exists $n \in \mathbb{N}_{>0}$ such that $1/n < \varepsilon$.

> ***Idea.*** If the positive integers had a real upper bound, completeness would provide a least one. But moving one integer step forward would cross that boundary.

Proof Suppose that $\mathbb{N}_{>0}$ were bounded above, and let $s = \sup \mathbb{N}_{>0}$. The number $s-1$ is not an upper bound, so there exists $n \in \mathbb{N}_{>0}$ with $n > s-1$. Then $n+1 \in \mathbb{N}_{>0}$ and $n+1 > s$, contradicting the fact that s is an upper bound. Thus $\mathbb{N}_{>0}$ is not bounded above.

For the equivalent formulation, apply the first statement to $x = 1/\varepsilon$. Then $n > 1/\varepsilon$, and hence $1/n < \varepsilon$. □

The following density property (稠密性) says that the real line contains rational and irrational numbers at every scale.

Proposition 1.4 (Density of Rational and Irrational Numbers)

If $a, b \in \mathbb{R}$ and $a < b$, then there exist $r \in \mathbb{Q}$ and $u \in \mathbb{R} \setminus \mathbb{Q}$ such that

$$a < r < b, \qquad a < u < b.$$

Proof By Proposition 1.3, choose $n \in \mathbb{N}_{>0}$ such that

$$n(b-a) > 1.$$

Let

$$m = \lfloor na \rfloor + 1.$$

Here $\lfloor t \rfloor$ denotes the greatest integer not exceeding t. Such an integer exists by the Archimedean property and the well-ordering of the positive integers. Then $na < m \leq na + 1 < nb$, so

$$a < \frac{m}{n} < b.$$

Thus $r = m/n$ is a rational number in (a, b).

Apply the rational-density statement to a nonempty open subinterval of

$$\left(\frac{a}{\sqrt{2}}, \frac{b}{\sqrt{2}} \right)$$

that does not contain 0. It contains a nonzero rational number q. Then $u = q\sqrt{2}$ lies in (a, b) and is irrational, because $q \neq 0$ and $\sqrt{2}$ is irrational. □

> ***Comment.*** Density and completeness express different facts. The rational numbers are dense in $\mathbb{R}$, yet $\mathbb{Q}$ is not complete. Having points arbitrarily close to every location does not mean that all limiting boundary points belong to the set.

1.1.3 The real line, intervals, and neighbourhoods

After choosing an origin, a positive direction, and a unit length, every real number corresponds to one point on a line and every point corresponds to one real number. We therefore identify $\mathbb{R}$ with the *real line* (数轴). The inequality $a < b$ means that the point a lies to the left of the point b.

For $a, b \in \mathbb{R}$ with $a < b$, the open interval (开区间), closed interval (闭区间), and two half-open intervals

(半开半闭区间) are

$$(a,b) = \{x \in \mathbb{R} \mid a < x < b\},$$
$$[a,b] = \{x \in \mathbb{R} \mid a \le x \le b\},$$
$$[a,b) = \{x \in \mathbb{R} \mid a \le x < b\},$$
$$(a,b] = \{x \in \mathbb{R} \mid a < x \le b\}.$$

Infinite intervals include

$$(-\infty, a), \quad (-\infty, a], \quad (a, +\infty), \quad [a, +\infty),$$

and the whole real line is $(-\infty, +\infty)$. The symbols $+\infty$ and $-\infty$ are not real numbers and are never included as endpoints.

Definition 1.3

Let $a \in \mathbb{R}$ and $r > 0$. The r-neighbourhood (邻域) *of a is*

$$U_r(a) = (a - r, a + r) = \{x \in \mathbb{R} \mid |x - a| < r\}.$$

The corresponding punctured r-neighbourhood (去心邻域) *is*

$$U_r(a) \setminus \{a\} = \{x \in \mathbb{R} \mid 0 < |x - a| < r\}.$$

♣

Neighbourhood notation translates a geometric statement into an inequality. The condition $x \in U_r(a)$ says exactly that the distance from x to a is less than r. Punctured neighbourhoods will become essential for function limits, where the behaviour near a may be independent of the value at a itself.

1.1.4 Absolute value and distance estimates

Definition 1.4

For $x \in \mathbb{R}$, its absolute value (绝对值) *is*

$$|x| = \begin{cases} x, & x \ge 0, \\ -x, & x < 0. \end{cases}$$

♣

Geometrically, $|x|$ is the distance from x to 0, while $|x - y|$ is the distance between x and y. The following elementary equivalences should become automatic: for $r > 0$,

$$|x| < r \iff -r < x < r,$$
$$|x| \le r \iff -r \le x \le r,$$
$$|x - a| < r \iff a - r < x < a + r.$$

Theorem 1.1

For all $x, y \in \mathbb{R}$, the triangle inequality (三角不等式) *and reverse triangle inequality* (反三角不等式) *state that*

$$|x + y| \le |x| + |y|, \tag{1.1}$$
$$\big||x| - |y|\big| \le |x - y|. \tag{1.2}$$

More generally, for all $a, b, c \in \mathbb{R}$,

$$|a - c| \le |a - b| + |b - c|.$$

♡

> ***Idea.*** The first inequality says that travelling from 0 to $x+y$ directly cannot be longer than travelling first from 0 to x and then by the displacement y. The reverse inequality is obtained by applying the same principle in both directions.

Proof Since $-|x| \le x \le |x|$ and $-|y| \le y \le |y|$, addition gives

$$-\big(|x|+|y|\big) \le x+y \le |x|+|y|\,.$$

This is equivalent to (1.1).

Applying the triangle inequality to $x=(x-y)+y$ gives

$$|x| \le |x-y|+|y|\,,$$

so $|x|-|y| \le |x-y|$. Interchanging x and y gives $|y|-|x| \le |x-y|$. Together these inequalities yield (1.2).

Finally, set $x=a-b$ and $y=b-c$ in the triangle inequality. □

Example 1.3 Turning a distance condition into an interval Solve

$$|2x-3|<5.$$

Solution *The inequality is equivalent to*

$$-5<2x-3<5,$$

and hence to

$$-1<x<4.$$

Thus the solution set is $(-1,4)$.

> ***Geometric Interpretation.*** Equivalently,
>
> $$|2x-3| = 2\left|x-\frac{3}{2}\right| < 5$$
>
> says that x lies within distance $5/2$ of $3/2$.

Example 1.4 A reusable estimate Prove that $|x|<b+|a|$ whenever $|x-a|<b$.

Proof Then

$$|x| = |x-a+a| \le |x-a|+|a| < b+|a|\,.$$

□

> ***Technique.*** This simple observation is repeatedly used in limit proofs: first restrict x or a sequence term to a convenient neighbourhood, then replace a variable factor by a constant bound.

> ***Question.*** Why is the estimate $|x-y| \ge \big||x|-|y|\big|$ called a reverse triangle inequality?

1.1.5 Selected inequalities for later use

The following estimates will reappear in sequence limits. Their value is that they replace nonlinear expressions by quantities that are easier to control.

Proposition 1.5 (Arithmetic–Geometric Mean Inequality)

For $u,v\ge 0$,

$$\sqrt{uv} \le \frac{u+v}{2},$$

with equality if and only if $u=v$. *This is the two-variable arithmetic–geometric mean inequality* (算术平均数与几何平均数不等式)*, or AM–GM.*

Proof The inequality $(\sqrt{u}-\sqrt{v})^2 \ge 0$ gives $u+v \ge 2\sqrt{uv}$. Equality holds exactly when $\sqrt{u}=\sqrt{v}$. □

Example 1.5 A bound for an iteration Let $a>0$. Show that the map

$$T(x)=\frac{1}{2}\left(x+\frac{a}{x}\right)$$

sends $[\sqrt{a},+\infty)$ into itself and satisfies $T(x)\le x$ there.

Proof If $x>0$, AM–GM gives

$$\frac{1}{2}\left(x+\frac{a}{x}\right)\ge\sqrt{a}.$$

Furthermore, when $x\ge\sqrt{a}$,

$$\frac{1}{2}\left(x+\frac{a}{x}\right)\le x.$$

□

> ***Looking Ahead.*** Thus the averaging operation sends $[\sqrt{a},+\infty)$ into itself and does not increase its input. These two observations will prove boundedness and monotonicity of a recursive sequence.

Proposition 1.6 (Bernoulli's Inequality)

For $t\ge -1$ and $n\in\mathbb{N}_{>0}$, Bernoulli's inequality(伯努利不等式)*is*

$$(1+t)^n\ge 1+nt.$$

♠

Proof The assertion is an equality for $n=1$. If it holds for n, multiplying by $1+t\ge 0$ gives

$$(1+t)^{n+1}\ge(1+nt)(1+t)=1+(n+1)t+nt^2\ge 1+(n+1)t.$$

Induction completes the proof. □

Example 1.6 Control square roots by rationalizing For $u,v\ge 0$ with $u+v>0$, prove that

$$\left|\sqrt{u}-\sqrt{v}\right|=\frac{|u-v|}{\sqrt{u}+\sqrt{v}}.$$

If $u,v\ge c>0$, then

$$\left|\sqrt{u}-\sqrt{v}\right|\le\frac{|u-v|}{2\sqrt{c}}.$$

Even without a positive lower bound, we still have

$$\left|\sqrt{u}-\sqrt{v}\right|\le\sqrt{|u-v|}.$$

Proof Multiplying by the conjugate gives

$$\left|\sqrt{u}-\sqrt{v}\right|=\frac{|(\sqrt{u}-\sqrt{v})(\sqrt{u}+\sqrt{v})|}{\sqrt{u}+\sqrt{v}}=\frac{|u-v|}{\sqrt{u}+\sqrt{v}}.$$

If $u,v\ge c>0$, then $\sqrt{u}+\sqrt{v}\ge 2\sqrt{c}$, which yields the first estimate.

For the last estimate, assume $u\ge v$. Then $\sqrt{u}+\sqrt{v}\ge\sqrt{u-v}$; rationalization gives the result when $u>v$, and the case $u=v$ is immediate. □

> ***Technique.*** The useful estimate depends on whether the denominator can be kept away from zero.

1.1.6 Key Concepts, Methods, and Further Remarks

This section has two complementary layers. Absolute values and inequalities provide quantitative control, while completeness and the Archimedean property provide existence. Keeping these layers distinct helps identify which part of an argument is computational and which part depends on the structure of the real numbers.

1. *Find the structural contradiction.* In an irrationality proof, put a hypothetical fraction in lowest terms before using divisibility.

2. *Translate geometry into inequalities.* Distances on the real line are absolute values, and neighbourhoods are interval conditions.
3. *Control a complicated expression by a simpler one.* Split sums by the triangle inequality, rationalize square-root differences, and use AM–GM or Bernoulli when their hypotheses fit. For a homogeneous expression with positive variables, normalization can reduce the number of variables: dividing $u+v \geq 2\sqrt{uv}$ by $v>0$ reduces it to $t+1 \geq 2\sqrt{t}$ with $t=u/v$.
4. *Choose an integer backward from the desired error.* To make $C/n < \varepsilon$ with $C>0$, choose an integer $N > C/\varepsilon$. Then the estimate holds for every $n \geq N$, not just one index.
5. *Identify where completeness is used.* Algebra and order alone are shared by $\mathbb{Q}$ and $\mathbb{R}$. Existence theorems involving limits or least upper bounds require the additional completeness of $\mathbb{R}$.

> ***Caution.*** Do not treat $\pm\infty$ as real numbers. Expressions such as $a < +\infty$ are useful notation, but ordinary field operations with ∞ are not defined in $\mathbb{R}$.

> ***Looking Ahead.*** Suprema need not be attained, but they can be approached from below. This approximation property is the decisive step in proving that a bounded nondecreasing sequence converges.

Exercise 1.1

1. Prove that $\sqrt{6}$ is irrational.
2. Let $r \in \mathbb{Q}$ and $u \in \mathbb{R} \setminus \mathbb{Q}$. Prove that $r+u$ is irrational, and prove that ru is irrational when $r \neq 0$.
3. Generalize Proposition 1.1 by proving that $\sqrt{p}$ is irrational whenever p is a positive integer that is not a perfect square.

   > ***Hint.*** Use uniqueness of prime factorization: what can be said about the exponents in the prime factorization of a perfect square? If $\sqrt{p} = m/n$ with $m,n \in \mathbb{N}_{>0}$, compare prime exponents on the two sides of $m^2 = pn^2$.

4. Prove that every nonempty open interval contains infinitely many rational numbers and infinitely many irrational numbers.
5. Write each set in interval notation:
$$\{x \in \mathbb{R} \mid |x-2| \leq 3\}, \qquad \{x \in \mathbb{R} \mid 0 < |x+1| < 2\}.$$
6. Solve the inequalities
$$|3x+1| \geq 2, \qquad |x-1| < |x+2|\,.$$
7. Prove that
$$|x+y+z| \leq |x|+|y|+|z|$$
for all $x,y,z \in \mathbb{R}$.
8. If $|x-a| < r$ and $|y-b| < s$, prove that
$$|(x+y)-(a+b)| < r+s.$$
Give a geometric interpretation.
9. Let $E \subseteq \mathbb{R}$ be nonempty and bounded above. Prove directly from the definition that $\sup E$ is unique.
10. Explain why the density of $\mathbb{Q}$ in $\mathbb{R}$ does not imply the completeness of $\mathbb{Q}$.
11. Let
$$A = \{x > 0 \mid x^2 < 5\}.$$

Determine $\sup A$, $\inf A$, and whether either extremum is attained. Justify every assertion from the definitions.

12. Let $E \subseteq \mathbb{R}$ be nonempty and bounded, let $c \in \mathbb{R}$, and let $\lambda > 0$. Prove that

$$\sup(c+E) = c + \sup E, \qquad \sup(\lambda E) = \lambda \sup E,$$

where $c + E = \{c + x : x \in E\}$ and $\lambda E = \{\lambda x : x \in E\}$. State the corresponding formulas for infima.

13. Let $a < b$. Find the minimum value of $|x-a| + |x-b|$ for $x \in \mathbb{R}$, and determine all points at which it is attained. Give both an algebraic and a geometric explanation.

14. Prove that

$$x(1-x) \leq \frac{1}{4}$$

for every $x \in [0,1]$, and determine the equality case.

15. Prove by rationalization that, for $t \geq 0$,

$$0 \leq \sqrt{1+t} - 1 \leq \frac{t}{2}.$$

Explain why the constant $1/2$ cannot be replaced by a smaller positive constant valid for every sufficiently small $t > 0$.

Supplement Exercises

S1. Let $E, F \subseteq \mathbb{R}$ be nonempty and bounded above, and define $E + F = \{x + y : x \in E,\ y \in F\}$. Prove directly from the least-upper-bound property that

$$\sup(E+F) = \sup E + \sup F.$$

S2. If E and F are nonempty and bounded, prove that

$$\sup(E \cup F) = \max\{\sup E, \sup F\}, \qquad \inf(E \cup F) = \min\{\inf E, \inf F\}.$$

Which analogous formulas for $E \cap F$ fail without additional hypotheses?

S3. Prove the finite Cauchy–Schwarz inequality

$$\left(\sum_{k=1}^{n} a_k b_k\right)^2 \leq \left(\sum_{k=1}^{n} a_k^2\right)\left(\sum_{k=1}^{n} b_k^2\right)$$

by considering $\sum_{k=1}^{n}(ta_k - b_k)^2$ as a quadratic polynomial in t. Determine the equality condition.

S4. For $m \in \mathbb{N}$, prove by induction that the arithmetic–geometric mean inequality holds for 2^m positive numbers:

$$\frac{x_1 + \cdots + x_{2^m}}{2^m} \geq (x_1 \cdots x_{2^m})^{1/2^m}.$$

S5. Find the best constant C such that

$$(x+y+z)^2 \leq C(x^2+y^2+z^2)$$

for all $x, y, z \in \mathbb{R}$, and prove that your constant is optimal.

S6. Let

$$B = \{q \in \mathbb{Q} : q \geq 0 \text{ and } q^2 < 2\}.$$

Prove that B has a supremum as a subset of $\mathbb{R}$, but has no supremum as a subset of the ordered field $\mathbb{Q}$.

1.2 Variables and Functions

核心概念与图像方法

- *Understand a function as a mapping with a specified domain, codomain, and assignment rule.*
- *Move between formulas, tables, verbal rules, piecewise definitions, and graphs.*
- *Combine functions by arithmetic operations and composition while tracking every domain restriction.*
- *Review elementary and non-elementary functions without confusing an expression with a function.*
- *Describe monotonicity, parity, periodicity, boundedness, and graph transformations.*
- *Test injectivity before constructing an inverse and interpret inverse graphs geometrically.*

Many quantities vary, and one quantity often depends on another. The distance travelled at constant speed depends on time, the area of a circle depends on its radius, and the length of a metal rod depends on temperature. Calculus studies such dependence, so its basic object is the function.

The essential point is not that a function has a formula. The essential point is that every allowed input has one, and only one, output.

1.2.1 Functions as mappings

Definition 1.5

Let $X, Y \subseteq \mathbb{R}$. A function(函数) *$f$ from X to Y, written*

$$f : X \to Y,$$

is a rule that assigns to every $x \in X$ a unique number $f(x) \in Y$. The set X is the domain(定义域)*, and Y is the* codomain(陪域)*. The* range(值域)*, or* image(像)*, of f is*

$$f(X) = \{f(x) \mid x \in X\}.$$

Thus $f(X) \subseteq Y$, and equality need not hold. ♣

A function is also called a *mapping*(映射).

The input x is often called the *independent variable*(自变量).

The value $f(x)$ is the *dependent variable*(因变量). These names emphasize variation, but the definition itself is set-theoretic: a function consists of a domain together with an assignment rule.

> ***Important Distinction.*** The domain is part of the function. The formulas
>
> $$f(x) = x^2 \quad (x \in \mathbb{R}) \qquad \text{and} \qquad g(x) = x^2 \quad (x \in [0, +\infty))$$
>
> define different functions, even though the displayed expressions are identical. Their ranges are also different if the first domain is changed further.

Definition 1.6

Two functions $f : X \to Y$ and $g : X' \to Y'$ are equal if $X = X'$, $Y = Y'$, and

$$f(x) = g(x)$$

for every $x \in X$. ♣

A function may be represented by a formula, a table, a graph, an algorithm, or a verbal rule. For example, a timetable assigning a departure time to each train is a function even if no useful algebraic formula describes it.

The *graph*(图像)of $f\colon X \to \mathbb{R}$ is the set

$$\{(x, f(x)) \in \mathbb{R}^2 \mid x \in X\}.$$

A plane curve is the graph of a function of x precisely when every vertical line meets it in at most one point. This vertical-line test is simply the uniqueness condition in Definition 1.5 expressed geometrically. The horizontal projection of the graph gives the domain, and the vertical projection gives the range. Zeros are the points where the graph meets the x-axis, while the portions above and below that axis show the sign of the function.

1.2.2 Domains, ranges, and piecewise definitions

When a function is given by a formula and no domain is stated, its *natural domain*(自然定义域)is the largest subset of $\mathbb{R}$ on which the formula gives a real value. This convention must be replaced by the actual physical or mathematical domain when context imposes additional restrictions.

Example 1.7 Determining a natural domain Find the natural domain of

$$f(x) = \sqrt{\frac{1+x}{1-x}}.$$

Solution *We need $x \neq 1$ and*

$$\frac{1+x}{1-x} \geq 0.$$

The critical points are -1 and 1. A sign analysis gives

$$-1 \leq x < 1.$$

Thus the natural domain is $[-1, 1)$.

> ***Check.*** Notice that solving only $1 + x \geq 0$ would miss the sign of the denominator.

Definition 1.7

A piecewise-defined function(分段函数)*is a single function whose values are specified by different formulas on different parts of its domain.* ♣

For instance,

$$T(x) = \begin{cases} 0, & 0 \leq x \leq 5000, \\ 0.03(x - 5000), & 5000 < x \leq 8000, \end{cases}$$

defines one function on $[0, 8000]$. At an endpoint where two pieces might meet, the inequalities must make clear which formula applies.

Example 1.8 Range depends on the domain Let $f(x) = x^2$. Determine its range when the domain is respectively $\mathbb{R}$, $[-2, 1]$, and $(2, +\infty)$.

Solution *On $\mathbb{R}$, the range is $[0, +\infty)$. On $[-2, 1]$, the range is $[0, 4]$. On $(2, +\infty)$, the range is $(4, +\infty)$.*

> ***Interpretation.*** The formula alone does not determine the range; the domain matters.

1.2.3 Operations and composition

Let $f\colon X \to \mathbb{R}$ and $g\colon Z \to \mathbb{R}$. Their sum, difference, and product are defined on $X \cap Z$ by

$$(f+g)(x) = f(x) + g(x),$$
$$(f-g)(x) = f(x) - g(x),$$
$$(fg)(x) = f(x)g(x).$$

The quotient is defined by

$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}$$

on

$$\{x \in X \cap Z \mid g(x) \neq 0\}.$$

Definition 1.8

Let $g\colon X \to Y$ and $f\colon Y_0 \to Z$. The composition (复合函数) *of f after g is*

$$(f \circ g)(x) = f(g(x)).$$

Its domain is

$$\{x \in X \mid g(x) \in Y_0\}.$$

♣

Composition is performed from right to left: first apply g, then apply f. It is associative whenever the compositions are defined, but it is generally not commutative.

Example 1.9 Composition changes the domain Let

$$f(t) = \sqrt{t}, \qquad g(x) = x^2 - 1.$$

Determine both compositions and their domains.

Solution *Then*

$$(f \circ g)(x) = \sqrt{x^2 - 1}$$

has domain

$$(-\infty, -1] \cup [1, +\infty).$$

In the other order,

$$(g \circ f)(x) = x - 1$$

has domain $[0, +\infty)$*, not all of* $\mathbb{R}$*.*

> ***Domain Check.*** Although the final formula $x - 1$ makes sense for every real x, the composition remembers that $\sqrt{x}$ was evaluated first.

> ***Important Distinction.*** Algebraic simplification does not automatically enlarge the domain of a function. For example,
>
> $$\frac{x^2 - 1}{x - 1} = x + 1$$
>
> only for $x \neq 1$. The expression on the left and the function $x \mapsto x + 1$ on $\mathbb{R}$ are not the same function.

1.2.4 Basic elementary functions

The standard building blocks of one-variable calculus are the basic elementary functions. Their domains must always be kept in view.

Table 1.1: Basic elementary functions and their standard real domains.

Type	Typical function	Standard domain
Constant	$f(x) = c$	$\mathbb{R}$
Power	$f(x) = x^\alpha$	depends on α
Exponential	$f(x) = a^x$, $a > 0$, $a \neq 1$	$\mathbb{R}$
Logarithmic	$f(x) = \log_a x$, $a > 0$, $a \neq 1$	$(0, +\infty)$
Trigonometric	$\sin x$, $\cos x$	$\mathbb{R}$
Trigonometric	$\tan x$	$\mathbb{R} \setminus \{\frac{\pi}{2} + k\pi \mid k \in \mathbb{Z}\}$
Inverse trigonometric	$\arcsin x$, $\arccos x$	$[-1, 1]$
Inverse trigonometric	$\arctan x$	$\mathbb{R}$

For real powers, x^α is always defined for $x > 0$ by

$$x^\alpha = \exp(\alpha \ln x).$$

The maximal real domain may be larger for special exponents. For example, $x^{1/3}$ is defined on $\mathbb{R}$, $x^{1/2}$ on $[0, +\infty)$, and x^{-2} on $\mathbb{R} \setminus \{0\}$. When $\alpha = m/n$ is in lowest terms, the parity of n and the sign of m determine whether negative inputs and the input 0 are allowed.

The natural logarithm is denoted by $\ln x$. The constant e and its construction from a sequence are developed in Section 1.3. Angles in trigonometric functions are measured in radians.

Definition 1.9

An elementary function(初等函数)*is a function obtained from basic elementary functions by finitely many arithmetic operations and compositions, on the domain where the resulting expression is defined.* ♣

Two useful functions built from exponentials are the hyperbolic sine and cosine:

$$\sinh x = \frac{e^x - e^{-x}}{2}, \qquad \cosh x = \frac{e^x + e^{-x}}{2}.$$

Direct calculation gives

$$\cosh^2 x - \sinh^2 x = 1.$$

These functions will later appear naturally in differential equations and integration.

Not every function is elementary, and not every function is represented by a smooth curve. The following examples are especially useful.

Example 1.10 Floor and fractional-part functions The *floor function*(取整函数)is

$$x \longmapsto \lfloor x \rfloor,$$

where $\lfloor x \rfloor$ is the greatest integer not exceeding x. Its defining property is

$$\lfloor x \rfloor \le x < \lfloor x \rfloor + 1.$$

The *fractional part*(小数部分)of x is

$$\{x\} = x - \lfloor x \rfloor \in [0, 1).$$

For negative inputs, this differs from simply retaining the digits after the decimal point; for example, $\{-3.14\} = 0.86$.

Example 1.11 Sign and Dirichlet functions The *sign function* (符号函数) is

$$\operatorname{sgn} x = \begin{cases} 1, & x > 0, \\ 0, & x = 0, \\ -1, & x < 0. \end{cases}$$

The *Dirichlet function* (狄利克雷函数) is

$$D(x) = \begin{cases} 1, & x \in \mathbb{Q}, \\ 0, & x \in \mathbb{R} \setminus \mathbb{Q}. \end{cases}$$

By Proposition 1.4, every open interval contains both rational and irrational numbers. Thus the graph of D consists of dense points on the two horizontal lines $y = 0$ and $y = 1$; it is not a curve in the usual visual sense. This example shows why a function must be defined as a rule rather than as a formula or a drawn curve.

1.2.5 Qualitative properties and graph transformations

Formulas give exact values, but calculus also studies the overall behaviour of a function. The following properties describe that behaviour without requiring us to calculate every value.

Definition 1.10

Let $f\colon X \to \mathbb{R}$.

(i) On an interval $I \subseteq X$, the function is nondecreasing *if*

$$x_1 < x_2 \quad \Longrightarrow \quad f(x_1) \le f(x_2),$$

and nonincreasing *if the final inequality is reversed. A function of either type is a* monotone function (单调函数) *on I. Replacing $\le$ or $\ge$ by a strict inequality defines* strictly increasing *and* strictly decreasing *functions.*

(ii) Suppose X is symmetric about 0. The function is an even function (偶函数) *if $f(-x) = f(x)$ for every $x \in X$, and an* odd function (奇函数) *if $f(-x) = -f(x)$ for every $x \in X$.*

(iii) A number $T > 0$ is a period *of f if*

$$x \in X \iff x + T \in X, \qquad f(x+T) = f(x) \quad (x \in X).$$

A function having a period is a periodic function (周期函数).

♣

Even functions have graphs symmetric about the y-axis, while odd functions have graphs symmetric about the origin. The function $x \mapsto |x|$ is even, nonincreasing on $(-\infty, 0]$, and nondecreasing on $[0, +\infty)$. The sine function is odd and 2π-periodic; the cosine function is even and 2π-periodic.

These properties are logically independent. For example, $\tan x$ is periodic but unbounded on its domain. The function

$$g(x) = \frac{1}{1+x^2}$$

is bounded but not periodic: if $T > 0$, then $g(T) < g(0)$, so T cannot be a period.

Graph transformations can be read by tracking what happens to a typical point $(u, f(u))$. If the domain of f is D, then the following table records both the geometry and the new domain.

When $a < 0$, the third transformation also reflects the graph across the x-axis, and the fourth reflects it across the y-axis. The domain must be transformed together with the graph; a visually correct shift with an unchanged domain may define the wrong function.

Table 1.2: Basic transformations of the graph $y = f(x)$.

New function	Effect on the graph	Domain
$f(x) + c$	vertical shift by c	D
$f(x - c)$	horizontal shift by c	$\{u + c \mid u \in D\}$
$af(x)$	vertical scale by $\lvert a\rvert$	D
$f(ax)$, $a \neq 0$	horizontal scale by $1/\lvert a\rvert$	$\{x \mid ax \in D\}$

> ***Question.*** Does either implication hold: periodic $\Rightarrow$ bounded or bounded $\Rightarrow$ periodic? Can one implication be repaired by adding a natural hypothesis?

1.2.6 Injectivity, surjectivity, and inverse functions

Definition 1.11

Let $f\colon X \to Y$.

(i) The function is injective (单射) *if*

$$f(x_1) = f(x_2) \implies x_1 = x_2.$$

(ii) It is surjective (满射) *if $f(X) = Y$.*

(iii) It is bijective (双射) *if it is both injective and surjective.* ♣

Injectivity says that distinct inputs never merge into the same output. Surjectivity depends on the chosen codomain: the function $x \mapsto x^2$ is surjective from $\mathbb{R}$ to $[0, +\infty)$ but not from $\mathbb{R}$ to $\mathbb{R}$.

Theorem 1.2

A function $f\colon X \to Y$ has an inverse function (反函数) *$f^{-1}\colon Y \to X$ if and only if f is bijective. In that case,*

$$f^{-1}(f(x)) = x \quad (x \in X), \qquad f(f^{-1}(y)) = y \quad (y \in Y).$$ ♡

> ***Idea.*** Surjectivity guarantees that each proposed inverse input has a preimage, while injectivity guarantees that this preimage is unique. Together they make the inverse assignment a function.

Proof Suppose first that f is bijective. For every $y \in Y$, surjectivity gives at least one $x \in X$ with $f(x) = y$, and injectivity makes this x unique. Define $f^{-1}(y)$ to be that unique x. The two displayed identities follow directly.

Conversely, suppose that an inverse $f^{-1}\colon Y \to X$ exists. For every $y \in Y$, the element $x = f^{-1}(y)$ satisfies $f(x) = y$, so f is surjective. If $f(x_1) = f(x_2)$, applying f^{-1} gives $x_1 = x_2$, so f is injective. Thus f is bijective. □

An injective function $f\colon X \to Y$ is automatically a bijection from X onto its range $f(X)$, and hence has an inverse

$$f^{-1}\colon f(X) \to X.$$

For example, $x \mapsto x^2$ is not injective on $\mathbb{R}$, but its restriction to $[0, +\infty)$ is injective and has inverse $x \mapsto \sqrt{x}$. Likewise, $\sin x$ is not injective on $\mathbb{R}$. Its restriction to $[-\pi/2, \pi/2]$ is bijective onto $[-1, 1]$, and the inverse of this restricted function is $\arcsin x$.

Example 1.12 Constructing an inverse while tracking its domain Let

$$f\colon (0, +\infty) \to \mathbb{R}, \qquad f(x) = \frac{2}{x} - \frac{x}{2}.$$

Prove that f is bijective and find f^{-1}.

Proof If $0 < x_1 < x_2$, then

$$\frac{2}{x_1} > \frac{2}{x_2} \qquad \text{and} \qquad -\frac{x_1}{2} > -\frac{x_2}{2},$$

so $f(x_1) > f(x_2)$. Thus f is injective.

For any $y \in \mathbb{R}$, the equation $y = f(x)$ is equivalent to

$$x^2 + 2yx - 4 = 0.$$

Its roots are $-y \pm \sqrt{y^2+4}$. Exactly one is positive, namely

$$x = -y + \sqrt{y^2+4},$$

because $\sqrt{y^2+4} > |y|$. Hence every $y \in \mathbb{R}$ has exactly one preimage in $(0, +\infty)$, and

$$f^{-1}(y) = -y + \sqrt{y^2+4} \qquad (y \in \mathbb{R}).$$

□

The graph of an inverse is obtained by interchanging the coordinates in every ordered pair. It is therefore the reflection of the original graph across the line $y = x$.

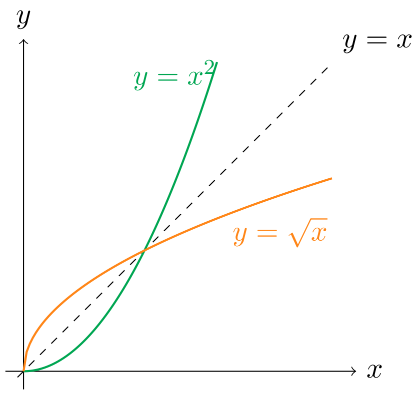


Figure 1.1: The graphs of $x \mapsto x^2$ on $[0, +\infty)$ and its inverse $x \mapsto \sqrt{x}$ are reflections across $y = x$.

> ***Comment.*** The notation f^{-1} denotes an inverse function, not a reciprocal. In general,
>
> $$f^{-1}(x) \neq \frac{1}{f(x)}.$$

1.2.7 Sequences and bounded functions

A sequence

$$a_1, a_2, a_3, \ldots$$

is a function whose domain is $\mathbb{N}_{>0}$:

$$a\colon \mathbb{N}_{>0} \to \mathbb{R}, \qquad a(n) = a_n.$$

The order of the terms matters, so a sequence is not merely the set $\{a_n \mid n \geq 1\}$. This viewpoint connects the present section directly with sequence limits in Section 1.3.

Definition 1.12

Let $f\colon X \to \mathbb{R}$. The function f is bounded above *on X if there exists $M \in \mathbb{R}$ such that*

$$f(x) \leq M$$

for every $x \in X$*. It is* bounded below *on* X *if there exists* $m \in \mathbb{R}$ *such that*

$$m \le f(x)$$

for every $x \in X$*. It is a* bounded function(有界函数) *on* X *if it is bounded both above and below.* ♣

Boundedness always refers to a specified domain. The function $f(x) = 1/x$ is bounded on $[1, +\infty)$ but unbounded on $(0, 1]$. The fact that $f(x)$ is a finite real number for every x in its domain does not supply one common bound valid for all inputs.

Proposition 1.7

A function $f : X \to \mathbb{R}$ *is bounded if and only if there exists* $C \ge 0$ *such that*

$$|f(x)| \le C$$

for every $x \in X$*.* ♠

Proof If f is bounded, choose $m, M \in \mathbb{R}$ such that

$$m \le f(x) \le M$$

for every $x \in X$. With $C = \max\{|m|, |M|\}$, we have $|f(x)| \le C$.

Conversely, if $|f(x)| \le C$, then

$$-C \le f(x) \le C$$

for every $x \in X$, so f is bounded. □

1.2.8 Section Summary and Problem-solving Strategies

The essential data of a function are its domain, codomain, and assignment rule. Formulas, graphs, and tables are representations of that data, not replacements for it. When working with a function, use the following order.

1. Identify the domain before simplifying or calculating.
2. Distinguish the codomain from the actual range.
3. For a composition, work from the inside out and impose every intermediate domain condition.
4. For an inverse, first check injectivity and determine the range that will become the inverse domain.
5. For qualitative properties, combine algebraic tests with graph symmetry, but verify conclusions on the stated domain.
6. For a graph transformation, transform the domain together with the points of the graph.
7. For boundedness, search for one constant valid for every input in the stated domain.

A compact trigonometric and hyperbolic toolkit

The formulas below are most useful when they are organized by their source rather than memorized as unrelated identities. The addition formulas generate the product-to-sum and sum-to-product formulas by addition, subtraction, and a change of variables.

For real u, v (and with nonzero denominators where tangent occurs),

Example 1.13 Derive a transformation instead of memorizing it Derive one product-to-sum formula and its corresponding sum-to-product formula.

Table 1.3: Trigonometric addition and subtraction formulas.

Function	Formula
Sine	$\sin(u \pm v) = \sin u \cos v \pm \cos u \sin v$
Cosine	$\cos(u \pm v) = \cos u \cos v \mp \sin u \sin v$
Tangent	$\tan(u \pm v) = \dfrac{\tan u \pm \tan v}{1 \mp \tan u \tan v}$

Table 1.4: Trigonometric product-to-sum and sum-to-product formulas.

Product to sum	Sum to product
$\sin u \sin v = \dfrac{\cos(u-v) - \cos(u+v)}{2}$	$\sin u + \sin v = 2 \sin \dfrac{u+v}{2} \cos \dfrac{u-v}{2}$
$\cos u \cos v = \dfrac{\cos(u-v) + \cos(u+v)}{2}$	$\sin u - \sin v = 2 \cos \dfrac{u+v}{2} \sin \dfrac{u-v}{2}$
$\sin u \cos v = \dfrac{\sin(u+v) + \sin(u-v)}{2}$	$\cos u + \cos v = 2 \cos \dfrac{u+v}{2} \cos \dfrac{u-v}{2}$
$\cos u \sin v = \dfrac{\sin(u+v) - \sin(u-v)}{2}$	$\cos u - \cos v = -2 \sin \dfrac{u+v}{2} \sin \dfrac{u-v}{2}$

Proof Adding the two sine addition formulas gives

$$\begin{aligned} \sin(u+v) + \sin(u-v) &= (\sin u \cos v + \cos u \sin v) \\ &\quad + (\sin u \cos v - \cos u \sin v) \\ &= 2 \sin u \cos v. \end{aligned}$$

Therefore

$$\sin u \cos v = \frac{\sin(u+v) + \sin(u-v)}{2}.$$

Now put $u = (x+y)/2$ and $v = (x-y)/2$. Then $u+v = x$ and $u-v = y$, so the same identity becomes

$$\sin x + \sin y = 2 \sin \frac{x+y}{2} \cos \frac{x-y}{2}.$$

□

Technique. The other entries in Table 1.4 follow from the same two operations: add or subtract a pair of angle-sum identities, then make a half-sum and half-difference substitution.

Inverse trigonometric functions require attention to their principal ranges. For example,

$$\arcsin x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right], \qquad \arccos x \in [0, \pi], \qquad \arctan x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right).$$

Consequently,

$$\begin{aligned} \arcsin x + \arccos x &= \frac{\pi}{2} && (-1 \le x \le 1), \\ \arctan x + \arctan(1/x) &= \begin{cases} \pi/2, & x > 0, \\ -\pi/2, & x < 0. \end{cases} \end{aligned}$$

Although

$$\tan(\arctan x + \arctan y) = \frac{x+y}{1-xy}$$

when $xy \neq 1$, replacing the left side by a single $\arctan$ may require adding or subtracting π. This branch issue is why inverse trigonometric formulas should not be manipulated as if the inverse functions were globally periodic. There are no literal product-to-sum formulas for $\arcsin$, $\arccos$, or $\arctan$; those transformations belong to the direct trigonometric functions.

The hyperbolic functions are defined by

$$\sinh x = \frac{e^x - e^{-x}}{2}, \qquad \cosh x = \frac{e^x + e^{-x}}{2}, \qquad \tanh x = \frac{\sinh x}{\cosh x}.$$

Their addition formulas have the same algebraic pattern as the trigonometric ones, with the decisive sign change in the cosine analogue:

$$\begin{aligned}
\sinh(u \pm v) &= \sinh u \cosh v \pm \cosh u \sinh v, \\
\cosh(u \pm v) &= \cosh u \cosh v \pm \sinh u \sinh v, \\
\tanh(u \pm v) &= \frac{\tanh u \pm \tanh v}{1 \pm \tanh u \tanh v}.
\end{aligned}$$

Table 1.5: Hyperbolic product-to-sum and sum-to-product formulas.

Product to sum	Sum to product
$\sinh u \sinh v = \dfrac{\cosh(u+v) - \cosh(u-v)}{2}$	$\sinh u + \sinh v = 2 \sinh \dfrac{u+v}{2} \cosh \dfrac{u-v}{2}$
$\cosh u \cosh v = \dfrac{\cosh(u+v) + \cosh(u-v)}{2}$	$\sinh u - \sinh v = 2 \cosh \dfrac{u+v}{2} \sinh \dfrac{u-v}{2}$
$\sinh u \cosh v = \dfrac{\sinh(u+v) + \sinh(u-v)}{2}$	$\cosh u + \cosh v = 2 \cosh \dfrac{u+v}{2} \cosh \dfrac{u-v}{2}$
$\cosh u \sinh v = \dfrac{\sinh(u+v) - \sinh(u-v)}{2}$	$\cosh u - \cosh v = 2 \sinh \dfrac{u+v}{2} \sinh \dfrac{u-v}{2}$

The principal inverse hyperbolic functions have especially useful logarithmic forms:

$$\begin{aligned}
\operatorname{arsinh} x &= \ln\left(x + \sqrt{x^2+1}\right) && (x \in \mathbb{R}), \\
\operatorname{arcosh} x &= \ln\left(x + \sqrt{x^2-1}\right) && (x \geq 1), \\
\operatorname{artanh} x &= \frac{1}{2} \ln\left(\frac{1+x}{1-x}\right) && (|x| < 1).
\end{aligned}$$

As with inverse trigonometric functions, these identities include domain and branch information; they are not obtained by formally placing "arc" in front of a direct-function identity.

Example 1.14 Why the functions are called hyperbolic Show that $(\cosh t, \sinh t)$ lies on the unit hyperbola and derive a hyperbolic addition formula directly from exponentials.

Proof From the definitions,

$$\begin{aligned}
\cosh^2 t - \sinh^2 t &= \frac{(e^t + e^{-t})^2 - (e^t - e^{-t})^2}{4} \\
&= 1.
\end{aligned}$$

Thus the point $(x, y) = (\cosh t, \sinh t)$ parametrizes the right-hand branch of

$$x^2 - y^2 = 1,$$

just as $(\cos t, \sin t)$ parametrizes the unit circle $x^2 + y^2 = 1$. Moreover,

$$\begin{aligned}
\sinh(u+v) &= \frac{e^{u+v} - e^{-(u+v)}}{2} \\
&= \frac{(e^u - e^{-u})(e^v + e^{-v}) + (e^u + e^{-u})(e^v - e^{-v})}{4} \\
&= \sinh u \cosh v + \cosh u \sinh v.
\end{aligned}$$

Adding and subtracting the formulas for $u + v$ and $u - v$ then produces Table 1.5. $\square$

Geometric and Complex Viewpoints. The word "hyperbolic" therefore reflects geometry, not merely notation. There is also a complex-variable viewpoint: Euler's formulas give

$$\cos(ix) = \cosh x, \qquad \sin(ix) = i \sinh x.$$

Thus the circle and hyperbola identities are connected by the imaginary substitution $x \mapsto ix$; this is a useful structural explanation, although complex exponentials are not needed to use the real formulas above.

Common Pitfall. An expression and a function are not the same thing. The expression describes an assignment rule, but the function also includes its domain. Many apparent contradictions about ranges, inverses, and compositions disappear once the domain is written explicitly.

Looking Ahead. A function limit will ask what $f(x)$ does when x is near a point, possibly without using the value at that point. The distinction between formula and function remains essential because a limit can depend on which domain points are available near the point of approach.

Exercise 1.2

1. Find the natural domains of

$$\ln(x^2 - 1), \qquad \frac{\sqrt{2x+1}}{x-3}, \qquad \arccos(2\sin x).$$

2. For $f(x) = x^2 - 2x$ on each of the domains $\mathbb{R}$, $[0, 3]$, and $(2, +\infty)$, determine the range.
3. Let $f(x) = 1/(x - 1)$ and $g(x) = \sqrt{x+2}$. Find formulas and domains for $f + g$, fg, f/g, $f \circ g$, and $g \circ f$.
4. Give two functions represented by the same formula but having different ranges. Explain why they are not equal as functions.
5. Prove that composition of functions is associative whenever all compositions involved are defined. Give an example showing that composition need not be commutative.
6. Determine $\lfloor x \rfloor$ and $\{x\}$ for

$$x = 3.7, \quad x = -0.2, \quad x = -4.$$

   Prove that $\{x + 1\} = \{x\}$ for all $x \in \mathbb{R}$.
7. Determine the parity, periods, and intervals of monotonicity that apply to

$$f(x) = x^2, \qquad g(x) = \cos(2x), \qquad h(x) = \frac{1}{1+x^2}.$$

   Give a bounded nonperiodic function and a periodic unbounded function.
8. Starting from the graph of $y = f(x)$, describe the transformations and domains of

$$y = 2f(x-1) - 3, \qquad y = f(-2x).$$

9. Determine whether each map is injective, surjective, or bijective:

$$x \mapsto x^2 : \mathbb{R} \to [0, +\infty), \qquad x \mapsto x^3 : \mathbb{R} \to \mathbb{R}, \qquad x \mapsto e^x : \mathbb{R} \to (0, +\infty).$$

10. Find the inverse of

$$f(x) = \frac{x-1}{x+2}$$

   after specifying a domain and codomain for which f is bijective.
11. Prove directly from the definitions that if f and g are bounded on the same nonempty set X, then $f + g$ and fg are bounded on X.

12. Give an example of a function that is bounded above but not bounded below, and another that is bounded on one interval but unbounded on a smaller-looking interval with a missing endpoint.
13. Explain why the Dirichlet function cannot be represented by a continuous curve, using Proposition 1.4.
14. Determine the natural domains, ranges, parity, and monotonicity of
$$\sinh x, \qquad \cosh x, \qquad \tanh x.$$
Use only their exponential definitions and elementary inequalities.
15. Starting with $y = \sinh x$ and $y = \tanh x$, solve algebraically for x and thereby verify
$$\operatorname{arsinh} y = \ln\left(y + \sqrt{y^2+1}\right), \qquad \operatorname{artanh} y = \frac{1}{2}\ln\left(\frac{1+y}{1-y}\right).$$
State the domain of each inverse function.
16. Derive the product-to-sum formulas for $\sin u \sin v$ and $\sinh u \sinh v$ from the corresponding addition formulas. Then use them to rewrite
$$\sin(3x)\sin x, \qquad \sinh(3x)\sinh x$$
as differences of two cosine-type functions.
17. Sketch the graph of
$$f(x) = |x-1| + |x+1|$$
by first writing a piecewise formula. Determine its range, parity, and intervals of monotonicity, and find all x for which $f(x) = 3$.
18. Consider $f(x) = x + 1/x$. Show that its restrictions to $(0, 1]$ and $[1, +\infty)$ are one-to-one, determine their ranges, and find an inverse formula for each restriction.
19. Solve on the stated domains:
$$\arcsin(2x-1) = \frac{\pi}{6}, \qquad \arctan x + \arctan\frac{1}{x} = \frac{\pi}{2} \quad (x > 0).$$
Explain why branch information is essential in the second equation.
20. Let $f(x) = \lfloor x \rfloor$ and $g(x) = \{x\}$. Determine formulas and ranges for $f \circ g$, $g \circ f$, and $g \circ g$. Which of these functions are periodic, and what are their periods?

Supplement Exercises

S1. For $x \notin 2\pi\mathbb{Z}$, prove the finite-sum identity
$$\sum_{k=1}^{n} \cos(kx) = \frac{\sin\big((2n+1)x/2\big) - \sin(x/2)}{2\sin(x/2)}.$$
Obtain it by converting each cosine into a difference of two sines so that the sum telescopes.

S2. Prove that, for $x \ge 0$,
$$2\arctan\sqrt{x} - \arcsin\frac{x-1}{x+1} = \frac{\pi}{2}.$$
Your proof must justify the ranges of all inverse trigonometric functions involved.

S3. Derive the tangent half-angle identities
$$\sin\theta = \frac{2t}{1+t^2}, \qquad \cos\theta = \frac{1-t^2}{1+t^2}, \qquad t = \tan\frac{\theta}{2},$$
and describe precisely which values of θ require separate treatment.

S4. Let D be the Dirichlet function, equal to 1 on $\mathbb{Q}$ and 0 on $\mathbb{R}\setminus\mathbb{Q}$. Determine the ranges of $xD(x)$, $D(x^2)$, and $D(D(x))$. Decide which of these functions are even, odd, periodic, or bounded.

S5. Let

$$f(x) = \frac{ax+b}{cx-a}, \qquad a^2 + bc \neq 0.$$

Choose the natural domain and codomain so that f is a bijection, and prove that $f^{-1} = f$.

> ***Optional Hint.*** Associate f with the matrix $\begin{pmatrix} a & b \\ c & -a \end{pmatrix}$ and square the matrix.

S6. Determine all real constants a, b for which

$$f(x) = \ln(a + be^x)$$

is well-defined on a nondegenerate interval I, maps I onto itself, and satisfies $f(f(x)) = x$ for every $x \in I$. State the maximal interval in each case.

1.3 Limits of Sequences

重点内容与证明方法

- *Understand a sequence as an ordered function on the positive integers.*
- *Read the quantifiers in the ε–N definition and construct direct proofs in the correct order.*
- *Establish uniqueness, boundedness, order properties, and arithmetic laws for limits.*
- *Use domination, squeeze arguments, normalization, and subsequences to prove convergence or divergence.*
- *Apply completeness through the monotone sequence theorem and define the constant e.*
- *Explore the Cauchy criterion and the Bolzano–Weierstrass theorem as completeness-based extensions.*

Limits turn approximation into an exact mathematical method. The area of a circle may be approached by areas of inscribed polygons, an irrational number by rational decimals, and a solution of an equation by repeated numerical approximations. In each case we obtain an ordered list of numbers and ask whether the terms settle near one fixed real number.

1.3.1 Sequences and the meaning of convergence

Definition 1.13

A real sequence (数列) *is a function*

$$a\colon \mathbb{N}_{>0} \to \mathbb{R}.$$

Its value at n is denoted by a_n, and the sequence is written $(a_n)_{n\geq 1}$ or simply (a_n). ♣

The index records order. For example, $(1, -1, 1, -1, \dots)$ and $(-1, 1, -1, 1, \dots)$ have the same set of values but are different sequences. Some sequences are given by an explicit formula, while others are defined recursively from earlier terms.

Consider three examples:

$$a_n = \frac{1}{n}, \qquad b_n = 2 + \frac{(-1)^n}{n}, \qquad c_n = (-1)^n.$$

The first approaches 0 from one side. The second approaches 2 while alternating around it. The third continues to jump between -1 and 1 and does not approach any single number. Monotonicity is therefore not part of convergence; eventual closeness is.

Definition 1.14

Let $(a_n)_{n\geq 1}$ be a sequence and let $L \in \mathbb{R}$. We say that (a_n) converges (收敛) *to L if for every $\varepsilon > 0$ there exists $N \in \mathbb{N}_{>0}$ such that*

$$n \geq N \quad \Longrightarrow \quad |a_n - L| < \varepsilon.$$

In this case we write

$$\lim_{n\to\infty} a_n = L \qquad \text{or} \qquad a_n \to L.$$

The number L is the limit (极限) *of the sequence. A sequence that has no real limit is called* divergent (发散). ♣

The definition has the logical form

$$\forall \varepsilon > 0 \quad \exists N \in \mathbb{N}_{>0} \quad \forall n \geq N : \quad |a_n - L| < \varepsilon.$$

The number N may depend on ε, but it must not depend on the later index n. We only need to find a suitable N; it need not be the smallest possible one.

The exact negation of the proposed limit $a_n \to L$ is

$$\exists \varepsilon_0 > 0 \quad \forall N \in \mathbb{N}_{>0} \quad \exists n \geq N : \quad |a_n - L| \geq \varepsilon_0.$$

Thus a divergence proof may fix one positive tolerance and find bad terms arbitrarily far out in the sequence. The tolerance must be independent of N; choosing a new tolerance for each tail would not negate convergence.

> ***Disproof Pattern.*** To show that a_n does not converge to a proposed value L, look for one number $\varepsilon_0 > 0$ and infinitely late indices whose terms remain at least ε_0 away from L. Two subsequences with different limits are a particularly efficient way to obtain such terms.

Geometrically, $|a_n - L| < \varepsilon$ means

$$L - \varepsilon < a_n < L + \varepsilon.$$

No matter how narrow the horizontal band around L is chosen, all sufficiently late terms must lie inside it.

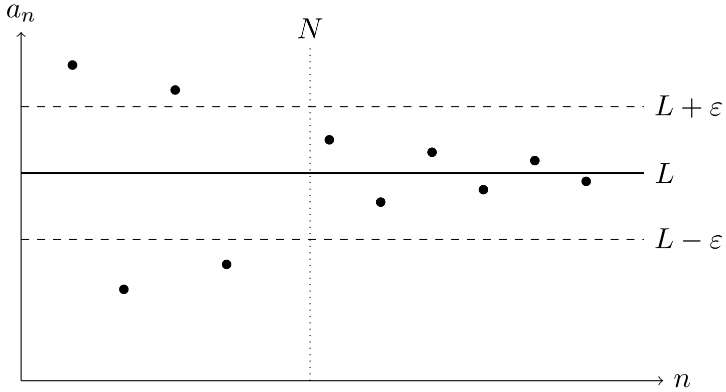


Figure 1.2: After a suitable index N, every term lies in the ε-band around L; the finitely many earlier terms are irrelevant to convergence.

> ***Question.*** Why does the definition require all $n \geq N$, rather than merely infinitely many values of n?

1.3.2 Direct proofs from the definition

An ε–N proof is usually discovered backward and written forward. We first ask what condition on n would guarantee $|a_n - L| < \varepsilon$. After determining N from this condition, the formal proof starts with an arbitrary $\varepsilon > 0$, chooses that N, and verifies the implication.

Example 1.15 A direct ε–N proof Prove that

$$a_n = 2 + \frac{(-1)^n}{n} \quad \longrightarrow \quad 2.$$

> ***Idea.*** The oscillation causes no difficulty because its magnitude is controlled by $1/n$:
> $$|a_n - 2| = \frac{1}{n}.$$
> Thus it is enough to make $n > 1/\varepsilon$.

Proof Let $\varepsilon > 0$. Choose

$$N = \left\lceil \frac{1}{\varepsilon} \right\rceil + 1.$$

If $n \geq N$, then $n > 1/\varepsilon$, and therefore

$$|a_n - 2| = \frac{|(-1)^n|}{n} = \frac{1}{n} < \varepsilon.$$

Hence $a_n \to 2$. □

Exact solution of the target inequality is often unnecessary. A simpler upper bound may reveal a workable N.

Example 1.16 Estimate first, choose N second Prove that

$$\lim_{n\to\infty} \frac{n+2}{n^2+1} = 0.$$

Proof For $n \geq 2$, we have $n + 2 \leq 2n$ and $n^2 + 1 \geq n^2$, so

$$0 \leq \frac{n+2}{n^2+1} \leq \frac{2}{n}.$$

Given $\varepsilon > 0$, choose

$$N = \max\left\{2, \left\lceil \frac{2}{\varepsilon} \right\rceil + 1\right\}.$$

Then, for $n \geq N$,

$$0 \leq \frac{n+2}{n^2+1} \leq \frac{2}{n} < \varepsilon.$$

This proves the limit directly from the definition. □

The following basic limits will be used repeatedly.

Proposition 1.8

Let $p > 0$, let $q \in \mathbb{R}$ satisfy $|q| < 1$, and let $a > 0$. Then

$$\lim_{n\to\infty} \frac{1}{n^p} = 0,$$

$$\lim_{n\to\infty} q^n = 0,$$

$$\lim_{n\to\infty} a^{1/n} = 1.$$

Proof Given $\varepsilon > 0$, the Archimedean property provides N such that

$$N > \varepsilon^{-1/p}.$$

Then $n \geq N$ implies $1/n^p < \varepsilon$, proving the first limit.

The second limit is immediate when $q = 0$. If $0 < |q| < 1$, write

$$\frac{1}{|q|} = 1 + h$$

with $h > 0$. Bernoulli's inequality gives $(1 + h)^n \geq 1 + nh$, so

$$0 < |q|^n = \frac{1}{(1+h)^n} \leq \frac{1}{1+nh} < \frac{1}{nh}.$$

The first limit, with $p = 1$, now proves $q^n \to 0$.

Suppose first that $a \geq 1$, and set $h_n = a^{1/n} - 1 \geq 0$. Since

$$a = (1 + h_n)^n \geq 1 + nh_n,$$

we obtain

$$0 \leq h_n \leq \frac{a-1}{n} \to 0.$$

Thus $a^{1/n} = 1 + h_n \to 1$. If $0 < a < 1$, put $b = 1/a > 1$. The proved case gives $b^{1/n} \to 1$, and

$$0 \le 1 - a^{1/n} = \frac{b^{1/n} - 1}{b^{1/n}} \le b^{1/n} - 1 \to 0.$$

Hence $a^{1/n} \to 1$ without using the quotient law, which will be proved later. □

1.3.3 First structural consequences

Theorem 1.3

A convergent sequence has exactly one limit.

> ***Idea.*** Two distinct candidate limits have disjoint sufficiently small neighbourhoods. A late term of the sequence cannot belong to both neighbourhoods at once.

Proof Suppose that $a_n \to L$ and $a_n \to M$. If $L \ne M$, set

$$\varepsilon = \frac{|L - M|}{3} > 0.$$

There exist $N_1, N_2 \in \mathbb{N}_{>0}$ such that

$$n \ge N_1 \implies |a_n - L| < \varepsilon, \qquad n \ge N_2 \implies |a_n - M| < \varepsilon.$$

For $n \ge \max\{N_1, N_2\}$, the triangle inequality gives

$$|L - M| \le |L - a_n| + |a_n - M| < 2\varepsilon = \frac{2}{3}|L - M|,$$

a contradiction. Therefore $L = M$. □

Proposition 1.9

Changing, adding, or deleting finitely many initial terms does not affect whether a sequence converges or what its limit is. In particular, if $a_n \to L$ and $k \in \mathbb{N}$ is fixed, then

$$a_{n+k} \to L.$$

Proof Suppose two sequences agree for every $n \ge n_0$. For a given $\varepsilon > 0$, choose N for the convergent sequence and replace it by $\max\{N, n_0\}$. The same ε-estimate then holds for the other sequence. The statement about shifts follows by taking a tail and reindexing it. □

Definition 1.15

A sequence (a_n) is a bounded sequence(有界数列) *if there exists $M \ge 0$ such that*

$$|a_n| \le M$$

for every $n \ge 1$.

Theorem 1.4

Every convergent sequence is bounded.

Proof Suppose $a_n \to L$. Taking $\varepsilon = 1$, choose N such that

$$n \ge N \quad\implies\quad |a_n - L| < 1.$$

For such n,

$$|a_n| \le |a_n - L| + |L| < 1 + |L|.$$

Now choose

$$M = \max\{|a_1|, \dots, |a_{N-1}|, 1 + |L|\},$$

with the finite list omitted if $N = 1$. Then $|a_n| \le M$ for every $n \ge 1$. □

The converse is false. The sequence $((-1)^n)$ is bounded but divergent. Boundedness prevents escape to infinity, but it does not prevent persistent oscillation.

1.3.4 Arithmetic laws and squeeze arguments

Theorem 1.5

Suppose that $a_n \to A$ and $b_n \to B$, and let $c \in \mathbb{R}$. Then

$$a_n + b_n \to A + B,$$
$$a_n - b_n \to A - B,$$
$$ca_n \to cA,$$
$$a_n b_n \to AB.$$

If $B \neq 0$, then $b_n \neq 0$ for all sufficiently large n and

$$\frac{a_n}{b_n} \to \frac{A}{B}.$$

Proof Strategy. For sums, split the final error into two errors. For products, add and subtract a convenient intermediate term and use boundedness of a convergent sequence. For quotients, first prove that a denominator converging to a nonzero number stays away from zero.

Proof Given $\varepsilon > 0$, choose N so that for $n \ge N$,

$$|a_n - A| < \frac{\varepsilon}{2}, \qquad |b_n - B| < \frac{\varepsilon}{2}.$$

Then

$$|(a_n + b_n) - (A + B)| \le |a_n - A| + |b_n - B| < \varepsilon.$$

The difference and constant-multiple laws follow by the same estimate.

By Theorem 1.4, choose $M \ge 0$ such that $|a_n| \le M$ for all n. Let

$$C = M + |B| + 1.$$

Choose N so that, for $n \ge N$,

$$|a_n - A| < \frac{\varepsilon}{C}, \qquad |b_n - B| < \frac{\varepsilon}{C}.$$

Then

$$\begin{aligned} |a_n b_n - AB| &= |a_n(b_n - B) + B(a_n - A)| \\ &\le M\,|b_n - B| + |B|\,|a_n - A| \\ &< \frac{M + |B|}{C}\varepsilon < \varepsilon. \end{aligned}$$

Thus $a_n b_n \to AB$.

Finally suppose $B \neq 0$. For all sufficiently large n,

$$|b_n - B| < \frac{|B|}{2},$$

so the reverse triangle inequality gives

$$|b_n| > \frac{|B|}{2} > 0.$$

For these indices,

$$\left|\frac{1}{b_n} - \frac{1}{B}\right| = \frac{|b_n - B|}{|b_n|\,|B|} \le \frac{2}{|B|^2}\,|b_n - B| \to 0.$$

Therefore $1/b_n \to 1/B$, and the quotient law follows from the product law. □

Example 1.17 A rational expression Evaluate

$$\lim_{n\to\infty} \frac{n^2 + 3n + 10}{3n^2 - n + 2}.$$

Proof Using the arithmetic laws,

$$\lim_{n\to\infty} \frac{n^2 + 3n + 10}{3n^2 - n + 2} = \lim_{n\to\infty} \frac{1 + 3/n + 10/n^2}{3 - 1/n + 2/n^2} = \frac{1}{3}.$$

□

> ***Technique.*** Dividing by the highest power of n turns the expression into combinations of sequences with known limits.

Example 1.18 Rationalization reveals the dominant scale Evaluate

$$\lim_{n\to\infty} \left(\sqrt{n^2 + n} - n\right).$$

> ***Idea.*** The two terms both have size n, so their leading parts cancel. Multiplying by the conjugate turns this hidden cancellation into a quotient with a nonzero limiting denominator.

Proof Rationalization gives

$$\sqrt{n^2 + n} - n = \frac{n}{\sqrt{n^2 + n} + n} = \frac{1}{\sqrt{1 + 1/n} + 1}.$$

Moreover,

$$0 \le \sqrt{1 + \frac{1}{n}} - 1 = \frac{1/n}{\sqrt{1 + 1/n} + 1} \le \frac{1}{2n}.$$

Hence $\sqrt{1 + 1/n} \to 1$ by the squeeze theorem. The quotient law now yields

$$\lim_{n\to\infty} \left(\sqrt{n^2 + n} - n\right) = \frac{1}{2}.$$

□

> ***Caution.*** The hypotheses in Theorem 1.5 matter. From convergence of $(a_n + b_n)$ one cannot conclude that (a_n) and (b_n) converge. For example, $a_n = n$ and $b_n = -n$ both diverge, while $a_n + b_n = 0$. Likewise, the theorem concerns finitely many arithmetic operations; it does not justify termwise limiting of sums whose number of terms grows with n.

The following squeeze theorem(夹逼定理)turns a useful two-sided estimate into a limit.

Theorem 1.6 (Squeeze Theorem)

Suppose that for all sufficiently large n,

$$p_n \le q_n \le r_n.$$

If $p_n \to L$ and $r_n \to L$, then $q_n \to L$. ♡

Proof Let $\varepsilon > 0$. For all sufficiently large n,

$$L - \varepsilon < p_n \le q_n \le r_n < L + \varepsilon.$$

Therefore $|q_n - L| < \varepsilon$ for all sufficiently large n, which is exactly $q_n \to L$. □

Example 1.19 An oscillating sequence controlled by its size Prove that

$$\lim_{n\to\infty} \frac{\sin n}{n} = 0.$$

Proof Since

$$-\frac{1}{n} \le \frac{\sin n}{n} \le \frac{1}{n}$$

and both bounding sequences converge to 0, Theorem 1.6 gives

$$\lim_{n\to\infty} \frac{\sin n}{n} = 0.$$

□

> ***Interpretation.*** The numerator itself does not converge; the shrinking factor $1/n$ controls the oscillation.

Example 1.20 Dominate an oscillating term Prove that the sequence

$$a_n = \frac{n^{2/3}\sin n}{n+1}$$

converges to 0.

> ***Idea.*** Do not try to track the oscillation of $\sin n$; replace it by the uniform bound $|\sin n| \le 1$.

Proof

$$|a_n| \le \frac{n^{2/3}}{n+1} \le \frac{1}{n^{1/3}}.$$

The right-hand side tends to 0, so Theorem 1.6 gives $a_n \to 0$. □

> ***Technique.*** The reusable step is to replace a complicated factor by a majorant that has a known limit.

Example 1.21 Normalize a root of a sum Let $m \ge 2$ be a fixed integer and set

$$s_n = (2^n + 3^n + \cdots + m^n)^{1/n}.$$

Find $\lim_{n\to\infty} s_n$.

Proof The largest exponential supplies the natural scale:

$$m^n \le 2^n + 3^n + \cdots + m^n \le (m-1)m^n.$$

Taking positive nth roots gives

$$m \le s_n \le m(m-1)^{1/n}.$$

Since $(m-1)^{1/n} \to 1$, the squeeze theorem yields $s_n \to m$. □

> ***Technique.*** This is a typical instance of normalizing by the dominant term before estimating.

Proposition 1.10

Let $a > 1$ and let $k \in \mathbb{N}_{>0}$. Then

$$\lim_{n\to\infty} \frac{n^k}{a^n} = 0, \qquad \lim_{n\to\infty} \frac{a^n}{n!} = 0.$$

Thus exponentials eventually dominate fixed powers, while factorials eventually dominate fixed exponentials. ♠

Proof Write $a = 1 + h$ with $h > 0$. For sufficiently large n, the binomial theorem gives

$$\begin{aligned} a^n &\geq \binom{n}{k+1} h^{k+1} \\ &= \frac{n(n-1)\cdots(n-k)}{(k+1)!} h^{k+1} \\ &\geq \frac{(n/2)^{k+1}}{(k+1)!} h^{k+1}. \end{aligned}$$

Consequently,

$$0 \leq \frac{n^k}{a^n} \leq \frac{2^{k+1}(k+1)!}{h^{k+1}} \frac{1}{n} \to 0.$$

Choose $N_0 \in \mathbb{N}_{>0}$ with $N_0 > 2a$. For $n > N_0$,

$$0 \leq \frac{a^n}{n!} = \frac{a^{N_0}}{N_0!} \prod_{j=N_0+1}^{n} \frac{a}{j} \leq \frac{a^{N_0}}{N_0!} \left(\frac{1}{2}\right)^{n-N_0}.$$

The right-hand side tends to 0 by Proposition 1.8, so the squeeze theorem completes the proof. □

Corollary 1.1 (A hierarchy of dominant scales)

Let $\alpha > 0$ and $a > 1$. As $n \to \infty$,

$$\ln n = o(n^\alpha), \qquad n^\alpha = o(a^n), \qquad a^n = o(n!), \qquad n! = o(n^n),$$

where $u_n = o(v_n)$ means $u_n/v_n \to 0$. Equivalently, one may remember the scale ordering

$$\ln n \ll n^\alpha \ll a^n \ll n! \ll n^n.$$

♡

Proof For the first comparison, let $m = \lfloor \log_2 n \rfloor$. Then $2^m \leq n < 2^{m+1}$, and hence

$$0 \leq \frac{\ln n}{n^\alpha} \leq \frac{(m+1)\ln 2}{2^{\alpha m}} \longrightarrow 0$$

by Proposition 1.10, applied to the base $2^\alpha > 1$. Choose an integer $k > \alpha$. For $n \geq 1$, $n^\alpha \leq n^k$, so

$$0 \leq \frac{n^\alpha}{a^n} \leq \frac{n^k}{a^n} \longrightarrow 0.$$

The third comparison is already contained in Proposition 1.10. Finally,

$$0 \leq \frac{n!}{n^n} = \prod_{j=1}^{n} \frac{j}{n} \leq \left(\frac{1}{2}\right)^{\lfloor n/2 \rfloor} \longrightarrow 0,$$

because the first $\lfloor n/2 \rfloor$ factors are at most $1/2$ and all remaining factors are at most 1. □

> ***How to use the hierarchy.*** In a sum of positive quantities with different scales, divide by the largest plausible scale. If every other normalized term tends to 0, the dominant term determines the asymptotic behaviour. The parameter restrictions matter: the displayed hierarchy assumes $\alpha > 0$ and $a > 1$.

Example 1.22 The slowly varying sequence $n^{1/n}$ Prove that $n^{1/n} \to 1$.

Proof Put $h_n = n^{1/n} - 1 \geq 0$. For $n \geq 2$, the binomial theorem gives

$$n = (1 + h_n)^n \geq \binom{n}{2} h_n^2 = \frac{n(n-1)}{2} h_n^2.$$

Hence

$$0 \leq h_n \leq \sqrt{\frac{2}{n-1}} \to 0.$$

Therefore

$$n^{1/n} \to 1, \qquad n^{1/n} - 1 \to 0.$$

□

> ***Why a new estimate is needed.*** The exponent and the base both vary, so this conclusion does not follow from the fixed-base limit in Proposition 1.8; the binomial estimate supplies the missing control.

Proposition 1.11 (Cesàro means)

If $a_n \to L$, then the arithmetic means satisfy
$$\frac{a_1 + a_2 + \cdots + a_n}{n} \longrightarrow L.$$

> ***Proof Strategy.*** The sum contains a number of terms that grows with n, so the finite-sum limit law does not apply directly. Separate a fixed initial block from a tail on which every error is uniformly small.

Proof Let $\varepsilon > 0$. Choose N such that $|a_k - L| < \varepsilon/2$ whenever $k \geq N$, and set
$$C = \sum_{k=1}^{N-1} |a_k - L|,$$
with $C = 0$ when $N = 1$. For all sufficiently large $n \geq N$, one has $C/n < \varepsilon/2$, and then
$$\begin{aligned} \left| \frac{1}{n} \sum_{k=1}^{n} a_k - L \right| &\leq \frac{1}{n} \sum_{k=1}^{n} |a_k - L| \\ &\leq \frac{C}{n} + \frac{1}{n} \sum_{k=N}^{n} \frac{\varepsilon}{2} < \varepsilon. \end{aligned}$$
Thus the means converge to L. □

1.3.5 Order and subsequences

Theorem 1.7

Suppose that $a_n \to A$ and $b_n \to B$.

(i) If $A < B$, then $a_n < b_n$ for all sufficiently large n.

(ii) If $a_n \leq b_n$ for all sufficiently large n, then $A \leq B$.

In particular, if $A > 0$, then $a_n > 0$ for all sufficiently large n.

Proof For (i), let $\varepsilon = (B - A)/3$. For all sufficiently large n,
$$a_n < A + \varepsilon < B - \varepsilon < b_n.$$

For (ii), suppose instead that $A > B$. Part (i), with the two sequences interchanged, would give $a_n > b_n$ for all sufficiently large n, contradicting the hypothesis. Thus $A \leq B$. The final statement follows by comparing (a_n) with the constant zero sequence. □

Strict inequalities need not survive in the limit. Although $1/n < 2/n$ for every n, both sequences have limit 0.

Definition 1.16

Let (a_n) be a sequence, and let
$$n_1 < n_2 < n_3 < \cdots$$
be positive integers. The sequence $(a_{n_j})_{j \geq 1}$ is called a subsequence (子列) *of (a_n).*

Theorem 1.8

If $a_n \to L$, then every subsequence (a_{n_j}) also converges to L. ♡

Proof Let $\varepsilon > 0$. Choose N such that $n \geq N$ implies $|a_n - L| < \varepsilon$. Since $n_j \geq j$, every $j \geq N$ satisfies $n_j \geq N$, and hence

$$\left|a_{n_j} - L\right| < \varepsilon.$$

Thus $a_{n_j} \to L$. □

The contrapositive is a powerful divergence test: if a sequence has two subsequences converging to different limits, then the original sequence diverges. For $a_n = (-1)^n$, the even subsequence is constantly 1 and the odd subsequence is constantly -1.

> ***Comment.*** Finding one convergent subsequence does not prove that the original sequence converges. The sequence $((-1)^n)$ already has two convergent subsequences. What convergence requires is that every sufficiently late term, not merely a selected infinite collection, lie close to one common limit.

1.3.6 Monotone sequences and completeness

Definition 1.17

A sequence (a_n) is nondecreasing (单调不减) *if $a_n \leq a_{n+1}$ for every n, and* nonincreasing (单调不增) *if $a_n \geq a_{n+1}$ for every n. A sequence of either type is called a* monotone sequence (单调数列). ♣

Theorem 1.9 (Monotone Sequence Theorem)

Every nondecreasing sequence that is bounded above converges. Its limit is the supremum of its set of values. Every nonincreasing sequence that is bounded below converges, with limit equal to the corresponding infimum. ♡

> ***Idea.*** Monotonicity prevents endless oscillation, while boundedness prevents escape to infinity. Completeness supplies the boundary value toward which the sequence is forced to move.

Proof Suppose that (a_n) is nondecreasing and bounded above. Let

$$S = \{a_n \mid n \geq 1\}, \qquad L = \sup S.$$

The number L exists by the least-upper-bound property.

Let $\varepsilon > 0$. Since $L - \varepsilon$ is not an upper bound of S, there exists $N \in \mathbb{N}_{>0}$ such that

$$a_N > L - \varepsilon.$$

For every $n \geq N$, monotonicity and the fact that L is an upper bound give

$$L - \varepsilon < a_N \leq a_n \leq L < L + \varepsilon.$$

Hence $|a_n - L| < \varepsilon$, so $a_n \to L$.

The nonincreasing case follows by applying the proved result to $(-a_n)$, or directly by using the infimum. □

Example 1.23 A recursively defined sequence Let

$$x_1 = 1, \qquad x_{n+1} = \sqrt{2 + x_n}.$$

Prove that (x_n) converges and determine its limit.

> ***Idea.*** Establish convergence from monotonicity and boundedness before solving the fixed-point equation for a possible limit.

Proof First, induction gives $1 \le x_n < 2$ for every n. Indeed, the assertion holds for $x_1 = 1$; if $1 \le x_n < 2$, then

$$1 < \sqrt{3} \le x_{n+1} = \sqrt{2 + x_n} < 2.$$

For $1 \le x_n < 2$,

$$x_{n+1}^2 - x_n^2 = 2 + x_n - x_n^2 = (2 - x_n)(x_n + 1) > 0.$$

Since both terms are positive, $x_{n+1} > x_n$. Thus (x_n) is increasing and bounded above by 2, so Theorem 1.9 gives a limit $L \in [1, 2]$.

The shifted sequence (x_{n+1}) has the same limit by Proposition 1.9. Taking limits in

$$x_{n+1}^2 = 2 + x_n$$

and using Theorem 1.5, we obtain

$$L^2 = 2 + L.$$

Hence $L = 2$ or $L = -1$. Because $L \in [1, 2]$, the limit is $L = 2$. □

> ***Common Pitfall.*** For a recursive sequence, solving the fixed-point equation gives only possible limits. It does not prove that the sequence converges. One must first establish convergence, often by monotonicity and boundedness, and only then pass to the limit in the recurrence.

1.3.7 The natural constant e

The monotone sequence theorem produces the *natural constant*(自然常数) e, one of the central constants of mathematics.

Theorem 1.10 (The Fundamental Sequence Defining e)

The sequence

$$e_n = \left(1 + \frac{1}{n}\right)^n$$

is strictly increasing and bounded above. Consequently, it converges. Its limit is denoted by

$$e = \lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n. \tag{1.3}$$

Moreover, $2 < e < 3$. ♡

> ***Proof Strategy.*** The binomial expansion separates e_n into positive terms. For each fixed position in the expansion, increasing n enlarges the corresponding factor, which proves monotonicity. Replacing all those factors by 1 gives a convergent geometric upper estimate.

Proof By the binomial theorem,

$$\begin{aligned} e_n &= \sum_{k=0}^{n} \binom{n}{k} \frac{1}{n^k} \\ &= \sum_{k=0}^{n} \frac{1}{k!} \prod_{j=0}^{k-1} \left(1 - \frac{j}{n}\right), \end{aligned}$$

where the empty product for $k = 0$ equals 1. For every fixed $k \le n$, replacing n by $n + 1$ increases each nonconstant factor in the product, and the expansion of e_{n+1} also has one additional positive term. Thus $e_{n+1} >$

e_n.

Every factor in each product lies in $[0, 1]$, so

$$e_n \le \sum_{k=0}^{n} \frac{1}{k!}.$$

For $k \ge 3$, we have $k! \ge 6 \cdot 2^{k-3}$. Therefore

$$\begin{aligned} e_n &\le 1 + 1 + \frac{1}{2} + \sum_{k=3}^{\infty} \frac{1}{6 \cdot 2^{k-3}} \\ &= \frac{5}{2} + \frac{1}{6} \sum_{j=0}^{\infty} \frac{1}{2^j} \\ &= \frac{17}{6} < 3. \end{aligned}$$

Hence (e_n) is increasing and bounded above, so it converges by Theorem 1.9. Since $e_n > e_1 = 2$ for $n > 1$, its limit satisfies $2 < e < 3$. □

The constant e is the base of the natural logarithm. It first arose in the study of continuously compounded interest, but it now appears throughout calculus, differential equations, probability, and complex analysis.

Example 1.24 A companion limit Evaluate

$$\lim_{n\to\infty} \left(1 - \frac{1}{n}\right)^n.$$

Proof For $n \ge 2$, put $m = n - 1$. Then

$$\begin{aligned} \left(1 - \frac{1}{n}\right)^n &= \left(\frac{m}{m+1}\right)^{m+1} \\ &= \frac{1}{(1 + 1/m)^m} \cdot \frac{1}{1 + 1/m}. \end{aligned}$$

By (1.3), the shift property, and the quotient law,

$$\lim_{n\to\infty} \left(1 - \frac{1}{n}\right)^n = e^{-1}.$$

□

1.3.8 Completeness extensions: monotone subsequences and Cauchy sequences

The next two results are not needed for the definition of e, but they reveal further consequences of completeness and will be useful when continuous functions on closed intervals are studied.

Lemma 1.1 (Monotone Subsequence Lemma)

Every real sequence has a monotone subsequence.

Proof Call an index n a *peak* if

$$a_n \ge a_m \qquad \text{for every } m > n.$$

If there are infinitely many peaks, list them as

$$n_1 < n_2 < n_3 < \cdots.$$

Since each n_j is a peak, $a_{n_j} \ge a_{n_{j+1}}$, so (a_{n_j}) is nonincreasing.

If there are only finitely many peaks, choose n_1 larger than all of them. Then n_1 is not a peak, so some

$n_2 > n_1$ satisfies $a_{n_2} > a_{n_1}$. The same argument can be repeated after n_2, producing

$$n_1 < n_2 < n_3 < \cdots, \qquad a_{n_1} < a_{n_2} < a_{n_3} < \cdots.$$

Thus the selected subsequence is strictly increasing. □

Theorem 1.11 (Bolzano–Weierstrass Theorem)

Every bounded real sequence has a convergent subsequence.

> ***Idea.*** The preceding lemma supplies a monotone subsequence. Boundedness is inherited by subsequences, so completeness can be applied through the monotone sequence theorem.

Proof Let (a_n) be bounded. By Lemma 1.1, it has a monotone subsequence (a_{n_j}). This subsequence is still bounded. If it is nondecreasing, it is bounded above; if it is nonincreasing, it is bounded below. In either case, Theorem 1.9 shows that (a_{n_j}) converges. □

Definition 1.18

A sequence (a_n) is a Cauchy sequence (柯西数列) *if for every $\varepsilon > 0$ there exists $N \in \mathbb{N}_{>0}$ such that*

$$m, n \geq N \implies |a_m - a_n| < \varepsilon.$$

The definition compares late terms with one another and does not require a candidate limit. The following Cauchy convergence criterion (柯西收敛准则) says that, over $\mathbb{R}$, this internal closeness is exactly convergence.

Theorem 1.12 (Cauchy Convergence Criterion)

A real sequence converges if and only if it is a Cauchy sequence.

> ***Proof Strategy.*** A convergent sequence has late terms close to the same limit, so they are close to one another. Conversely, a Cauchy sequence is bounded; Bolzano–Weierstrass then gives one convergent subsequence, and the Cauchy condition pulls every late term toward that subsequential limit.

Proof Suppose first that $a_n \to L$. Given $\varepsilon > 0$, choose N such that

$$n \geq N \implies |a_n - L| < \frac{\varepsilon}{2}.$$

If $m, n \geq N$, then

$$|a_m - a_n| \leq |a_m - L| + |a_n - L| < \varepsilon.$$

Thus (a_n) is Cauchy.

Conversely, suppose that (a_n) is Cauchy. Taking $\varepsilon = 1$, choose N_0 such that

$$m, n \geq N_0 \implies |a_m - a_n| < 1.$$

For $n \geq N_0$, comparison with a_{N_0} gives

$$|a_n| \leq |a_{N_0}| + 1.$$

Together with the finitely many preceding terms, this proves that (a_n) is bounded.

By Theorem 1.11, some subsequence (a_{n_j}) converges to a number L. Given $\varepsilon > 0$, choose N_1 so that

$$m, n \geq N_1 \implies |a_m - a_n| < \frac{\varepsilon}{2}.$$

Choose j large enough that $n_j \geq N_1$ and

$$|a_{n_j} - L| < \frac{\varepsilon}{2}.$$

Then every $n \geq N_1$ satisfies

$$|a_n - L| \leq |a_n - a_{n_j}| + |a_{n_j} - L| < \varepsilon.$$

Hence $a_n \to L$. ☐

> ***Question.*** Where does completeness of $\mathbb{R}$ enter the proof of the Cauchy criterion, and why can the corresponding statement fail for a sequence considered only inside $\mathbb{Q}$?

1.3.9 Key Concepts, Strategies, and Common Pitfalls

The core concepts are convergence, boundedness, subsequences, monotonicity, and completeness. The extension through Cauchy sequences shows that convergence can also be detected without knowing the limit in advance. For a proposed sequence limit, ask the following questions in order.

1. Can the target error $|a_n - L|$ be simplified or bounded by a familiar quantity?
2. Do the arithmetic laws apply, or are some component sequences divergent?
3. Can an oscillating or complicated term be squeezed by its absolute value?
4. Would two subsequences expose different limiting behaviours?
5. For a recursive sequence, can monotonicity and boundedness be proved before solving the fixed-point equation?
6. For a sum or an nth root, what is the dominant scale, and what happens after normalization by it?
7. If the number of summands grows with n, can the expression be split into a fixed head and a uniformly controlled tail?

Keep the following logical points explicit.

- In an ε–N proof, N may depend on ε but not on n.
- Finitely many exceptional terms do not affect a limit.
- Boundedness is necessary for convergence but not sufficient.
- An eventual strict inequality may become equality after taking limits.
- Limit laws for a finite number of operations do not justify an increasing number of additions or multiplications.

> ***Looking Ahead.*** The definition of a function limit has the same logical architecture as sequence convergence. The discrete tail condition $n \geq N$ will be replaced by the geometric condition $0 < |x - a| < \delta$, while estimation and backward choice of parameters remain the central proof methods.

Exercise 1.3

1. Use the definition to prove
$$\lim_{n\to\infty} \frac{3n+1}{2n-3} = \frac{3}{2}.$$
Give an explicit valid choice of N in terms of ε.
2. Use the definition to prove that $n^{-p} \to 0$ for $p > 0$ without citing Proposition 1.8.
3. Prove directly from the definition that if $a_n \to L$, then $|a_n| \to |L|$. Is the converse true?
4. Prove that every convergent sequence is eventually contained in the interval $(L-1, L+1)$, but explain why this alone does not establish boundedness of the entire sequence.
5. Evaluate
$$\lim_{n\to\infty} \frac{4n^3 - n + 7}{2n^3 + 5n^2 - 1}, \qquad \lim_{n\to\infty} \left(\sqrt{n+1} - \sqrt{n}\right).$$

State which limit law or estimate is used at each step.

6. Use the squeeze theorem to find
$$\lim_{n\to\infty} \frac{\cos(n^2)}{\sqrt{n}}, \qquad \lim_{n\to\infty} \left(2^{2n} + 3^n\right)^{1/n}.$$
7. Show that $a_n = (-1)^n + 1/n$ diverges by finding two subsequences with different limits.
8. Prove that a sequence converges to L if and only if its even and odd subsequences both converge to L.
9. Let $x_1 = 2$ and
$$x_{n+1} = \frac{1}{2}\left(x_n + \frac{3}{x_n}\right).$$
Prove that (x_n) converges and determine its limit.
10. Prove that
$$\lim_{n\to\infty} \frac{n^{10}}{2^n} = 0$$
by the method of Proposition 1.10.
11. Prove that
$$\left(1 + \frac{1}{n}\right)^{n+1}$$
is decreasing and bounded below by e.
12. Explain precisely why the calculation
$$\lim_{n\to\infty} \left(\frac{1}{n+1} + \frac{1}{n+2} + \cdots + \frac{1}{2n}\right) = 0$$
cannot be justified by saying that each summand tends to 0. Find a positive lower bound for the displayed sequence.
13. Suppose that (a_n) and (b_n) are Cauchy sequences. Prove directly from the definition that $(a_n + b_n)$ is Cauchy, and explain where completeness is used when one concludes that the sum sequence converges.
14. If $a_n \to L$ and $m \in \mathbb{N}$ is fixed, prove directly from the definition that $a_{n+m} \to L$. Explain why the conclusion need not hold when the shift m is replaced by an integer sequence m_n satisfying $n + m_n \geq 1$ but for which $n + m_n$ does not tend to $+\infty$.
15. Suppose that (a_n) is bounded and $b_n \to 0$. Prove, without invoking the product law for limits, that $a_n b_n \to 0$.
16. For $a, b > 0$, prove that
$$\lim_{n\to\infty} (a^n + b^n)^{1/n} = \max\{a, b\}.$$
17. Evaluate
$$\lim_{n\to\infty} \sqrt[n]{n^2 + n}.$$
Give upper and lower bounds that make the squeeze theorem applicable.
18. Let $H_n = 1 + 1/2 + \cdots + 1/n$. Prove that
$$\lim_{n\to\infty} \sqrt[n]{H_n} = 1$$
using only the elementary bounds $1 \leq H_n \leq n$ and previously established basic limits.
19. If $a_n \to a$ and $b_n \to b$, prove that
$$\max\{a_n, b_n\} \to \max\{a, b\}, \qquad \min\{a_n, b_n\} \to \min\{a, b\}.$$
You may use formulas involving $|a_n - b_n|$.

20. If $a_n \to L$, prove the weighted Cesàro conclusion

$$\frac{a_1 + 2a_2 + \cdots + na_n}{1 + 2 + \cdots + n} \longrightarrow L.$$

> ***Hint.*** Apply the fixed-head/small-tail idea from Proposition 1.11 to the errors $a_k - L$. The weighted contribution of the fixed head is divided by $1 + 2 + \cdots + n$, while every tail error is bounded by the same prescribed tolerance.

Supplement Exercises

S1. For each $n \geq 2$, let $x_n \in (0, 1)$ be the unique solution of

$$x + x^2 + \cdots + x^n = 1.$$

Prove that (x_n) is monotone and bounded, and determine its limit.

S2. Define

$$r_n = \sqrt{1 + \sqrt{2 + \sqrt{3 + \cdots + \sqrt{n}}}}.$$

Prove that (r_n) is increasing and bounded above, and hence convergent. Can the simple bound used in your proof determine the limit exactly?

S3. Let $a > 0$ and choose $x_1 > \sqrt{a}$. Define

$$x_{n+1} = \frac{1}{2}\left(x_n + \frac{a}{x_n}\right).$$

Prove that $x_n \to \sqrt{a}$ and obtain an identity relating $x_{n+1} - \sqrt{a}$ to $(x_n - \sqrt{a})^2$. What does this identity say about the speed of convergence?

S4. Let $F_1 = F_2 = 1$ and $F_{n+1} = F_n + F_{n-1}$. By studying the even and odd subsequences of F_{n+1}/F_n, prove that

$$\lim_{n\to\infty} \frac{F_{n+1}}{F_n} = \frac{1 + \sqrt{5}}{2}.$$

S5. Prove the following subsequence criterion: a sequence (a_n) converges to L if and only if every subsequence of (a_n) has a further subsequence that converges to L.

> ***Hint.*** For the nontrivial direction, negate the ε–N definition and construct a subsequence that stays a fixed positive distance from L.

S6. Let

$$s_n = \sum_{k=1}^{n} \frac{1}{k^2}.$$

Prove directly that (s_n) is a Cauchy sequence by comparing $1/k^2$ with $1/(k(k-1))$ for $k \geq 2$. Explain exactly where completeness is used to conclude convergence.

S7. Let $a_1, \ldots, a_m > 0$ be fixed. Prove that

$$\lim_{n\to\infty} (a_1^n + \cdots + a_m^n)^{1/n} = \max_{1\leq k\leq m} a_k.$$

1.4 Limits of Functions

重点内容与证明方法

- *Interpret the quantifiers in a function limit.*
- *Distinguish one-sided, two-sided, and infinite approaches.*
- *Prove limits using local estimates, limit laws, and sequences.*
- *Establish the fundamental trigonometric and exponential limits.*
- *Compare infinitesimals and recognize when cancellation needs finer estimates.*
- *Choose methods for direct proofs, calculations, and counterexamples.*

A sequence approaches its limit along the ordered inputs $1, 2, 3, \ldots$. For a function, the input may approach a point through all admissible real values. The problem is therefore to control many possible approaches at once. For example,

$$f(x) = \frac{x^2 - 1}{x - 1} \qquad (x \neq 1)$$

has no value at 1, but its values approach 2 as x approaches 1. The equality $f(x) = x + 1$ for $x \neq 1$ explains the limiting behaviour without filling the hole in the domain.

1.4.1 Punctured neighbourhoods and the definition of a limit

Recall that $U_r(a) = \{x \in \mathbb{R} : |x - a| < r\}$. In a limit at a, the input belongs to a punctured neighbourhood: $0 < |x - a| < r$. The exclusion of a is essential. Changing the value of a function at a, or leaving that value undefined, does not affect its limit at a.

To include endpoints and restricted domains, let $f\colon D \to \mathbb{R}$, where $D \subseteq \mathbb{R}$. We call a an *accumulation point*(聚点)of D if every punctured neighbourhood of a contains a point of D. The point a need not belong to D. This condition guarantees that there are inputs that can approach a; it prevents a limit statement from becoming vacuously true. Most examples below have a domain containing a whole punctured neighbourhood of the limiting point.

Definition 1.19

Let a be an accumulation point of D and let $L \in \mathbb{R}$. We say that f has limit(函数极限) *L as $x \to a$ through D, and write*

$$\lim_{\substack{x \to a \\ x \in D}} f(x) = L$$

if for every $\varepsilon > 0$ there exists $\delta > 0$ such that, for every $x \in D$,

$$0 < |x - a| < \delta \quad \Longrightarrow \quad |f(x) - L| < \varepsilon.$$

We usually abbreviate this to $\lim_{x \to a} f(x) = L$, with the domain understood. ♣

The tolerance ε is prescribed first. We must then find one δ that works for *every* admissible input in the punctured δ-neighbourhood. The number δ may depend on ε, the function, and the fixed point a, but it may not depend on the input x subsequently chosen. The exact negation of the proposed limit is

$$\exists \varepsilon_0 > 0 \quad \forall \delta > 0 \quad \exists x \in D : \quad 0 < |x - a| < \delta, \qquad |f(x) - L| \geq \varepsilon_0.$$

Thus failure means that one fixed output error recurs at inputs arbitrarily close to a. This quantifier pattern is the source of the bad sequence constructed in Heine's criterion below. Geometrically, the graph over a sufficiently narrow punctured vertical strip must lie inside the prescribed horizontal band, as in **Figure** 1.3.

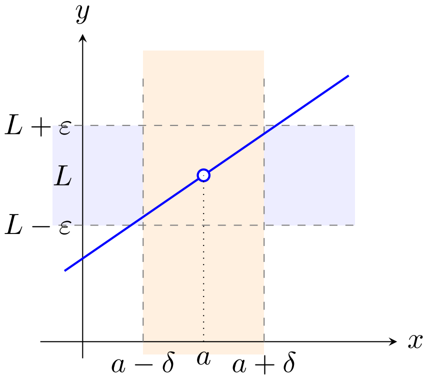


Figure 1.3: A horizontal tolerance band and a corresponding punctured input neighbourhood. The open circle does not affect the limit.

Example 1.25 Designing a neighbourhood Prove directly that $\lim_{x\to 1}(3x-2)=1$.

> ***Idea.*** Start with the desired conclusion and simplify its left side: $|(3x-2)-1| = 3|x-1|$. The output error is exactly three times the input error.

Proof Let $\varepsilon > 0$ and choose $\delta = \varepsilon/3$. If $0 < |x-1| < \delta$, then

$$|(3x-2)-1| = 3|x-1| < 3\delta = \varepsilon.$$

This proves the limit. □

Example 1.26 A temporary local bound Prove directly that $\lim_{x\to 2} x^2 = 4$.

> ***Idea.*** The factorization $\left|x^2-4\right| = |x-2|\,|x+2|$ exposes the small quantity $|x-2|$. The remaining factor still depends on x. First restrict $|x-2| < 1$; then $|x+2| < 5$.

Proof Given $\varepsilon > 0$, set $\delta = \min\{1, \varepsilon/5\}$. If $0 < |x-2| < \delta$, then $|x+2| \le |x-2| + 4 < 5$, and therefore

$$\left|x^2-4\right| < 5|x-2| < 5\delta \le \varepsilon.$$

□

The preliminary restriction $\delta \le 1$ is not part of the definition. It is a reusable proof device: first place all nonsmall factors under control, then choose a smaller neighbourhood to meet the requested error. The limit definition also shows that functions agreeing throughout some punctured neighbourhood have the same limit there. Unlike sequences, however, changing a function at infinitely many inputs approaching a may change its limit, even when those inputs form a very sparse set.

1.4.2 One-sided limits and different approaches

Definition 1.20

Suppose D has points arbitrarily close to a from the right. The right-hand limit (右极限) $\lim_{x\to a^+} f(x) = L$ *means that for every $\varepsilon > 0$ there is $\delta > 0$ such that*

$$x \in D, \quad 0 < x-a < \delta \quad \Longrightarrow \quad |f(x)-L| < \varepsilon.$$

The left-hand limit (左极限) $\lim_{x\to a^-} f(x) = L$ *is defined by replacing $0 < x-a < \delta$ with $0 < a-x < \delta$, assuming approach from the left is possible.* ♣

Proposition 1.12

If D has points arbitrarily close to a from both sides, then

$$\lim_{x\to a} f(x) = L \quad \Longleftrightarrow \quad \lim_{x\to a^-} f(x) = \lim_{x\to a^+} f(x) = L.$$

♠

Proof A two-sided δ works on each side separately. Conversely, for a given $\varepsilon > 0$, choose suitable left and right radii δ_- and δ_+. With $\delta = \min\{\delta_-, \delta_+\}$, every admissible x satisfying $0 < |x-a| < \delta$ lies on one of the two controlled sides. □

For $f(x) = |x|/x$ on $\mathbb{R}\setminus\{0\}$, the left-hand limit at 0 is -1 and the right-hand limit is 1. The two-sided limit does not exist. The value assigned at 0 cannot repair this disagreement. By contrast, for $f(x) = \sqrt{x}$ on $[0, +\infty)$, the limit at 0 relative to the domain is a right-hand limit. There is no left-hand approach in the domain, so Proposition 1.12 does not require a left-hand limit in this example.

Definition 1.21

If D is unbounded above, then $\lim_{x\to+\infty} f(x) = L$ means

$$\text{for every } \varepsilon > 0 \text{ there is } A > 0 \text{ such that } x \in D,\ x > A \implies |f(x) - L| < \varepsilon.$$

For $x \to -\infty$, assume D is unbounded below and use $x < -A$. For $|x| \to \infty$, assume D is unbounded and use $|x| > A$. When both tails of D are unbounded, the last limit exists exactly when the two tail limits exist and agree. ♣

For example, $1/x \to 0$ as $|x| \to \infty$, since $|x| > 1/\varepsilon$ implies $|1/x| < \varepsilon$. The notation $x \to \infty$ is sometimes used for $x \to +\infty$; we keep the sign explicit whenever the direction matters. In particular,

$$\lim_{x\to+\infty} \frac{x}{\sqrt{x^2+1}} = 1, \qquad \lim_{x\to-\infty} \frac{x}{\sqrt{x^2+1}} = -1.$$

The sign difference comes from $\sqrt{x^2} = |x|$, not from x. These two computations will follow from the limit laws below.

Definition 1.22

Let a be an accumulation point of D. We write $\lim_{x\to a} f(x) = +\infty$ if for every $M > 0$ there is $\delta > 0$ such that

$$x \in D, \quad 0 < |x - a| < \delta \quad \implies \quad f(x) > M.$$

The statement $\lim_{x\to a} f(x) = -\infty$ uses $f(x) < -M$ instead. One-sided infinite limits use the corresponding one-sided neighbourhoods. Infinite limits as $x \to +\infty$ or $x \to -\infty$ use a tail condition in place of $0 < |x - a| < \delta$. ♣

An *infinite limit*(无穷极限)describes eventual growth beyond every bound; it is not a finite real limit. For instance,

$$\lim_{x\to 0} \frac{1}{x^2} = +\infty, \qquad \lim_{x\to 0^+} \frac{1}{x} = +\infty, \qquad \lim_{x\to 0^-} \frac{1}{x} = -\infty.$$

For the first statement choose $\delta = 1/\sqrt{M}$. The last two statements do not combine into a signed two-sided infinite limit for $1/x$, although $|1/x| \to +\infty$. Nor does unboundedness near a point alone imply an infinite limit: a function may repeatedly return to 0 while also taking arbitrarily large values.

Proposition 1.13 (Basic rules for infinite limits)

The following assertions hold along any one fixed approach.

(i) If $g \to +\infty$ and $f \geq g$ eventually, then $f \to +\infty$. If $g \to -\infty$ and $f \leq g$ eventually, then $f \to -\infty$.

(ii) If $f \to +\infty$ and g is bounded below eventually, then $f + g \to +\infty$. If $f \to -\infty$ and g is bounded above eventually, then $f + g \to -\infty$.

(iii) If $f \to +\infty$ and $g \to c \neq 0$, then $fg \to +\infty$ when $c > 0$ and $fg \to -\infty$ when $c < 0$. Replacing $f \to +\infty$ by $f \to -\infty$ reverses these signs.

(iv) If $|f| \to +\infty$ and g is bounded eventually, then $g/f \to 0$.

(v) If $f \to 0$ and $f > 0$ eventually, then $1/f \to +\infty$; if $f < 0$ eventually, then $1/f \to -\infty$. Conversely, $f \to +\infty$ or $f \to -\infty$ implies $1/f \to 0$, with the corresponding eventual sign. ♠

Proof The comparison statements follow immediately by applying the defining threshold M to the smaller function in the positive case and to the larger function in the negative case. For (ii), if $g \geq C$ eventually and

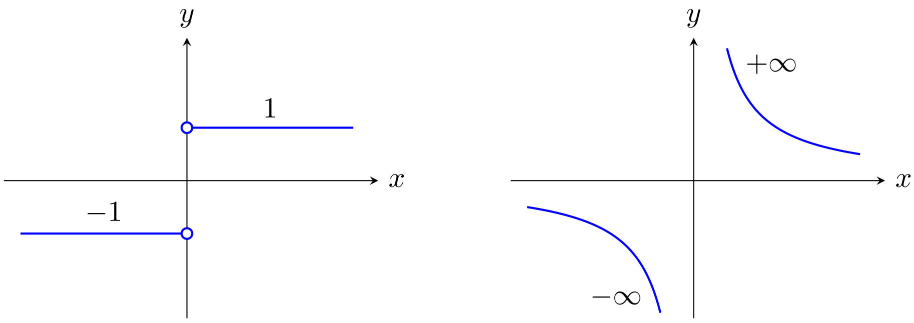
$M > 0$, take $f > \max\{M - C, 1\}$ sufficiently far along the approach; the other sign is analogous. For (iii), when $c > 0$ one eventually has $g > c/2$, and $f > 2M/c$ then gives $fg > M$. The remaining signs follow by applying this argument to $-f$ or $-g$. For (iv), choose an eventual bound $|g| \le C$ and then require $|f| > C/\varepsilon$; the case $C = 0$ is immediate. Finally, the inequalities

$$0 < f < \frac{1}{M} \iff \frac{1}{f} > M, \qquad f > \frac{1}{\varepsilon} \implies 0 < \frac{1}{f} < \varepsilon$$

prove (v), and the negative cases follow after changing signs. □

> ***Use sign information, not symbol pushing.*** The rules above apply only after eventual signs or one-sided bounds have been established. For example, $f \to +\infty$ and $g \to -\infty$ do not determine $f + g$; competing terms must first be compared or transformed.

In graph language, $x = a$ is a *vertical asymptote* (铅直渐近线) if at least one of the one-sided limits of f at a is $+\infty$ or $-\infty$. The line $y = L$ is a *horizontal asymptote* (水平渐近线) at a specified tail if the corresponding limit at infinity equals L. The function $1/x$ has both types of asymptote. A graph can cross a horizontal asymptote; an asymptote describes a limiting relation, not a barrier. The two graphs in **Figure** 1.4 contrast finite side limits with infinite side limits.



$f(x) = |x|/x$ $\qquad$ $g(x) = 1/x$

Figure 1.4: The two sides approach different finite levels on the left and grow with opposite signs on the right. Neither graph has a finite two-sided limit at 0.

1.4.3 Local properties and the sequential criterion

Theorem 1.13

Suppose $f(x) \to L \in \mathbb{R}$ as $x \to a$ through D. Then the following statements hold.

1. *The limit is unique.*
2. *The function is bounded on some punctured neighbourhood of a relative to D.*
3. *If $L > 0$, then $f(x) > 0$ throughout some such neighbourhood; if $L < 0$, then $f(x) < 0$ there.*
4. *If also $g(x) \to K$ on the same approach and $f(x) \le g(x)$ for all sufficiently close admissible $x \ne a$, then $L \le K$.*

♡

> ***Proof Strategy.*** Choose an output tolerance adapted to the desired conclusion: a fraction of the gap between alleged limits, the fixed tolerance 1, or a fraction of the distance from L to 0. The input neighbourhood then follows from the definition.

Proof If $L \ne K$ were two limits of f, use $\varepsilon = |L - K|/3$ in both limit definitions. There is an admissible

$x \neq a$ in the intersection of the resulting neighbourhoods, because a is an accumulation point. Then

$$|L-K| \leq |L-f(x)| + |f(x)-K| < \frac{2}{3}|L-K|,$$

a contradiction. For boundedness, use $\varepsilon = 1$ to obtain $|f(x)| < |L|+1$ near a. If $L > 0$, use $\varepsilon = L/2$ to obtain $f(x) > L/2 > 0$. Apply this result to $-f$ when $L < 0$. Finally, if $L > K$, use $\varepsilon = (L-K)/3$ for both f and g and restrict further to the neighbourhood where $f \leq g$. An admissible point there would satisfy

$$f(x) > L - \varepsilon > K + \varepsilon > g(x),$$

which is impossible. □

The second assertion is *local boundedness*(局部有界性): its bound need not be valid on all of D. The third is *local sign preservation*(局部保号性) when the limit is nonzero. Also, a strict inequality need not stay strict in the limit: $x^2 > 0$ for $x \neq 0$, but its limit at 0 is 0. All four arguments work on a one-sided neighbourhood or on a tail after the input condition is changed accordingly.

Theorem 1.14 (Heine's sequential criterion)

Let a be an accumulation point of D. Then $\lim_{x\to a} f(x) = L$ if and only if, for every sequence (x_n) in $D \setminus \{a\}$ satisfying $x_n \to a$, one has $f(x_n) \to L$. ♡

> ***Idea.*** A function limit controls every sufficiently close input, hence every sufficiently late term of any approaching sequence. For the converse, negate the function-limit definition and choose one bad input within distance $1/n$ for each n.

Proof Suppose first that $f(x) \to L$ and let $x_n \to a$ with $x_n \in D \setminus \{a\}$. For a prescribed $\varepsilon > 0$, choose $\delta > 0$ from the function-limit definition. For all sufficiently large n, $0 < |x_n - a| < \delta$, so $|f(x_n) - L| < \varepsilon$. Thus $f(x_n) \to L$.

Conversely, if f does not tend to L, there exists $\varepsilon_0 > 0$ such that for every $\delta > 0$ one can find $x \in D$ with

$$0 < |x-a| < \delta, \qquad |f(x) - L| \geq \varepsilon_0.$$

Choose such a point x_n with $\delta = 1/n$. Then $x_n \to a$, but $|f(x_n) - L| \geq \varepsilon_0$ for every n, contradicting the assumed sequential property. □

The criterion has one-sided and infinite-approach versions. For example, $f(x) \to L$ as $x \to +\infty$ exactly when $f(x_n) \to L$ for every sequence $x_n \in D$ tending to $+\infty$. In the converse proof, a failure supplies a bad point $x_n > n$ instead of a bad point within $1/n$ of a. For a right-hand limit use $0 < x_n - a < 1/n$, and for a negative tail use $x_n < -n$. The same selection argument gives the $|x| \to \infty$ version using $|x_n| > n$.

Example 1.27 Two ways to approach the same point Determine whether $\sin(1/x)$ and $x\sin(1/x)$ have limits as $x \to 0$.

Solution *For $n \geq 1$, set*

$$x_n = \frac{1}{2\pi n + \pi/2}, \qquad y_n = \frac{1}{2\pi n + 3\pi/2}.$$

Both sequences tend to 0, but $\sin(1/x_n) = 1$ and $\sin(1/y_n) = -1$. By Theorem 1.14, $\sin(1/x)$ has no limit at 0, even from the right. For the second function,

$$|x\sin(1/x)| \leq |x|.$$

Given $\varepsilon > 0$, the choice $\delta = \varepsilon$ proves that its limit is 0. Oscillation may persist while its amplitude tends to zero.

The graph in **Figure** 1.5 suggests the distinction, but a finite drawing cannot display all oscillations near 0. The sequences and the estimate provide the proof. Similarly, the Dirichlet function $D(x)$ from Section 1.2 has no limit at any $a \in \mathbb{R}$. By density, choose a rational r_n and an irrational s_n in $(a, a+1/n)$. Then both input

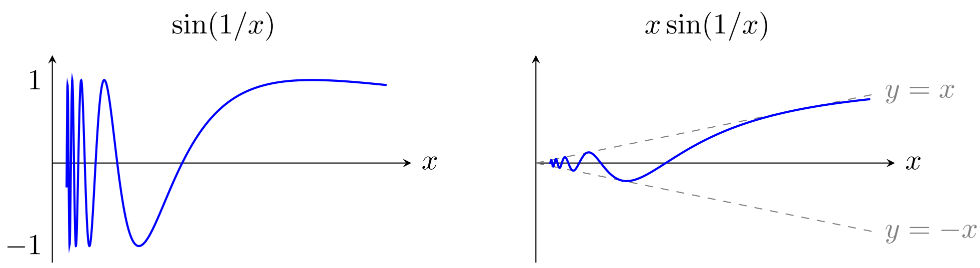


Figure 1.5: Right-hand portions of two oscillatory graphs. In the second graph, a shrinking envelope forces a limit.

sequences tend to a, whereas $D(r_n) = 1$ and $D(s_n) = 0$ for all n.

1.4.4 Limit laws and composition

Theorem 1.15

Suppose $f(x) \to A$ and $g(x) \to B$ along the same approach, with finite limits. For every $c \in \mathbb{R}$,

$$f(x) + g(x) \to A + B, \qquad cf(x) \to cA, \qquad f(x)g(x) \to AB.$$

If $B \neq 0$, then g is nonzero sufficiently far along the approach and

$$\frac{f(x)}{g(x)} \to \frac{A}{B}.$$

♡

Proof For every admissible approaching sequence (x_n), the sequences $f(x_n)$ and $g(x_n)$ tend to A and B by Theorem 1.14 and its corresponding approach version. The sequence arithmetic laws, Theorem 1.5, give the asserted limits along this arbitrary sequence. The converse sequential criterion gives the function limits. If $B \neq 0$, the estimate $|g(x) - B| < |B|/2$ holds eventually and implies $|g(x)| > |B|/2$. Thus the quotient is defined on the required final part of the approach. □

Theorem 1.16 (Squeeze theorem)

Suppose $u(x) \leq f(x) \leq v(x)$ eventually along an approach, and both $u(x)$ and $v(x)$ tend to the same finite limit L. Then $f(x) \to L$ along that approach.

♡

Proof Given $\varepsilon > 0$, restrict the inputs so that the two comparisons hold and

$$|u(x) - L| < \varepsilon, \qquad |v(x) - L| < \varepsilon.$$

Then $L - \varepsilon < u(x) \leq f(x) \leq v(x) < L + \varepsilon$. □

In particular, if $h(x) \to 0$ and k is bounded near the approach, then $h(x)k(x) \to 0$. Indeed, an eventual bound $|k(x)| \leq C$ yields $|h(x)k(x)| \leq C\,|h(x)|$. The bounded factor need not have a limit.

Example 1.28 Cancellation and rationalization Evaluate

$$\lim_{x\to 2} \frac{x^2-4}{x-2}, \qquad \lim_{x\to 0} \frac{\sqrt{1+x}-1}{x}, \qquad \lim_{x\to-\infty} \left(\sqrt{x^2+x}+x\right).$$

Solution *In the first limit, the quotient equals $x+2$ on the punctured domain, so the answer is 4. For the second, rationalization gives*

$$\frac{\sqrt{1+x}-1}{x} = \frac{1}{\sqrt{1+x}+1} \qquad (x \neq 0).$$

Here $\left|\sqrt{1+x}-1\right| = |x|/(\sqrt{1+x}+1) \leq |x|$ for $x \geq -1$. Thus $\sqrt{1+x} \to 1$, and the answer is $1/2$. For *sufficiently negative x, the third expression is defined and*

$$\sqrt{x^2+x}+x=\frac{x}{\sqrt{x^2+x}-x}=\frac{-1}{\sqrt{1+1/x}+1}\longrightarrow -\frac{1}{2}.$$

The last equality uses $\sqrt{x^2}=|x|=-x$ on this tail.

The same idea applies to higher roots without searching for a special trick. If $p=\sqrt[m]{A}$ and $q=\sqrt[m]{B}$ are real, then

$$A-B=p^m-q^m=(p-q)\bigl(p^{m-1}+p^{m-2}q+\cdots+pq^{m-2}+q^{m-1}\bigr).$$

Whenever the second factor has a nonzero limit, this identity replaces the difference of two mth roots by a quotient whose small factor is $A-B$.

> ***Generalized rationalization.*** For a difference of roots, set the roots equal to new symbols and factor a difference of powers. The familiar conjugate is exactly the case $m=2$. Before dividing, check that the resulting sum in the denominator stays away from zero.

The quotient law cannot be applied directly when the limiting denominator is 0. The commonly occurring forms

$$\frac{0}{0},\qquad \frac{\infty}{\infty},\qquad \infty-\infty,\qquad 0\cdot\infty,\qquad 1^\infty,\qquad 0^0,\qquad \infty^0$$

are *indeterminate forms*(未定式), not values. The notation records only the separate limiting tendencies of the visible parts; it does not determine how their rates compare. For example, x^2/x, x/x, and x/x^2 have very different limiting behaviours at 0. Likewise, as $x\to+\infty$, the three differences $x-x$, $(x+1)-x$, and $2x-x$ tend respectively to 0, 1, and $+\infty$. By contrast, a bounded numerator divided by a denominator whose absolute value tends to infinity is determined by Proposition 1.13: the quotient tends to 0. The correct response to an indeterminate form is therefore to transform or estimate the expression, not to perform arithmetic with the symbols 0 and ∞.

Theorem 1.17

Suppose $u(x)\to b$ as $x\to a$, and $v(t)\to L$ as $t\to b$ through the domain of v. Assume the composition is defined for all sufficiently close admissible $x\neq a$. Either of the following additional conditions ensures that $v(u(x))\to L$:

1. *$u(x)\neq b$ for all sufficiently close admissible $x\neq a$;*
2. *b is in the domain of v and $v(b)=L$.*

The same statement holds for a one-sided or infinite approach of x; the values $u(x)$ must follow the approach specified for v. ♡

> ***Idea.*** The outer limit gives control when $0<|t-b|<\eta$. The inner limit supplies $|u(x)-b|<\eta$. The only possible mismatch is $u(x)=b$, which the punctured outer limit does not control.

Proof Given $\varepsilon>0$, choose $\eta>0$ so that every admissible t with $0<|t-b|<\eta$ satisfies $|v(t)-L|<\varepsilon$. Choose a sufficiently small punctured neighbourhood of a on which the composition is defined and $|u(x)-b|<\eta$. Under the first condition, restrict further so that $u(x)\neq b$; the outer estimate then applies. Under the second condition, the same estimate applies whenever $u(x)\neq b$, and when $u(x)=b$ the output equals $v(b)=L$ exactly. In either case the required error is less than ε. The proof changes only its input neighbourhood condition for other approaches. □

Neither extra condition can simply be discarded. Let $u(x)=0$ for every x, and let $v(0)=1$ while $v(t)=0$ for $t\neq 0$. Then $u(x)\to 0$ and $\lim_{t\to 0}v(t)=0$, but $v(u(x))=1$. In the next section, continuity of the outer function will provide precisely the second condition.

> ***Substitution Checklist.*** When setting $t = \phi(x)$ in a limit, record all four points explicitly.
> (1) Determine the exact approach of t: two-sided, one-sided, or infinite.
> (2) Rewrite the entire expression in terms of t, including restrictions inherited from the original domain.
> (3) Verify that $t = \phi(x)$ follows the domain and approach required by the new limit.
> (4) Check the composition hypothesis at the puncture; in particular, either $\phi(x) \neq b$ eventually or the outer function must be controlled at b itself.
>
> A substitution such as $t = -x$ reverses sides, while $t = 1/x$ turns a tail into a one-sided approach to 0.

1.4.5 The fundamental trigonometric limit

The basic geometric estimate is obtained from the unit circle. Angles are measured in radians throughout. For $0 < t < \pi/2$, compare an inscribed triangle, a circular sector, and a triangle formed with the tangent line in **Figure** 1.6. Their areas are respectively $\frac{1}{2}\sin t$, $\frac{1}{2}t$, and $\frac{1}{2}\tan t$. Elementary circle geometry therefore gives

$$\sin t < t < \tan t, \qquad \cos t < \frac{\sin t}{t} < 1.$$

This argument uses the geometric meaning of radian measure and the sector-area formula; it does not use differentiation or integration.

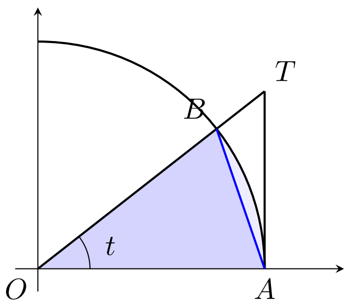


Figure 1.6: The inclusions $\triangle OAB \subset$ sector $OAB \subset \triangle OAT$ compare the sine, angle, and tangent.

For negative t, oddness of sine gives $|\sin t| \le |t|$ as well, at least for $|t| < \pi/2$. Consequently,

$$0 \le 1 - \cos t = 2\sin^2(t/2) \le \frac{t^2}{2}$$

for all sufficiently small t. Hence $\cos t \to 1$ as $t \to 0$.

Theorem 1.18
With angles measured in radians,

$$\lim_{x \to 0} \frac{\sin x}{x} = 1. \tag{1.4}$$

♡

Proof For $0 < x < \pi/2$, the geometric inequalities place $\sin x/x$ between $\cos x$ and 1. Both bounds tend to 1, so Theorem 1.16 gives the right-hand limit. Since $\sin(-x)/(-x) = \sin x/x$, the left-hand limit is the same. Apply Proposition 1.12. □

To apply (1.4), match the argument of the sine with the small quantity in the denominator. The addition formulas now give, for every fixed $a \in \mathbb{R}$,

$$\lim_{x \to a} \sin x = \sin a, \qquad \lim_{x \to a} \cos x = \cos a.$$

Indeed, write $x = a + h$, use $\sin h \to 0$ and $\cos h \to 1$, and apply the arithmetic laws to the addition formulas. Where the limiting cosine is nonzero, the quotient law also gives the limit of the tangent.

Example 1.29 Matching the small argument Evaluate

$$\lim_{x\to 0} \frac{\sin(3x)}{\tan(2x)}, \qquad \lim_{x\to 0} \frac{1-\cos x}{x^2}.$$

Solution *For the first limit, separate the two standard ratios:*

$$\frac{\sin(3x)}{\tan(2x)} = \frac{3}{2}\frac{\sin(3x)}{3x}\frac{2x}{\sin(2x)}\cos(2x) \longrightarrow \frac{3}{2}.$$

All denominators are nonzero for sufficiently small nonzero x. For the second, the half-angle identity gives

$$\frac{1-\cos x}{x^2} = \frac{1}{2}\left(\frac{\sin(x/2)}{x/2}\right)^2 \longrightarrow \frac{1}{2}.$$

1.4.6 The exponential limit and related limits

In Section 1.3, the sequence $(1+1/n)^n$ defined the number e. The next step is to allow a real input. Before applying logarithms to a limit, we record the elementary limit estimates they require. We use the real exponential and logarithmic functions and their order and algebraic properties introduced in Section 1.2.

Lemma 1.2

For $b > 0$ and $c \in \mathbb{R}$,

$$\lim_{x\to c} b^x = b^c.$$

For $b > 0$, $b \neq 1$, and $t > 0$,

$$\lim_{x\to t} \log_b x = \log_b t.$$

> ***Idea.*** Near exponent 0, trap a power between $b^{-1/n}$ and $b^{1/n}$ and use the root-sequence limit. For a logarithm, translate the desired output interval back to a positive input interval using monotonicity.

Proof Suppose first $b > 1$. By Proposition 1.8, $b^{1/n} \to 1$, and its reciprocal also tends to 1. For a prescribed $\varepsilon > 0$, choose a positive integer n with $b^{1/n} - 1 < \varepsilon$ and $1 - b^{-1/n} < \varepsilon$. If $|h| < 1/n$, monotonicity gives

$$b^{-1/n} < b^h < b^{1/n},$$

so $\left|b^h - 1\right| < \varepsilon$. Thus $b^h \to 1$ as $h \to 0$, and $b^{c+h} = b^c b^h \to b^c$. The case $b = 1$ is constant, and the case $0 < b < 1$ follows by writing $b^x = 1/(1/b)^x$.

For logarithms with $b > 1$, let $\varepsilon > 0$ and choose

$$\delta = \min\{t - tb^{-\varepsilon},\ tb^{\varepsilon} - t\} > 0.$$

If $|x - t| < \delta$, then $tb^{-\varepsilon} < x < tb^{\varepsilon}$, hence $\log_b t - \varepsilon < \log_b x < \log_b t + \varepsilon$. For $0 < b < 1$, use $\log_b x = -\log_{1/b} x$. □

Theorem 1.19

For the real-valued expression with $1 + x > 0$,

$$\lim_{x\to 0} (1+x)^{1/x} = e. \tag{1.5}$$

Equivalently, $\lim_{t\to+\infty}(1 + 1/t)^t = e$.

Proof Strategy. For a positive real t, place t between consecutive integers and trap the whole expression between two sequences already controlled by Theorem 1.10. Then transform the negative side of $x \to 0$ into a positive-side limit by an exact identity.

Proof Let $t \geq 1$ and $n = \lfloor t \rfloor$, so $n \leq t < n+1$. Monotonicity in the base and in a positive exponent gives

$$\left(1 + \frac{1}{n+1}\right)^n \leq \left(1 + \frac{1}{t}\right)^t \leq \left(1 + \frac{1}{n}\right)^{n+1}.$$

The left bound equals $\bigl(1 + 1/(n+1)\bigr)^{n+1} / \bigl(1 + 1/(n+1)\bigr)$, and the right bound equals $\bigl(1 + 1/n\bigr)^n \bigl(1 + 1/n\bigr)$. Both tend to e as $n \to \infty$. Since $\lfloor t \rfloor \to \infty$ as $t \to +\infty$, their values approach e along the real input t as well: the sequence estimates for every $n \geq N$ apply whenever $t \geq N$. The squeeze theorem proves the limit at $+\infty$. The substitution $t = 1/x$ gives (1.5) from the right.

For $-1 < x < 0$, put $y = -x/(1+x) > 0$. As $x \to 0^-$, one has $y \to 0^+$, and

$$1 + x = \frac{1}{1+y}, \qquad (1+x)^{1/x} = (1+y)^{(1+y)/y} = (1+y)^{1/y}(1+y) \longrightarrow e.$$

The one-sided limits agree. $\square$

Corollary 1.2

The following limits hold:

$$\lim_{x \to 0} \frac{\ln(1+x)}{x} = 1, \qquad \lim_{x \to 0} \frac{e^x - 1}{x} = 1. \tag{1.6}$$

More generally, for $b > 0$,

$$\lim_{x \to 0} \frac{b^x - 1}{x} = \ln b.$$

♡

Idea. Taking a logarithm converts the power limit into a quotient limit. Then the inverse relation between the exponential and logarithm transfers the result to $e^x - 1$.

Proof By Lemma 1.2, the logarithm tends to its value at e. The composition theorem therefore allows us to take the logarithm in (1.5):

$$\frac{\ln(1+x)}{x} = \ln\bigl((1+x)^{1/x}\bigr) \longrightarrow \ln e = 1.$$

For the second limit, set $u = e^x - 1$. The same lemma gives $u \to 0$, and $u \neq 0$ for $x \neq 0$ by strict monotonicity of the exponential. Since $x = \ln(1+u)$,

$$\frac{e^x - 1}{x} = \frac{u}{\ln(1+u)} \longrightarrow 1.$$

If $b \neq 1$, substitute $t = x \ln b$ into the second limit and use $b^x = e^{x \ln b}$. If $b = 1$, the numerator is identically zero and the limit is $0 = \ln 1$. $\square$

Example 1.30 Tracking direction under substitution Evaluate

$$\lim_{x \to 0^-} \frac{\ln(1-x)}{x}, \qquad \lim_{x \to +\infty} x\bigl(e^{1/x} - 1\bigr).$$

Solution *For the first limit, set $t = -x$. Then $x \to 0^-$ means $t \to 0^+$, not merely $t \to 0$, and*

$$\frac{\ln(1-x)}{x} = -\frac{\ln(1+t)}{t} \longrightarrow -1.$$

For the second, set $t = 1/x$. The tail $x \to +\infty$ becomes $t \to 0^+$, and

$$x\bigl(e^{1/x} - 1\bigr) = \frac{e^t - 1}{t} \longrightarrow 1.$$

In both cases the direction and the transformed domain agree with the relevant one-sided form of (1.6).

Proposition 1.14 (Logarithmic reduction of a power)

Along a fixed approach, suppose that $F(x) > 0$ eventually and

$$G(x)\ln F(x) \longrightarrow A \in \mathbb{R}.$$

Then

$$F(x)^{G(x)} \longrightarrow e^A.$$

In particular, if $u(x) \to 0$, $1 + u(x) > 0$ eventually, and $u(x)v(x) \to c \in \mathbb{R}$, then

$$\bigl(1 + u(x)\bigr)^{v(x)} \longrightarrow e^c.$$

This conclusion remains valid when $u(x) = 0$ at some inputs.

Proof The identity

$$F(x)^{G(x)} = \exp\bigl(G(x)\ln F(x)\bigr)$$

and continuity of the exponential give the first statement. For the special case, define

$$r(t) = \begin{cases} \dfrac{\ln(1+t)}{t}, & t \neq 0, \\ 1, & t = 0. \end{cases}$$

By (1.6), $r(t) \to 1$ as $t \to 0$, and the chosen value makes r continuous there. At every relevant input, including those with $u(x) = 0$,

$$v(x)\ln(1 + u(x)) = u(x)v(x)r(u(x)) \longrightarrow c.$$

Apply the first statement with $F = 1 + u$ and $G = v$. □

Example 1.31 A power with a hidden finite exponent Evaluate

$$\lim_{x\to 0}\bigl(1 + x\sin x\bigr)^{1/x^2}.$$

Solution *For sufficiently small x, one has $x\sin x \geq 0$, so the base is positive. Set $u(x) = x\sin x$ and $v(x) = 1/x^2$. Then $u(x) \to 0$ and*

$$u(x)v(x) = \frac{\sin x}{x} \longrightarrow 1.$$

Therefore Proposition 1.14 gives the limit e.

> ***Power-form decision.*** For a positive varying base, do not manipulate an apparent form such as 1^∞, 0^0, or ∞^0 directly. Take a logarithm and study the single product $G\ln F$. The positivity of F is a domain condition, and the resulting product may still require factoring, substitution, or a rate comparison.

1.4.7 Infinitesimals and their comparison

The words “small” and “large” become meaningful only after an approach has been specified. For example, $1/x$ tends to 0 as $x \to +\infty$, but its magnitude grows without bound as $x \to 0$.

Definition 1.23

Along a fixed approach, a function α is an infinitesimal (无穷小量) *if $\alpha(x) \to 0$. It is an* infinite quantity (无穷大量) *if $|\alpha(x)| \to +\infty$.*

An infinitesimal is a varying quantity with limit zero, not a particular very small positive number. A finite limit $f(x) \to L$ is equivalent to $f(x) = L + \alpha(x)$ with an infinitesimal α. Finite sums of infinitesimals are infinitesimal, as is the product of an infinitesimal and a bounded function. The quotient of two infinitesimals requires a separate comparison. If α is eventually nonzero, then

$$\alpha(x) \to 0 \quad \Longleftrightarrow \quad \frac{1}{|\alpha(x)|} \to +\infty.$$

Indeed, the inequalities $|\alpha(x)| < 1/M$ and $1/|\alpha(x)| > M$ are equivalent. The analogous reciprocal statement holds for an infinite quantity.

Definition 1.24

Let α and β be infinitesimals along the same approach, and assume β is eventually nonzero.

1. *If $\alpha/\beta \to 0$, we write $\alpha = o(\beta)$ and say that α is of* higher order(高阶无穷小) *than β.*
2. *If $\alpha/\beta \to c \neq 0$ with $c \in \mathbb{R}$, they are of the* same order(同阶无穷小).
3. *If $\alpha/\beta \to 1$, we write $\alpha \sim \beta$ and call them* equivalent infinitesimals(等价无穷小).

♣

For example, as $x \to 0$, $x^2 = o(x)$, $3x$ and x have the same order, and $\sin x \sim x$. The comparison concerns a ratio, so signs must be retained: $-x$ and x have the same order but are not equivalent. Also, not every pair is comparable by a ratio limit; $x\sin(1/x)$ and x provide an example. The assertion $\alpha \sim \beta$ is equivalent to $\alpha = \beta(1+\eta)$ with $\eta \to 0$, or to $\alpha - \beta = o(\beta)$.

Proposition 1.15

Suppose $\alpha \sim \widetilde{\alpha}$ and $\beta \sim \widetilde{\beta}$, with all four functions nonzero sufficiently far along the approach. Then

$$\frac{\alpha\beta}{\widetilde{\alpha}\widetilde{\beta}} \to 1, \qquad \frac{\alpha/\beta}{\widetilde{\alpha}/\widetilde{\beta}} \to 1.$$

In particular, if $\widetilde{\alpha}/\widetilde{\beta} \to L \in \mathbb{R}$, then $\alpha/\beta \to L$. The same replacement preserves a signed infinite limit of the quotient.

♠

Proof The first ratio is $(\alpha/\widetilde{\alpha})(\beta/\widetilde{\beta})$, and the second is $(\alpha/\widetilde{\alpha})/(\beta/\widetilde{\beta})$. Both tend to 1 by the arithmetic laws. Multiplication by a factor tending to 1 preserves a finite limit. For an infinite limit, this factor lies between $1/2$ and $3/2$ eventually, so the definition of the corresponding infinite limit gives the conclusion. □

The trigonometric estimates and (1.6) yield the useful list, all as $x \to 0$:

$$\sin x \sim x, \qquad \tan x \sim x, \qquad 1-\cos x \sim \frac{x^2}{2}, \qquad \ln(1+x) \sim x, \qquad e^x - 1 \sim x. \tag{1.7}$$

For a fixed $p \in \mathbb{R} \setminus \{0\}$, one also has

$$(1+x)^p - 1 \sim px.$$

To verify it, put $t = p\ln(1+x)$ and use

$$\frac{(1+x)^p - 1}{px} = \frac{e^t - 1}{t}\frac{\ln(1+x)}{x} \longrightarrow 1.$$

All expressions are taken for sufficiently small x with $1 + x > 0$. For $p = 0$, the numerator is zero; the ratio to px is undefined, so the equivalence must not be asserted.

Example 1.32 Replacement in a product Evaluate $\displaystyle\lim_{x\to 0}\frac{\sin(2x)\ln(1+3x)}{1-\cos(4x)}$.

Solution *By (1.7), the three factors have equivalents $2x$, $3x$, and $8x^2$, respectively. Thus Proposition 1.15 gives*

$$\lim_{x\to 0}\frac{\sin(2x)\ln(1+3x)}{1-\cos(4x)} = \lim_{x\to 0}\frac{(2x)(3x)}{8x^2} = \frac{3}{4}.$$

Example 1.33 Cancellation changes the required precision Evaluate $\displaystyle\lim_{x\to 0}\frac{\sqrt{1+x}-1-x/2}{x^2}$.

> ***Idea.*** The equivalence $\sqrt{1+x}-1 \sim x/2$ describes the leading term. That term is subtracted in the numerator, so the equivalence alone does not determine the remaining error. Use exact identities before passing to a limit.

Solution *For sufficiently small nonzero x, let $s=\sqrt{1+x}$. Then*

$$s-1-\frac{x}{2}=\frac{x}{s+1}-\frac{x}{2}=\frac{x(1-s)}{2(s+1)}=-\frac{x^2}{2(s+1)^2}.$$

Since $s \to 1$, the required limit is $-1/8$. Replacing $s-1$ by $x/2$ inside the difference would have erased the entire nonzero answer.

Equivalence is compatible with multiplication and division in the stated setting; it does not authorize term-by-term substitution in an arbitrary sum or difference. For the elementary counterexample $\alpha = x + x^2 \sim x$, the difference $\alpha - x = x^2$ is not equivalent to the result 0 of such a substitution. Whenever leading terms cancel, return to an identity or prove a sharper error estimate.

1.4.8 Methods, pitfalls, and further questions

A useful order of work is to determine the domain and approach, test whether direct substitution is justified, and only then choose a transformation. For a direct proof, isolate a small factor and bound the other factors locally. For algebraic indeterminate forms, factor or rationalize. For a bounded oscillation multiplied by a small factor, use an absolute-value estimate and squeeze. For a change of variables, track the transformed domain and the direction of approach before applying the outer limit. For a positive power with varying base and exponent, take a logarithm and study the product of the exponent with the logarithm of the base. For infinite quantities, establish eventual signs and compare rates before using the rules in Proposition 1.13. For a suspected failure of a limit, try to find two admissible approaching sequences with incompatible output limits. For products or quotients of standard small quantities, check the hypotheses of equivalent replacement.

Three distinctions deserve particular attention. The value $f(a)$ and the limit at a answer different questions. Local boundedness is weaker than existence of a limit, and unboundedness is weaker than a signed infinite limit. Finally, a finite collection of numerical samples or approaching sequences can disprove a proposed limit but cannot establish the universal requirement in Theorem 1.14.

> ***Further Questions.*** Can a function have a limit at every point while being discontinuous at some points? What additional condition makes substitution of the limiting input legitimate? The next section answers these questions by comparing the limit near a point with the value at the point itself.

Exercise 1.4

1. Write the full quantified definitions of $\lim_{x\to a^-} f(x) = L$, $\lim_{x\to-\infty} f(x) = L$, and $\lim_{x\to a^+} f(x) = -\infty$. State the corresponding assumptions on the domain.
2. Let $m, b, a \in \mathbb{R}$. Prove directly that $\lim_{x\to a}(mx + b) = ma + b$. For a fixed $\varepsilon > 0$ and $m \neq 0$, find the largest possible δ in the definition; treat $m = 0$ separately.
3. Prove directly that $\lim_{x\to 2} 1/x = 1/2$. Explain why a preliminary restriction such as $|x - 2| < 1$ is useful.
4. Determine both one-sided limits at 0 for each function, and decide whether the two-sided limit exists:
$$\frac{\sqrt{1+x}-1}{|x|}, \qquad \frac{|\sin x|}{x}, \qquad \lfloor x \rfloor, \qquad \frac{1}{x^3}.$$
Distinguish finite and infinite limits.
5. Evaluate
$$\lim_{x\to a} \frac{x^n - a^n}{x - a} \quad (n \geq 1,\ a \in \mathbb{R}), \qquad \lim_{x\to 0} \frac{\sqrt{1+2x} - \sqrt{1-x}}{x}.$$
In the first limit, handle $n = 1$ separately so that no convention for 0^0 is needed.
6. Evaluate the limits of $\sqrt{x^2+3x} - \sqrt{x^2 - x}$ as $x \to +\infty$ and as $x \to -\infty$. Identify where the sign of x affects the rationalized expression.
7. Find all vertical and horizontal asymptotes of $f(x) = (2x + 1)/(x - 1)$, stating the relevant limits.
8. Evaluate
$$\lim_{x\to 0} x\lfloor 1/x \rfloor, \qquad \lim_{x\to+\infty} \frac{\sin x}{x + \cos x}.$$
Use an absolute-value estimate in each case, and justify the denominator bound in the second expression.
9. Let $h(x) = \sin(\pi/x)$ for $x \neq 0$. Since $h(1/n) = 0$ for all $n \geq 1$, a proposed argument claims that $\lim_{x\to 0^+} h(x) = 0$. Explain the logical error and construct another positive input sequence that disproves the claimed limit.
10. Let $f(x) = xD(x)$, where D is the Dirichlet function. Prove that $f(x) \to 0$ as $x \to 0$, and that f has no limit at any $a \neq 0$.
11. Evaluate
$$\lim_{x\to 0} \frac{\sin(x^2)}{x \sin x}, \qquad \lim_{x\to 0} \frac{\sin(3x) - \sin x}{x}, \qquad \lim_{x\to 0} \frac{1 - \cos(3x)}{x \sin(2x)}.$$
12. Evaluate
$$\lim_{x\to 0}(1 + 2x)^{3/x}, \qquad \lim_{x\to+\infty} \left(1 - \frac{2}{x}\right)^x.$$
State where the real powers are defined before transforming them.
13. Evaluate
$$\lim_{x\to 0} \frac{\ln(1 + 4x)}{\sin(3x)}, \qquad \lim_{x\to 0} \frac{5^x - 1}{\tan(2x)}, \qquad \lim_{x\to 0} \frac{(1 + x)^p - 1}{x} \quad (p \in \mathbb{R}).$$
14. Order x^4, $1 - \cos x$, and $\sin x$ by their relative smallness as $x \to 0$. Express the comparisons using o and give an equivalent for $1 - \cos x$.
15. Decide which statements follow from $f(x) \to L$ as $x \to a$, and give a proof or counterexample: f is bounded on its whole domain; f is bounded near a; $f(x) < L$ eventually; $|f(x)| \to |L|$. Interpret "near a" as punctured if $f(a)$ is undefined.
16. Prove the squeeze theorem using the sequential criterion and the sequence squeeze theorem. Identify the part of the argument that requires the approaching sequence to be arbitrary.
17. Let $u(x) \to 0$ as $x \to 0$, and define $v(0) = 2$ and $v(t) = 1$ for $t \neq 0$. Give one u for which $v(u(x)) \to 1$, one for which it tends to 2, and one for which it has no limit at 0.

18. Evaluate by exact rationalization
$$\lim_{x\to 0}\frac{\sqrt{1+x}+\sqrt{1-x}-2}{x^2}.$$
Explain why substituting $\sqrt{1+x}-1\sim x/2$ and $\sqrt{1-x}-1\sim -x/2$ inside the sum does not give the answer.
19. Determine all real c, d for which the function
$$f(x)=\begin{cases}(e^{cx}-1)/x, & x<0,\\ d, & x=0,\\ \ln(1+x)/\sin(2x), & x>0\end{cases}$$
has a finite limit at 0, on the domain where the displayed formulas are defined. What is that limit, and what role does d play?
20. Suppose $\alpha\sim\beta$ and $\gamma=o(\beta)$, with β eventually nonzero. Prove that $\alpha+\gamma\sim\beta$. Construct $\alpha\sim\beta$ for which $\alpha-\beta$ is not identically zero, and explain why its order cannot be recovered from the equivalence alone.

Supplement Exercises

S1. Prove the sequential criterion for $\lim_{x\to a} f(x)=+\infty$. Write out the negation of the infinite-limit definition in the converse argument.
S2. For $p>0$, study $f_p(x)=|x|^p\sin(1/x)$ as $x\to 0$. Determine its limit and find all $q>0$ for which $f_p=o(|x|^q)$.
S3. Suppose f is defined on a punctured neighbourhood of 0 and $f(x)^2\to 1$ as $x\to 0$. Show that $|f(x)|\to 1$. Must f have a limit? Give a counterexample and then a sufficient condition involving its eventual sign.
S4. Suppose $f(x)\to A$ and $f(x)g(x)\to B$ as $x\to a$, with $A\neq 0$. Prove that $g(x)\to B/A$. Show by examples that no such conclusion follows when $A=0$, even if g is bounded.
S5. Let f be increasing and bounded on (a,b), where $a<b$ are finite. Prove that its one-sided limits at the endpoints exist and equal $\inf f((a,b))$ and $\sup f((a,b))$, respectively. Here "increasing" may be read as nondecreasing.
S6. Along $x\to 0^+$, let $v(x)=1/x$. Construct choices of $u(x)\to 0$, with $1+u(x)>0$, for which $(1+u(x))^{v(x)}$ tends respectively to 1, e, and e^2. Construct one further choice for which the power has no limit.

> ***Hint.*** Use Proposition 1.14 and design the product $u(x)v(x)$ first. For the nonconvergent case, make that product oscillate between two values along suitable sequences.

S7. Construct a function on $(0,1)$ which is unbounded in every right-hand neighbourhood of 0 but has neither limit $+\infty$ nor limit $-\infty$ there. Then construct one whose absolute value tends to $+\infty$ while the function has no signed infinite limit.
S8. For integers $m,n\geq 1$, evaluate
$$\lim_{x\to 0}\frac{\sqrt{1+\sin(mx)}-\sqrt{1-\sin(mx)}}{\sin(nx)}.$$
Use only identities and limits proved in this section.
S9. Let $E=\{1/n : n\geq 1\}$ and define $f(x)=1$ on E and $f(x)=0$ on $\mathbb{R}\setminus E$. Determine every $a\in\mathbb{R}$ at which $\lim_{x\to a} f(x)$ exists. Distinguish a point of E from the accumulation point 0.
S10. Prove that $\alpha\sim\beta$ and $\beta\sim\gamma$ imply $\alpha\sim\gamma$, under the eventual nonzero conditions. Is $\alpha+\beta\sim\gamma+\eta$ implied by $\alpha\sim\gamma$ and $\beta\sim\eta$? Give a counterexample with both sums eventually nonzero, and explain how cancellation causes the failure.

1.5 Continuous Functions

重点内容与证明方法

- *Relate continuity to values, nearby limits, and domain endpoints.*
- *Apply algebra, composition, and quantitative estimates.*
- *Prove inverse continuity with the domain made explicit.*
- *Classify discontinuities and construct continuous extensions.*
- *Use sequences and dense sets to test continuity.*
- *Distinguish continuity from stronger forms of local control.*

A limit describes what the values near a point approach. Continuity asks whether that limiting behaviour agrees with the value at the point. For instance, $(x^2-1)/(x-1)$ tends to 2 at 1. Assigning the value 2 fills its hole consistently with the surrounding graph; assigning any other value creates a mismatch. This agreement, rather than any particular formula or way of drawing a graph, is the central idea.

1.5.1 Continuity at a point and on a domain

Definition 1.25

Let $f: D \to \mathbb{R}$ and $a \in D$. The function is continuous at a (在一点连续) *if for every $\varepsilon > 0$ there exists $\delta > 0$ such that*

$$x \in D, \quad |x-a| < \delta \quad \Longrightarrow \quad |f(x)-f(a)| < \varepsilon.$$

The function is continuous on D (在集合上连续) *if it is continuous at every point of D.* ♣

The exact negation of continuity at a is

$$\exists \varepsilon_0 > 0 \quad \forall \delta > 0 \quad \exists x \in D: \quad |x-a| < \delta, \qquad |f(x)-f(a)| \geq \varepsilon_0.$$

Unlike the negation of a punctured function limit, the selected input is allowed to equal a in principle; however, it cannot do so because the output error would then be zero. Choosing one such input for each $\delta = 1/n$ produces the bad sequence used in the sequential criterion for continuity.

This definition includes the input $x = a$. At that input the output error is zero, so no exclusion is needed. When a is an accumulation point of D, continuity is equivalent to

$$\lim_{\substack{x \to a \\ x \in D}} f(x) = f(a).$$

Thus the familiar limit test checks three things: the value $f(a)$ is defined, the finite limit exists, and that limit equals $f(a)$.

The domain is part of the statement. For example, $1/x$ is continuous at every point of $\mathbb{R} \setminus \{0\}$. It is not a function on all of $\mathbb{R}$ with the displayed formula. We may discuss its singular behaviour or possible extension at 0, but continuity at 0 requires a value there. At an isolated point of D, continuity is automatic: choose a neighbourhood containing no other point of D. The accumulation-point assumption in the function-limit definition prevents us from using a vacuous limit test at such a point.

If f is defined on $[a,b]$ with $a < b$, continuity at a means

$$\lim_{x \to a^+} f(x) = f(a),$$

and continuity at b means the corresponding equality for the left-hand limit. These are *right continuity*(右连续) and *left continuity*(左连续), respectively. Continuity on $[a, b]$ requires continuity at every interior point and these one-sided conditions at the endpoints. For half-open intervals, impose the one-sided condition only at the included endpoint.

Example 1.34 An endpoint estimate Prove that $f(x) = \sqrt{x}$ is continuous on $[0, +\infty)$.

Solution *For nonnegative x, y, assume first that $x \geq y$. Then*

$$x - y = (\sqrt{x} - \sqrt{y})(\sqrt{x} + \sqrt{y}) \geq (\sqrt{x} - \sqrt{y})^2.$$

Interchanging x, y when necessary gives

$$\left|\sqrt{x} - \sqrt{y}\right| \leq \sqrt{|x - y|}. \tag{1.8}$$

For any $a \geq 0$ and $\varepsilon > 0$, choose $\delta = \varepsilon^2$. If $x \geq 0$ and $|x - a| < \delta$, then $|\sqrt{x} - \sqrt{a}| < \varepsilon$. This argument also covers the endpoint $a = 0$.

> ***Reading a graph.*** The informal description “draw the graph without lifting the pencil” can guide intuition on an interval, but it does not specify domains, endpoints, or the required input control. For proofs, use the definition or one of the criteria established below.

1.5.2 Algebra and composition of continuous functions

Theorem 1.20

If $f, g\colon D \to \mathbb{R}$ are continuous at $a \in D$, then $f + g$, $f - g$, and fg are continuous at a. Every constant multiple of f is continuous there. If $g(a) \neq 0$, then f/g is defined on a neighbourhood of a relative to D and is continuous at a on its domain. ♡

Proof If a is an accumulation point of D, apply Theorem 1.15 to the limits $f(x) \to f(a)$ and $g(x) \to g(a)$. The resulting limits equal the values of the corresponding functions at a. When $g(a) \neq 0$, continuity gives $|g(x) - g(a)| < |g(a)|/2$ near a, so $|g(x)| > |g(a)|/2$ there. If a is isolated in D, every displayed function that is defined at a is continuous there directly from the definition. □

Restricting a continuous function to a smaller domain preserves continuity at every point retained in that domain. Indeed, the same δ controls all inputs in the smaller set. This observation lets us apply the algebraic rules on a common domain even when the original functions have different domains.

Theorem 1.21

Let $f\colon D \to E$ and $g\colon E \to \mathbb{R}$, where $D, E \subseteq \mathbb{R}$. If f is continuous at $a \in D$ and g is continuous at $f(a)$ relative to E, then $g \circ f$ is continuous at a. ♡

> ***Idea.*** Work backwards from the desired output error. Continuity of g supplies a tolerance for its input, and continuity of f turns that tolerance into a neighbourhood of a. Unlike a punctured outer limit, continuity also controls the case $f(x) = f(a)$.

Proof Let $\varepsilon > 0$. Since g is continuous at $f(a)$, there is $\eta > 0$ such that

$$y \in E, \quad |y - f(a)| < \eta \quad \Longrightarrow \quad |g(y) - g(f(a))| < \varepsilon.$$

Continuity of f at a gives $\delta > 0$ such that

$$x \in D, \quad |x - a| < \delta \quad \Longrightarrow \quad |f(x) - f(a)| < \eta.$$

Because $f(x) \in E$, substitution into the first implication proves the required estimate for $g(f(x))$. □

Example 1.35 Continuity without limits of all factors Determine all points of continuity of

$$F(x) = \begin{cases} x\sin(1/x), & x \neq 0, \\ 0, & x = 0. \end{cases}$$

Solution *At every nonzero point, continuity follows from arithmetic and composition, using the sine limits* *established in Section 1.4. At* 0*, the inner function* $1/x$ *has no finite limit, so that composition argument does not apply. Instead,* $|F(x) - F(0)| \le |x|$ *proves continuity directly. Thus* F *is continuous on all of* $\mathbb{R}$*. Theorems give sufficient conditions; failure of one theorem's hypotheses does not establish discontinuity.*

1.5.3 Inverse functions and elementary functions

An inverse function interchanges inputs and outputs. To establish its continuity, an input interval around a must produce a usable interval of output control around $f(a)$. Strict monotonicity provides such control without a formula for the inverse.

Theorem 1.22

Let I *be an interval and let* $f\colon I \to \mathbb{R}$ *be strictly monotone. Then the inverse* $f^{-1}\colon f(I) \to I$ *is continuous relative to its domain* $f(I)$ *and is strictly monotone in the same direction as* f*. In particular, this applies when* f *is continuous and strictly monotone on* I*.* ♡

> ***Proof Strategy.*** For an increasing function, points to the left and right of a have images strictly below and above $f(a)$. Keep the new output between those images; monotonicity forces its inverse image to remain near a. At an included endpoint, only the available side needs a bracket.

Proof Suppose f is strictly increasing, fix $a \in I$, and write $b = f(a)$. The inverse exists on $f(I)$ by injectivity, and order preservation of the inverse follows directly from strict increase of f. Let $\varepsilon > 0$. If I has a point less than a, the interval property allows us to choose $u \in I$ with $a - \varepsilon < u < a$. Likewise, if I has a point greater than a, choose $v \in I$ with $a < v < a + \varepsilon$. For every available bracket, the numbers $b - f(u)$ and $f(v) - b$ are strictly positive. Choose δ to be the minimum of the available positive numbers; if neither side is available, $I = \{a\}$ and the inverse is continuous automatically.

Let $y \in f(I)$ satisfy $|y - b| < \delta$, and set $x = f^{-1}(y)$. If u was chosen, then $y > f(u)$, so $x > u > a - \varepsilon$. If no left bracket was available, every point of I is at least a, which gives the same required lower bound. Similarly, either $y < f(v)$ forces $x < v < a + \varepsilon$, or every point of I is at most a. Thus $\left|f^{-1}(y) - a\right| < \varepsilon$, proving continuity at b. For strictly decreasing f, apply the increasing case to $-f$ and compose the resulting inverse with $y \mapsto -y$. □

The theorem asserts continuity on the *actual range* $f(I)$. It does not assert that this range is an interval when f is discontinuous. For example, the strictly increasing function

$$f(x) = \begin{cases} x, & x < 0, \\ x + 1, & x \ge 0 \end{cases}$$

has range $(-\infty, 0) \cup [1, +\infty)$. Its inverse is continuous on that range, including at 1 relative to the range, although f itself jumps at 0. In the next section, the intermediate value theorem will show that the range of a *continuous* function on an interval is an interval.

Corollary 1.3

Every elementary function is continuous on its natural domain. ♡

Proof Constant functions and the identity are continuous directly from the definition. The arithmetic rules give continuity of polynomials and of rational functions wherever their denominators are nonzero. The trigonometric limits in Section 1.4 give continuity of sine and cosine, and of their quotients wherever defined. The exponential and logarithmic functions are continuous by Lemma 1.2.

For $x > 0$, a real power $x^p = e^{p \ln x}$ is continuous by composition. When $p > 0$ and the value $0^p = 0$ is included, continuity at 0 follows from $0 \le x < \varepsilon^{1/p} \implies x^p < \varepsilon$. Integer powers already follow from arithmetic. Roots on their real domains are continuous inverses of the corresponding strictly monotone power functions; odd roots include negative inputs, and even roots use the nonnegative interval. The inverse trigonometric functions are continuous by Theorem 1.22, using the standard intervals of strict monotonicity specified in Section 1.2. This also handles their included endpoints. Finally, finitely many arithmetic operations and compositions preserve continuity on the domain where the resulting expression is defined. □

Example 1.36 Determine the domain first Find the natural domain of

$$h(x) = \frac{\ln(1 - x^2)}{\sqrt{x + 2}}$$

and determine where it is continuous.

Solution *The logarithm requires $1 - x^2 > 0$, so $-1 < x < 1$. The denominator requires $x + 2 > 0$, which already holds on this interval. Thus the natural domain is $(-1, 1)$, and h is continuous at every point of it by Corollary 1.3. The formula has no values at -1 and 1. Endpoint extension is a separate question, to be decided by limits rather than by a direct substitution into the formula.*

1.5.4 Discontinuities and continuous extensions

When a function is defined on a neighbourhood of a but fails to be continuous there, a is a *point of discontinuity* (间断点). The same terminology is often used when the original formula is defined only on a punctured neighbourhood, in order to discuss whether it can be extended across the missing point. We will explicitly distinguish a missing value from an assigned value that fails the continuity test.

Definition 1.26

Suppose f is defined on a punctured neighbourhood of a.

1. *A* removable discontinuity (可去间断点) *occurs if $\lim_{x\to a} f(x) = L \in \mathbb{R}$, but $f(a)$ is missing or differs from L.*
2. *A* jump discontinuity (跳跃间断点) *occurs if both one-sided limits are finite but unequal.*
3. *An* infinite discontinuity (无穷间断点) *occurs if at least one one-sided limit is $+\infty$ or $-\infty$.*
4. *An* oscillatory discontinuity (振荡间断点) *describes a failure of a one-sided limit caused by persistent oscillation, as for $\sin(1/(x - a))$.*

♣

Removable and jump discontinuities are traditionally called discontinuities of the *first kind* (第一类间断点), since both one-sided finite limits exist. If at least one finite one-sided limit does not exist, the discontinuity is of the *second kind* (第二类间断点). The last two descriptions in Definition 1.26 identify common mechanisms of the second kind; a more complicated function can combine different behaviours on its two sides. Neither a jump nor persistent oscillation can be repaired by changing just $f(a)$.

The schematics in **Figure** 1.7 separate a mismatched value, unequal side limits, and unbounded growth. The following examples make the distinctions exact.

Example 1.37 Classifying singular behaviour Classify the behaviour at 0 of the following formulas, initially defined for $x \neq 0$:

$$\frac{\sin x}{x}, \qquad \frac{|x|}{x}, \qquad \frac{1}{x^2}, \qquad \sin(1/x).$$

For each, determine whether assigning a value at 0 can make it continuous there.

Solution *The first has limit 1, so assigning that value gives continuity; its missing point is removable. The*

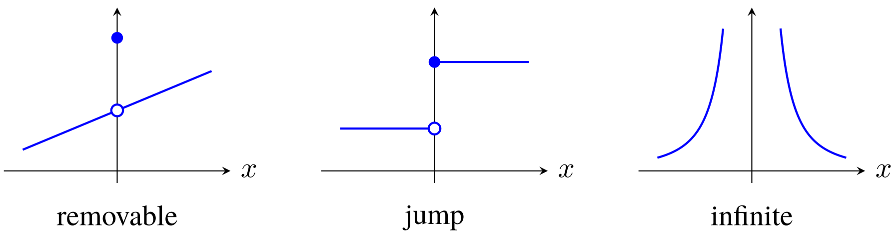


Figure 1.7: Three ways in which nearby values can fail to agree with a value at the origin. Oscillation gives another mechanism, illustrated in **Figure** 1.5.

second has side limits -1 and 1, hence a jump. The third tends to $+\infty$, so it has an infinite discontinuity under any real-valued assignment at 0. The fourth has distinct sequential output limits even from one side, hence oscillatory behaviour. Only the first admits a continuous extension.

Proposition 1.16

Let a be an accumulation point of D, with $a \notin D$, and let $f\colon D \to \mathbb{R}$. A function $F\colon D \cup \{a\} \to \mathbb{R}$ agreeing with f on D can be continuous at a if and only if $\lim_{x\to a,\, x\in D} f(x)$ exists as a finite real number. In that case the unique possible value is

$$F(a) = \lim_{\substack{x\to a\\ x\in D}} f(x).$$

If f is also continuous on D, this defines a continuous extension (连续延拓) *on $D \cup \{a\}$.*

Proof Continuity at a requires the limit to equal $F(a)$, so necessity and uniqueness follow from the definition and uniqueness of limits. Conversely, assign the finite limit to $F(a)$. The limit estimate gives continuity at a, including the trivial case $x = a$. At a point $b \in D$, restrict to a neighbourhood of radius less than $|b - a|$; within it the extension agrees with f, so any continuity at b is preserved. □

Example 1.38 Gluing three pieces Find all $A, B \in \mathbb{R}$ for which

$$f(x) = \begin{cases} \sin x/x, & x < 0,\\ A, & x = 0,\\ B + \ln(1+x)/x, & x > 0 \end{cases}$$

is continuous on $\mathbb{R}$.

Solution *Each open piece is continuous on its domain, so only 0 needs a special check. The left-hand limit is 1, and the right-hand limit is $B + 1$. Both must equal $f(0) = A$. Thus $A = 1$ and $B = 0$, and these choices do give continuity everywhere. Matching the two side limits would determine B but would leave the value A unchecked.*

Example 1.39 A floor-function composition Determine the discontinuities of $g(x) = \lfloor x^2 \rfloor$.

Solution *If x_0^2 is not an integer, then it lies strictly between two consecutive integers. Continuity of x^2 keeps nearby squared values in that same interval, so g is locally constant at x_0. At 0, one has $g(x) = 0$ for $|x| < 1$, so g is continuous there as well. For a positive integer n, at $x_0 = \sqrt{n}$ the left and right limits are $n - 1$ and n. At $x_0 = -\sqrt{n}$, their order is reversed: the limits are n and $n - 1$. Thus the discontinuities are exactly $\pm\sqrt{n}$ for $n \geq 1$, and each is a jump. An integer value of the inner function alone does not guarantee a discontinuity of the composition; the way nearby inputs reach that value matters.*

1.5.5 Sequential continuity and quantitative control

Theorem 1.23

Let $f\colon D \to \mathbb{R}$ and $a \in D$. Then f is continuous at a if and only if every sequence (x_n) in D converging to a satisfies $f(x_n) \to f(a)$. The sequence is allowed to contain the point a. ♡

> ***Idea.*** The argument resembles Heine's criterion, but the input neighbourhood is no longer punctured and the proposed output limit is fixed to be $f(a)$. Failure of continuity supplies inputs arbitrarily close to a whose outputs stay a fixed positive distance from $f(a)$.

Proof If f is continuous at a, choose δ for a given $\varepsilon > 0$. Since $x_n \to a$, eventually $|x_n - a| < \delta$, and hence $|f(x_n) - f(a)| < \varepsilon$. Conversely, if continuity fails, there is $\varepsilon_0 > 0$ such that for every $n \geq 1$ we can choose $x_n \in D$ with

$$|x_n - a| < \frac{1}{n}, \qquad |f(x_n) - f(a)| \geq \varepsilon_0.$$

This sequence tends to a but its output sequence does not tend to $f(a)$, contradicting the stated property. □

Example 1.40 A function controlled at just one point Determine all points of continuity of $f(x) = xD(x)$ on $\mathbb{R}$.

Solution *Since $|f(x)| \leq |x|$ and $f(0) = 0$, the function is continuous at 0. At $a \neq 0$, choose rational and irrational sequences both tending to a. The rational outputs tend to a, and the irrational outputs equal 0. They cannot both tend to the single value $f(a)$. Thus the only point of continuity is 0. The factor x suppresses the dense-set disagreement exactly where that factor vanishes.*

Definition 1.27

A function $f\colon D \to \mathbb{R}$ is Lipschitz continuous (Lipschitz 连续) *on D if there is a constant $K \geq 0$ such that*

$$|f(x) - f(y)| \leq K\,|x - y| \qquad (x, y \in D).$$ ♣

Every Lipschitz function is continuous: if $K > 0$, choose $\delta = \varepsilon/K$ in the continuity definition at any point; if $K = 0$, the function is constant on a nonempty domain. Even a local estimate $|f(x) - f(a)| \leq C\,|x - a|$ near one point is enough for continuity at that point. Such estimates give more information than the bare existence of a suitable neighbourhood.

The reverse triangle inequality shows that $|x|$ is Lipschitz on $\mathbb{R}$ with $K = 1$. For $x, y \in [-R, R]$ with $R > 0$,

$$\left|x^2 - y^2\right| = |x - y|\,|x + y| \leq 2R\,|x - y|\,,$$

so the square function is Lipschitz on each bounded interval of this form. The square-root function is continuous at 0, but it is not Lipschitz on $[0, 1]$. Indeed, a Lipschitz estimate with $y = 0$ and $x > 0$ would require $1/\sqrt{x} \leq K$, which fails as $x \to 0^+$. The weaker estimate (1.8) still gives continuity.

1.5.6 Further reading: a dense set of discontinuities

The Dirichlet function is discontinuous everywhere because two dense sets retain a fixed output separation. Can the same rational–irrational split produce continuity at irrational inputs? The answer changes when the size of a rational value decreases with its denominator.

Example 1.41 Thomae's function On $[0,1]$, define

$$T(x) = \begin{cases} 1/q, & x = p/q \text{ in lowest terms}, \quad q \geq 1, \\ 0, & x \notin \mathbb{Q}. \end{cases}$$

Here $0 = 0/1$ and $1 = 1/1$. Prove that T is continuous at every irrational point and discontinuous at every rational point of $[0,1]$.

> ***Proof Strategy.*** For a fixed tolerance, only finitely many rationals in $[0,1]$ have denominators small enough to produce a large output. An irrational input can be separated from that finite exceptional set by a positive distance. At a rational input, an irrational sequence immediately detects the mismatch.

Proof Let $a \in [0,1]$ be irrational and let $\varepsilon > 0$. Choose a positive integer N with $1/N < \varepsilon$. There are only finitely many reduced rationals in $[0,1]$ with denominators $q \leq N$, since for each such q the numerator satisfies $0 \leq p \leq q$. Call this finite set E_N. It does not contain a, so the minimum distance from a to E_N is positive. Choose δ smaller than that distance. If $x \in [0,1]$ and $|x - a| < \delta$, then either x is irrational and $T(x) = 0$, or x has reduced denominator $q > N$ and $T(x) = 1/q < 1/N < \varepsilon$. In either case $|T(x) - T(a)| < \varepsilon$.

Now let $a = p/q$ be rational in lowest terms. By density, choose irrational points $x_n \in [0,1]$ tending to a, using the available one-sided interval at an endpoint. Then $T(x_n) = 0$ for every n, whereas $T(a) = 1/q > 0$. The sequential criterion proves discontinuity at a. □

In fact, the same finite-set argument proves that the limit of $T(x)$ at every point of $[0,1]$, through distinct domain points, is 0. At a rational a, omit a itself from E_N when choosing the distance; if no exceptional points remain, any small radius suffices. Thus each rational discontinuity is removable individually. Removing all of them produces the zero function. Continuity is local, and a complicated global pattern of exceptional points does not prevent a precise local proof.

1.5.7 Methods, pitfalls, and the local-to-global question

For a function given by elementary formulas on pieces, first identify the natural domain of each formula and the points where pieces meet. Away from those points, continuity often follows from the algebra and composition rules. At a meeting point, compute the available side limits and compare every one of them with the assigned value. At a missing point, a finite limit supplies the only possible continuous extension. When limits are difficult to compute, an error estimate or a carefully chosen sequence may settle the question directly.

Continuity of the outer function is a sufficient condition for continuity of a composition, not a necessary one. Likewise, Lipschitz control is sufficient but not necessary for continuity. Avoid turning a theorem's hypothesis into an automatic test for the opposite conclusion when that hypothesis fails. At an endpoint, all neighbourhoods are interpreted relative to the domain.

> ***Looking ahead.*** Continuity at one point controls nearby values, but the radius may depend on that point. What can be concluded when continuity holds at every point of a closed bounded interval? The next section explains why the function must attain its largest and smallest values, take every intermediate value, and admit a single input tolerance valid throughout the interval.

Exercise 1.5

1. State the definition of continuity at a for a function on $[a,b)$ with $a < b$. Explain why a left-hand limit at a is not required and why no continuity condition is imposed at the missing endpoint b.

2. Prove directly that $f(x) = x^2$ is continuous at every $a \in \mathbb{R}$. Give a possible δ in terms of a and ε.
3. Determine the natural domain of each function and justify continuity on that domain:
$$\sqrt{1-x^2}, \qquad \ln(x^2-1), \qquad \frac{\arcsin x}{1-x}, \qquad e^{1/(x-2)}.$$
4. Let $f(0) = c$ and $f(x) = \sin(3x)/x$ for $x \neq 0$. Determine the value of c giving continuity on $\mathbb{R}$. Classify the discontinuity at 0 for every other value of c.
5. Find $a, b \in \mathbb{R}$ such that the following function is continuous on $\mathbb{R}$:
$$f(x) = \begin{cases} x^2, & x \leq 1, \\ ax+b, & 1 < x < 2, \\ 3x-1, & x \geq 2. \end{cases}$$
6. Classify every discontinuity of $\lfloor x \rfloor$, $x - \lfloor x \rfloor$, and $1/(x^2-1)$. For the last formula, discuss its missing points and their one-sided behaviour.
7. For the formula
$$f(x) = \frac{\sin x}{x(x-1)} \qquad (x \in \mathbb{R} \setminus \{0,1\}),$$
decide which missing points admit continuous extensions and find their values. Determine the largest domain in $\mathbb{R}$ to which this particular function can be continuously extended.
8. Prove that if f is continuous at a and $f(a) > 0$, then f is positive on a neighbourhood of a relative to its domain. Why is a positive value at a alone insufficient?
9. Prove that if f is continuous, then so is $|f|$. Deduce continuity of $\max\{f, g\}$ and $\min\{f, g\}$ whenever f and g are continuous on the same domain.
10. Give an example where $g \circ f$ is continuous at 0 but g is discontinuous at $f(0)$. Give another where $g \circ f$ is continuous at 0 but f is discontinuous at 0.
11. Determine all discontinuities of $f(x) = \lfloor \cos x \rfloor$ and classify each one. Pay particular attention to inputs where $\cos x$ equals 1, 0, or -1.
12. Use sequences to determine all points of continuity of
$$f(x) = \begin{cases} x^2, & x \in \mathbb{Q}, \\ -x, & x \notin \mathbb{Q}. \end{cases}$$
13. Prove that $f(x) = |2x-1|$ is Lipschitz on $\mathbb{R}$. Prove that $g(x) = 1/x$ is Lipschitz on $[a, +\infty)$ for each fixed $a > 0$, and give a valid constant.
14. Prove that the function $x \mapsto x^{1/3}$ is continuous on $\mathbb{R}$ by applying the inverse-continuity theorem. Show that it is not Lipschitz on any interval containing 0 as an interior point.
15. Let $f\colon [-1,1] \to \mathbb{R}$ be continuous and strictly increasing, and let $b = f(0)$. Given $0 < \varepsilon < 1$, give an explicit positive δ in terms of $f(-\varepsilon)$, $f(0)$, and $f(\varepsilon)$ which proves continuity of f^{-1} at b.
16. Let $D = \{0\} \cup [1,2]$, define $f(0) = 100$, and set $f(x) = x$ on $[1,2]$. Prove that f is continuous on D. Explain why continuity at the isolated point 0 should be checked using the neighbourhood definition.
17. Suppose $f\colon \mathbb{R} \to \mathbb{R}$ is continuous at a and $f(x) = 0$ for every rational x. Prove that $f(a) = 0$. Deduce that a continuous function on $\mathbb{R}$ is determined by its values on $\mathbb{Q}$.
18. Let $f\colon [-1,1] \to \mathbb{R}$ satisfy $|f(x) - f(0)| \leq C\,|x|^p$ for some constants $C > 0$ and $p > 0$. Prove continuity at 0 and give an explicit δ for every $\varepsilon > 0$. Does this assumption imply continuity at other points? Construct a counterexample if it does not.

Supplement Exercises

S1. Let $f, g\colon \mathbb{R} \to \mathbb{R}$ be continuous, and define $h = f$ on $\mathbb{Q}$ and $h = g$ on $\mathbb{R} \setminus \mathbb{Q}$. Prove that h is continuous at a if and only if $f(a) = g(a)$.

S2. Prove that if both $f + g$ and $f - g$ are continuous at a point, then both f and g are continuous there. Is the analogous claim with $f + g$ and fg true? Give a proof or counterexample.

S3. Let $f\colon \mathbb{R} \to \mathbb{R}$ satisfy $f(x+y) = f(x)+f(y)$ for all real x, y. If f is continuous at 0, prove that $f(x) = cx$ for all x, where $c = f(1)$.

> ***Hint.*** First establish the formula on $\mathbb{Q}$ and then pass to real inputs.

S4. Suppose $f\colon (0, +\infty) \to \mathbb{R}$ is continuous at 1 and satisfies $f(x^2) = f(x)$ for every $x > 0$. Prove that f is constant, using repeated square roots and a sequence converging to 1.

S5. Prove directly that the limit of Thomae's function from Example 1.41 at each rational $a \in [0, 1]$, through distinct domain points, is 0. Explain how the finite exceptional set must be modified when it contains a.

S6. Suppose $f\colon D \to \mathbb{R}$ satisfies $|f(x) - f(y)| \le C\,|x - y|^p$ for all $x, y \in D$, where $C > 0$ and $p > 0$. Prove continuity on D. If D is an interval and $p > 1$, prove that f is constant by dividing a segment into n equal parts.

S7. Let I be an interval, and suppose $f\colon I \to \mathbb{R}$ is strictly increasing and $f(I)$ is also an interval. Prove that f is continuous. You may apply Theorem 1.22 to f^{-1}; explain why its hypotheses hold.

S8. Give a continuous injective function on a domain that is not an interval whose inverse on its range is discontinuous.

> ***Hint.*** One possible domain is $[0, 1) \cup \{2\}$; arrange for a sequence approaching the missing point 1 to have images approaching the image of 2.

S9. Let $f\colon \mathbb{R} \to \mathbb{R}$ be continuous at a and suppose $|f(a)| < 1$. Prove that some neighbourhood of a admits a constant $q < 1$ with $|f(x)| \le q$ throughout that neighbourhood. Deduce that for every sequence $x_n \to a$, one has $f(x_n)^n \to 0$.

S10. Define $F(0) = 0$ and $F(x) = x\sin(1/x^2)$ for $x \ne 0$. Prove continuity on $\mathbb{R}$ and prove that F is not Lipschitz on any neighbourhood of 0.

> ***Hint.*** For the second assertion, compare inputs $x_n = (2\pi n+\pi/2)^{-1/2}$ and $y_n = (2\pi n+3\pi/2)^{-1/2}$.

1.6 Continuous Functions on Closed Intervals

重点内容与证明方法

- *Use intermediate values to prove existence, and test uniqueness separately.*
- *Distinguish bounds from attained extrema.*
- *Verify the hypotheses of closed-interval theorems.*
- *Study proofs using bisection and convergent subsequences.*
- *Compare continuity and uniform continuity.*
- *Prove Heine–Cantor and examine its hypotheses.*

Continuity is defined one point at a time. Nevertheless, on a closed bounded interval it forces several properties of the whole function: it takes every intermediate value, attains its extreme values, and admits a single input tolerance that controls all points at once. These conclusions connect the local language of limits with the completeness of the real line. They also make it possible to prove that equations have solutions without first finding a formula for those solutions.

Throughout this section, $a, b \in \mathbb{R}$ and $a < b$ unless stated otherwise. Continuity on $[a, b]$ includes right continuity at a and left continuity at b. We first state the main conclusions and use them in examples. Complete proofs are given in the deeper-reading subsection Section 1.6.3, using results already established in this chapter.

1.6.1 Intermediate values and zeros

Theorem 1.24 (Intermediate value theorem)

Let $f\colon [a, b] \to \mathbb{R}$ be continuous. If y lies between $f(a)$ and $f(b)$, including either endpoint value, then there exists $c \in [a, b]$ such that $f(c) = y$. If y lies strictly between $f(a)$ and $f(b)$, such a point lies in (a, b). ♡

The *intermediate value property*(介值性)expresses the absence of a gap in the values taken between two inputs. It does not assert monotonicity, and the point c need not be unique. For instance, a continuous oscillating graph may cross one horizontal level many times. The theorem applies to any continuous function on an interval by restricting it to the closed segment joining the two chosen inputs. Thus the interval need not itself be closed or bounded in order for intermediate values between two attained values to be attained.

Corollary 1.4 (Zero theorem)

If $f\colon [a, b] \to \mathbb{R}$ is continuous and $f(a)f(b) < 0$, then $f(c) = 0$ for some $c \in (a, b)$. If instead $f(a)f(b) \le 0$, then a zero exists in $[a, b]$. ♡

Proof With a strict sign change, 0 lies strictly between the endpoint values, so apply Theorem 1.24 with $y = 0$. If an endpoint value is zero, that endpoint supplies the conclusion in the nonstrict case. □

Conversely, applying the zero theorem to $g(x) = f(x) - y$ proves the intermediate value theorem for a strictly intermediate level y. Endpoint levels are already attained at the endpoints. Thus the two statements contain the same existence mechanism. The strict sign condition is convenient for locating a zero inside an interval, but it is sufficient rather than necessary: x^2 has a zero on $[-1, 1]$ although its endpoint values are both positive.

Example 1.42 Existence and uniqueness of a root Prove that $x = \cos x$ has exactly one solution in $[0, 1]$.

> ***Idea.*** To prove existence, move both sides to a continuous function with opposite endpoint signs. To prove uniqueness, use order: the left side is strictly increasing and the right side is decreasing on the interval. These are separate arguments.

Solution *Set $g(x) = x - \cos x$. It is continuous on $[0, 1]$ by the continuity rules, $g(0) = -1 < 0$, and $g(1) = 1 - \cos 1 > 0$. The last inequality follows from $1 - \cos 1 = 2\sin^2(1/2) > 0$. The zero theorem gives a solution in $(0, 1)$. If $0 \le x < y \le 1$, cosine is decreasing on $[0, 1] \subset [0, \pi]$, so $\cos x \ge \cos y$. The equalities $x = \cos x$ and $y = \cos y$ would therefore imply $x \ge y$, a contradiction. Thus the solution is unique. The elementary order property of cosine here can also be checked from $\cos x - \cos y = 2\sin((x + y)/2)\sin((y - x)/2) > 0$ for $x < y$ in this interval.*

The zero theorem supplies existence even when solving the equation exactly is impractical. It also supports a numerical procedure: once the endpoint signs are opposite, repeatedly halve the interval and keep a half whose endpoint values have opposite signs. The proof below shows why this *bisection method* (二分法) converges and gives its error bound.

1.6.2 Boundedness, extrema, and the range

Theorem 1.25

Every continuous function $f\colon [a, b] \to \mathbb{R}$ is bounded on $[a, b]$. That is, there exists $C \ge 0$ such that $|f(x)| \le C$ for every $x \in [a, b]$. ♡

Continuity at one point gives boundedness only near that point. The theorem combines the local behaviour over the entire interval. A point-dependent family of local bounds is not yet a proof of one global bound; the deeper-reading proof explains how the closed bounded domain supplies the missing step.

Definition 1.28

A function $f\colon D \to \mathbb{R}$ attains its maximum (取得最大值) *at $x_M \in D$ if $f(x) \le f(x_M)$ for every $x \in D$. It* attains its minimum (取得最小值) *at $x_m \in D$ if $f(x_m) \le f(x)$ for every $x \in D$. These are absolute extrema on the specified domain.* ♣

Theorem 1.26 (Extreme value theorem)

If $f\colon [a, b] \to \mathbb{R}$ is continuous, there exist $x_m, x_M \in [a, b]$ such that

$$f(x_m) \le f(x) \le f(x_M) \qquad (x \in [a, b]).$$

Thus f attains both a minimum and a maximum. The points of attainment need not be unique and may be endpoints. ♡

A supremum is a bound characterized by order; a maximum must also be a value of the function. For $f(x) = x$ on $(0, 1)$, the supremum is 1 and the infimum is 0, but neither is attained. On $[0, 1]$, both are attained. The distinction in **Figure** 1.8 is exactly what boundedness alone leaves unresolved.

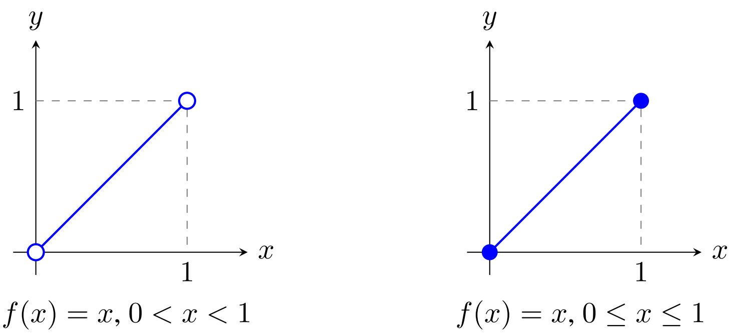


Figure 1.8: The same formula is bounded on both domains. Including the endpoints allows both extreme bounds to be attained.

Corollary 1.5

If $f\colon [a,b] \to \mathbb{R}$ is continuous and its minimum and maximum are m and M, then

$$f([a,b]) = [m, M].$$

In particular, if $f(x) > 0$ for every $x \in [a,b]$, then $f(x) \ge m > 0$ throughout the interval for some m. ♡

Proof Every value belongs to $[m, M]$ by the definitions of the extrema. If $m = M$, the function is constant and the range assertion is immediate. Otherwise, choose points x_m, x_M attaining the extrema and apply the intermediate value theorem on the segment with these two endpoints, placed in increasing order. Every number between m and M is attained there. If f is everywhere positive, its attained minimum $m = f(x_m)$ is positive. □

Pointwise positivity on an open interval gives no such uniform positive lower bound: $f(x) = x$ is positive on $(0,1)$, but its infimum is zero. On a closed bounded interval, the positive lower bound also shows that the reciprocal of an everywhere positive continuous function is bounded.

Example 1.43 Existence of a fixed point Let $f\colon [a,b] \to [a,b]$ be continuous. Prove that there is $c \in [a,b]$ with $f(c) = c$. Such a point is called a *fixed point* (不动点) of f.

> ***Idea.*** Subtract the input: the map-into-itself assumption gives the signs of $f(x) - x$ at both endpoints, even without knowing the actual endpoint values.

Solution *Let $g(x) = f(x) - x$. It is continuous on $[a,b]$, and*

$$g(a) = f(a) - a \ge 0, \qquad g(b) = f(b) - b \le 0.$$

If either endpoint value is zero, that endpoint is fixed. Otherwise the zero theorem gives a zero in (a,b). Uniqueness need not hold: the identity function fixes every point of the interval.

Proposition 1.17

If $f\colon I \to \mathbb{R}$ is continuous and injective on an interval I, then it is strictly monotone. Consequently, its inverse is continuous on the interval $f(I)$. ♠

> ***Proof Strategy.*** For three ordered inputs, the middle output must lie between the other two outputs. Otherwise an intermediate horizontal level would be attained twice, contradicting injectivity. Once this between-values property is known, the direction of increase or decrease cannot reverse.

Proof If $r < s < t$ are in I, injectivity makes their three outputs distinct. Suppose, for example, $f(s) > \max\{f(r), f(t)\}$. Choose y strictly between that maximum and $f(s)$. The intermediate value theorem on

$[r, s]$ and $[s, t]$ produces one preimage in each open segment, contradicting injectivity. The case $f(s) < \min\{f(r), f(t)\}$ is analogous. Hence $f(s)$ lies strictly between $f(r)$ and $f(t)$ for every ordered triple.

If I has at most one point, the conclusion is vacuous. Otherwise choose $u < v$ in I. Suppose $f(u) < f(v)$. For any $x < y$ in I, arrange the distinct points among u, v, x, y in increasing order. For each three consecutive points, the between-values property shows that the two consecutive output differences have the same sign. Thus all consecutive differences have one common sign. That sign is positive because $f(v) - f(u) > 0$ is a sum of some of those differences. It follows that $f(x) < f(y)$. The case $f(u) > f(v)$ similarly gives strict decrease. The intermediate value theorem shows that $f(I)$ is an interval, and Theorem 1.22 gives continuity of the inverse. □

1.6.3 Further reading: proofs from completeness

The main theorems are useful before one has mastered every detail of their proofs. The complete arguments below explain their common foundations and are intended for deeper study. They use the monotone sequence theorem, the Bolzano–Weierstrass theorem, and sequential continuity; no later calculus results are needed.

Lemma 1.3 (Nested intervals)

Let $I_n = [a_n, b_n]$, $n \geq 0$, be nonempty closed intervals with

$$I_{n+1} \subseteq I_n, \qquad b_n - a_n \to 0.$$

Then there is exactly one point c belonging to every I_n, and both endpoint sequences converge to c. ♡

Idea. The left endpoints move right, and the right endpoints move left. They remain bounded by the first interval, so monotone convergence gives limits; shrinking lengths force those limits to agree.

Proof The sequence (a_n) is nondecreasing and bounded above by b_0, while (b_n) is nonincreasing and bounded below by a_0. By Theorem 1.9, they converge to limits α and β. Since $b_n - a_n \to 0$, the sequence limit laws give $\beta = \alpha$; write the common value as c. For each fixed n, nesting gives $a_n \leq a_k \leq b_k \leq b_n$ for all $k \geq n$. Passing to the limit yields $a_n \leq c \leq b_n$, so $c \in I_n$ for every n. If d also belongs to every I_n, then $|c - d| \leq b_n - a_n$ for every n, hence $c = d$. □

Proof Strategy for the intermediate value theorem. It is enough to prove the zero theorem. Bisect a sign-changing interval and retain a sign-changing half. The nested intervals locate one common point, and continuity prevents that point's value from being strictly positive or strictly negative.

Proof of Theorem 1.24. First suppose $f(a) < 0 < f(b)$. Set $a_0 = a$ and $b_0 = b$. At stage n, let $m_n = (a_n + b_n)/2$. If $f(m_n) = 0$, a zero has been found. Otherwise retain $[m_n, b_n]$ when $f(m_n) < 0$, and retain $[a_n, m_n]$ when $f(m_n) > 0$. Unless the process has already found a zero, this constructs nested intervals satisfying

$$f(a_n) < 0 < f(b_n), \qquad b_n - a_n = \frac{b - a}{2^n}.$$

By Lemma 1.3, both endpoint sequences tend to a common $c \in [a, b]$. By Theorem 1.23, $f(a_n) \to f(c)$ and $f(b_n) \to f(c)$. Preservation of nonstrict order in sequence limits gives $f(c) \leq 0$ and $f(c) \geq 0$. Thus $f(c) = 0$, and the strict signs at the original endpoints imply $c \in (a, b)$. If the endpoint signs are reversed, apply the argument to $-f$.

For a level y strictly between $f(a)$ and $f(b)$, apply the established zero result to the continuous function $f - y$. If y equals an endpoint value, that endpoint supplies a preimage. This proves the full theorem. □

The proof also gives an explicit approximation guarantee. After n bisections, the retained interval has length $(b-a)/2^n$ and contains a zero c. Its midpoint m_n therefore satisfies

$$|m_n - c| \le \frac{b-a}{2^{n+1}}.$$

This bound concerns the input error, not necessarily the output error $|f(m_n)|$. If a midpoint is already a zero, the procedure may stop early. The intervals in **Figure** 1.9 illustrate the nested structure.

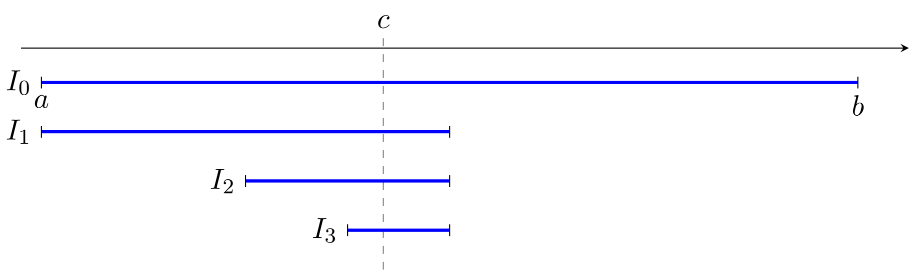


Figure 1.9: Successive sign-changing halves shrink to one point. The particular retained half depends on the sign at its midpoint.

> ***Proof Strategy for boundedness.*** Failure of a global bound would let us choose points where the outputs exceed $1, 2, 3, \ldots$ in magnitude. The inputs lie in a bounded interval, so a subsequence converges; closedness keeps its limit in the domain, where continuity contradicts the diverging outputs.

Proof of Theorem 1.25. Suppose f is not bounded on $[a,b]$. For each integer $n \ge 1$, choose $x_n \in [a,b]$ with $|f(x_n)| > n$. The sequence (x_n) is bounded, so Theorem 1.11 gives a subsequence $x_{n_k} \to c$. Because $a \le x_{n_k} \le b$, preservation of order gives $a \le c \le b$. Continuity at c implies $f(x_{n_k}) \to f(c)$, so this output subsequence is bounded by Theorem 1.4. But $|f(x_{n_k})| > n_k \ge k$, contradicting boundedness. □

Notice the three different roles in this proof. Boundedness of the interval supplies a convergent subsequence. Closedness keeps its limit in the domain. Continuity carries convergence from the inputs to the outputs. Dropping any one of these features may break the argument.

> ***Proof Strategy for attainment of extrema.*** Boundedness supplies a finite supremum M of the range. Approximate M by actual values $f(x_n)$, then extract a convergent input subsequence. Continuity turns approximation of the bound into exact attainment at the limiting input.

Proof of Theorem 1.26. The nonempty range $f([a,b])$ is bounded by Theorem 1.25. Let $M = \sup f([a,b])$. By Proposition 1.2, for every $n \ge 1$ there is $x_n \in [a,b]$ such that

$$M - \frac{1}{n} < f(x_n) \le M.$$

Thus $f(x_n) \to M$ by the squeeze theorem for sequences. By Bolzano–Weierstrass, some subsequence x_{n_k} converges to $x_M \in [a,b]$. Continuity and the subsequence limit property give

$$f(x_M) = \lim_{k\to\infty} f(x_{n_k}) = M.$$

Hence the supremum is attained. Apply the same argument to the continuous function $-f$ to obtain a point x_m at which f attains its minimum. □

The sequence (x_n) chosen in this proof is a *maximizing sequence*(极大化序列): its function values approach the supremum. The sequence itself need not converge, nor need it be monotone. Only a convergent subsequence is required. This proof pattern will recur in optimization problems beyond calculus.

1.6.4 Why the hypotheses matter

The examples below isolate different failures. On $(0,1)$, the continuous function $1/x$ is unbounded, and the continuous function x is bounded but attains neither extreme bound. On the closed unbounded interval $[0,+\infty)$, the continuous function x is unbounded, while $1/(1+x)$ is bounded but does not attain its infimum 0. Thus neither closedness alone nor boundedness of the domain alone gives the closed-interval conclusions.

Continuity is also essential. Define $f(0)=0$ and $f(x)=1/x$ for $0<x\le 1$. This real-valued function has the closed bounded domain $[0,1]$ but is unbounded because it is discontinuous at 0. For a bounded example, define

$$g(x)=\begin{cases} x, & 0\le x<1,\\ 0, & x=1.\end{cases}$$

Its supremum is 1, but no value equals 1; continuity fails at the endpoint. For failure of intermediate values, the step function equal to 0 on $[-1,0)$ and 1 on $[0,1]$ omits $1/2$ despite having endpoint values on either side of it.

Finally, the interval hypothesis is essential for the intermediate value conclusion. The function $f(x)=x$ is continuous on $[-2,-1]\cup[1,2]$, but it has no zero there. The domain has a gap through which an intermediate input would have to pass. Closed bounded domains retain subsequence properties relevant to extrema; they need not retain the interval property relevant to intermediate values.

1.6.5 Uniform continuity

For continuity at one point, the radius δ may depend on that point. The square function illustrates why this dependence can matter:

$$\left|x^2-y^2\right| = |x-y|\,|x+y|\,.$$

On a bounded interval, the second factor admits one common bound. On all of $\mathbb{R}$, inputs of very large magnitude may amplify a small input difference.

Definition 1.29

A function $f\colon D\to\mathbb{R}$ is uniformly continuous(一致连续) *on D if for every $\varepsilon>0$ there exists $\delta>0$ such that, for all $x,y\in D$,*

$$|x-y|<\delta \quad\Longrightarrow\quad |f(x)-f(y)|<\varepsilon.$$

♣

The same δ must work for every pair of points in the domain. The distinction is a change in quantifier order:

$$\begin{aligned} \text{continuity on } D: \quad & \forall a\in D\;\forall\varepsilon>0\;\exists\delta>0\;\forall x\in D,\\ & \qquad |x-a|<\delta \implies |f(x)-f(a)|<\varepsilon;\\ \text{uniform continuity on } D: \quad & \forall\varepsilon>0\;\exists\delta>0\;\forall x,y\in D,\\ & \qquad |x-y|<\delta \implies |f(x)-f(y)|<\varepsilon. \end{aligned}$$

Uniform continuity implies continuity by fixing one point of the pair. Every Lipschitz function is uniformly continuous, since $\delta=\varepsilon/K$ is independent of the base point when $K>0$. If $K=0$, any positive δ works.

Example 1.44 Different domains, different control Determine the uniform continuity of x^2 on bounded intervals and on $\mathbb{R}$, of $1/x$ on $(0,1)$, and of $\sqrt{x}$ on $[0,+\infty)$.

Solution *On $[-R,R]$ with $R>0$, the estimate $\left|x^2-y^2\right|\le 2R\,|x-y|$ gives uniform continuity of the square* *function. On $\mathbb{R}$, take $x_n = n$ and $y_n = n + 1/n$. Then*

$$|x_n - y_n| = \frac{1}{n} \to 0, \qquad |x_n^2 - y_n^2| = 2 + \frac{1}{n^2} > 2.$$

If one radius δ worked for $\varepsilon = 1$, these pairs would violate it for all sufficiently large n. Thus x^2 is not uniformly continuous on $\mathbb{R}$.

For $1/x$ on $(0,1)$, use $x_n = 1/(n+1)$ and $y_n = 1/(n+2)$ for $n \geq 1$. The input differences tend to 0, but $|1/x_n - 1/y_n| = 1$. The choice $\varepsilon = 1/2$ rules out uniform continuity. For the square root, (1.8) gives the single choice $\delta = \varepsilon^2$ for every pair of nonnegative inputs. Hence $\sqrt{x}$ is uniformly continuous on the unbounded interval $[0, +\infty)$, even though it is unbounded and is not Lipschitz near 0.

Proposition 1.18

A function $f\colon D \to \mathbb{R}$ is uniformly continuous on D if and only if for every pair of sequences $(x_n), (y_n)$ in D with $|x_n - y_n| \to 0$, one has

$$|f(x_n) - f(y_n)| \to 0.$$

The input sequences need not converge individually.

Proof If f is uniformly continuous, choose δ for a prescribed $\varepsilon > 0$. Eventually $|x_n - y_n| < \delta$, so the output difference is less than ε. Conversely, failure of uniform continuity means that there is $\varepsilon_0 > 0$ such that for every $\delta > 0$ some pair $x, y \in D$ satisfies

$$|x - y| < \delta, \qquad |f(x) - f(y)| \geq \varepsilon_0.$$

Choose such a pair (x_n, y_n) for $\delta = 1/n$. These sequences contradict the assumed sequential property. □

Theorem 1.27 (Heine–Cantor theorem)

Every continuous function on a closed bounded interval $[a, b]$ is uniformly continuous there.

> ***Proof Strategy.*** If no common radius worked, there would be arbitrarily close input pairs whose outputs stay a fixed positive distance apart. Extract a convergent subsequence of the first inputs; closeness makes the paired second inputs converge to the same point. Continuity at that point contradicts the output separation.

Proof Suppose the conclusion fails. By the negation in Proposition 1.18, there exist $\varepsilon_0 > 0$ and $x_n, y_n \in [a, b]$ such that

$$|x_n - y_n| < \frac{1}{n}, \qquad |f(x_n) - f(y_n)| \geq \varepsilon_0$$

for every $n \geq 1$. By Bolzano–Weierstrass, a subsequence x_{n_k} converges to some $c \in [a, b]$. The estimate

$$|y_{n_k} - c| \leq |y_{n_k} - x_{n_k}| + |x_{n_k} - c| < \frac{1}{n_k} + |x_{n_k} - c|$$

shows that $y_{n_k} \to c$ as well. Continuity gives $f(x_{n_k}) \to f(c)$ and $f(y_{n_k}) \to f(c)$, hence their difference tends to 0. This contradicts the lower bound ε_0. □

Heine–Cantor guarantees a common input radius without necessarily producing a convenient formula for it. When a direct estimate is available, it may give more quantitative information. For example, on $[a, b]$ with $0 < a < b$,

$$\left| \frac{1}{x} - \frac{1}{y} \right| = \frac{|x - y|}{xy} \leq \frac{1}{a^2} |x - y|,$$

so $\delta = a^2 \varepsilon$ suffices for the reciprocal function. This choice makes the distance from 0 explicit.

> ***Connection with compactness.*** The boundedness, extreme-value, and uniform-continuity proofs use the fact that every sequence in $[a, b]$ has a subsequence converging to a point still in $[a, b]$. This is the sequential form of *compactness*(紧致性)in the real line. Later study of compact sets will organize these three arguments into one general framework. Intermediate values use a different feature: an interval contains every point between two of its points. The disconnected-domain example above explains why these two features should be kept distinct.

1.6.6 Methods, pitfalls, and the next stage of calculus

For an existence problem, first build a continuous auxiliary function whose zero expresses the desired conclusion. Choose an interval and verify the endpoint signs explicitly. For uniqueness, add an independent order argument; the intermediate value theorem alone does not provide it. For extrema, specify the full domain and check continuity at included endpoints before invoking the theorem. For uniform continuity, seek an estimate independent of the base point, or use Heine–Cantor after checking that the domain is a closed bounded interval.

The proofs introduce three reusable constructions: nested sign-changing intervals, a sequence chosen from failure of boundedness, and a maximizing sequence chosen from the supremum property. In each construction, the sequence is a tool for converting an assertion about all inputs into a limit argument at one point. The paired-sequence test serves the same purpose when disproving uniform continuity.

Chapter 1 has moved from the order and completeness of $\mathbb{R}$ to functions, sequences, limits, and continuity. The next step studies the limiting ratio of an output change to an input change. Later, uniform control on closed intervals will help compare sums formed from many small subintervals. The local estimates and global theorems developed here supply the foundations for both differentiation and integration.

Exercise 1.6

1. State the intermediate value theorem, the extreme value theorem, and the Heine–Cantor theorem with all hypotheses. For each theorem, identify whether its conclusion concerns existence of a value, attainment of a bound, or control of input errors. Outline the contradiction proof of Heine–Cantor, identifying the separate roles of boundedness, closedness, and continuity.
2. Prove that $x^3 + x - 1 = 0$ has exactly one real root and that this root lies in $(0, 1)$. Use order rather than derivatives for uniqueness.
3. Prove that $e^x = 3 - 2x$ has exactly one solution in $(0, 1)$. State separately the existence and uniqueness arguments.
4. Prove that $x^4 - 3x^2 + 1 = 0$ has at least four distinct real roots by finding four disjoint sign-changing intervals. You need not solve the equation explicitly.

   > ***Hint.*** The polynomial is even. Locate two disjoint sign changes on the positive half-line and then reflect the intervals across the origin.
5. Suppose f is continuous on $[a, b]$ and $f(a) = 2$, $f(b) = 5$. Which of the following must hold: f takes the value 3; f takes the value 0; f is strictly increasing; f has a maximum; $f([a, b]) = [2, 5]$? Prove the necessary conclusions and give counterexamples for the others.
6. Find the supremum, infimum, and any attained extrema of each function on the stated domain:
$$x \text{ on } (0, 1], \qquad x^2 \text{ on } [-1, 2], \qquad \frac{1}{1+x} \text{ on } [0, +\infty).$$
7. Let $f\colon [a, b] \to \mathbb{R}$ be continuous and satisfy $f(x) \neq 0$ everywhere. Prove that $|f(x)| \geq m > 0$ throughout the interval for some m. Deduce boundedness of $1/f$.

8. If $f, g\colon [a,b] \to \mathbb{R}$ are continuous and $f(x) < g(x)$ for every x, prove that there exists $\eta > 0$ such that $f(x) + \eta \le g(x)$ for every x. Give a counterexample on $(0,1)$.
9. Prove that a continuous function on an interval which never vanishes has the same sign throughout that interval. Identify where the interval hypothesis enters the proof.
10. Let $f\colon [0,1] \to [0,1]$ be continuous and decreasing, where decreasing may be read as nonincreasing. Prove that f has exactly one fixed point.
11. Apply bisection to $f(x) = x^2 - 2$ on $[1,2]$ and write down the first three retained half-intervals. How many bisections ensure that the midpoint of the retained interval differs from its contained root by at most 10^{-3}?
12. Explain why $f(x) = \sin x$ attains a minimum and maximum on $[0, 2\pi]$, and determine its range there. Could the extreme value theorem alone tell you the numerical values of those extrema?
13. Prove uniform continuity of $f(x) = 1/(1+x)$ on $[0, +\infty)$ by an explicit estimate. Explain why Heine–Cantor on one closed bounded interval does not directly give this conclusion on the whole domain.
14. Prove that $\sin(1/x)$ is not uniformly continuous on $(0,1)$, even though it is bounded there. In contrast, prove uniform continuity of $\sin x$ on $\mathbb{R}$ by using a trigonometric identity to control the output difference.
15. Prove uniform continuity of $x^{1/3}$ on $\mathbb{R}$ and show that it is unbounded there. What does this example say about a possible implication from uniform continuity to boundedness on an arbitrary domain?

    > ***Hint.*** For real u, v, use $u^2 + uv + v^2 \ge (u-v)^2/4$ to compare $\left|u^3 - v^3\right|$ with $|u-v|^3$.

16. Suppose f and g are uniformly continuous on the same domain D. Prove that $f + g$ is uniformly continuous. If both functions are bounded, prove that fg is uniformly continuous. Show that the boundedness assumption cannot simply be omitted from the product statement.
17. Let $f\colon [a,b] \to \mathbb{R}$ be continuous, and let (x_n) be a sequence in $[a,b]$ satisfying $f(x_n) \to \max_{x\in[a,b]} f(x)$. Prove that some subsequence of (x_n) converges to a point where f attains its maximum. Must the whole sequence converge? Give an example.
18. Construct examples showing that each of the following is possible: a bounded continuous function on an open interval that has no maximum; an unbounded function on a closed bounded interval; a continuous function on a disconnected closed bounded domain that omits a value between two attained values. State which hypothesis is absent in each example.

Supplement Exercises

S1. Let $f\colon [a,b] \to \mathbb{R}$ be continuous and suppose $f(a) = f(b)$. For each positive integer n, prove that there exists $x \in [a, b - (b-a)/n]$ such that

$$f\left(x + \frac{b-a}{n}\right) = f(x).$$

> ***Hint.*** Consider the continuous difference function and the sum of its values at equally spaced inputs.

S2. Let $f\colon [a,b] \to [a,b]$ satisfy $|f(x) - f(y)| \le q\,|x-y|$ for some $0 \le q < 1$. Prove that f has a unique fixed point c. If $x_{n+1} = f(x_n)$ with $x_0 \in [a,b]$, prove $|x_n - c| \le q^n\,|x_0 - c|$ and deduce convergence. Treat $q = 0$ separately if needed.

S3. Suppose $f\colon (a,b) \to \mathbb{R}$ is uniformly continuous, where $a < b$ are finite. Prove that f has a unique continuous extension to $[a,b]$.

> ***Hint.*** First show that f sends Cauchy sequences to Cauchy sequences; then compare two sequences approaching the same endpoint.

S4. Conversely, if $f\colon (a,b) \to \mathbb{R}$ extends continuously to $[a,b]$, prove that it is uniformly continuous on (a,b). Use this criterion to decide uniform continuity of $\sin(1/x)$ and $x\sin(1/x)$ on $(0,1)$.

S5. Prove that every uniformly continuous function on a bounded interval is bounded, even when the interval is open.

> ***Hint.*** Divide the interval into finitely many pieces short enough for the output tolerance 1.

S6. Suppose $f: \mathbb{R} \to \mathbb{R}$ is continuous and has finite limits as $x \to +\infty$ and as $x \to -\infty$. Prove that f is uniformly continuous on $\mathbb{R}$.

> ***Hint.*** Use tail control together with Heine–Cantor on a central closed interval, taking care of pairs near the boundaries of that interval.

S7. Prove that a continuous periodic function $f: \mathbb{R} \to \mathbb{R}$ is bounded and uniformly continuous on $\mathbb{R}$.

> ***Hint.*** If $T > 0$ is a period, use boundedness on $[0, T]$ and uniform continuity on an interval slightly longer than one period.

S8. Let $f: [a, b] \to \mathbb{R}$ be continuous and suppose its maximum is attained at exactly one point c. Prove that every maximizing sequence converges to c. Contrast this with a continuous function whose maximum is attained at more than one point.

S9. Let $f, g: [a, b] \to \mathbb{R}$ be continuous, with f strictly increasing and g strictly decreasing. Suppose $f(a) < g(a)$ and $f(b) > g(b)$. Prove that their graphs meet at exactly one point. Then give an example with both functions strictly increasing and the same endpoint inequalities for which there are at least three intersections.

S10. Let $f: [0, 1] \to \mathbb{R}$ be continuous. Prove that for every $\varepsilon > 0$ there is a continuous function P that is linear on each of finitely many subintervals and satisfies

$$|f(x) - P(x)| < \varepsilon \qquad (0 \leq x \leq 1).$$

> ***Hint.*** Use uniform continuity to choose a partition and let P join the function values at consecutive partition points by line segments.

Chapter 1 Problem-Solving Toolkit

This toolkit summarizes how to choose among the ideas developed in Chapter 1. It is not a list of automatic substitutions: domain, direction of approach, and theorem hypotheses must be settled before a calculation begins. A reliable solution usually has two layers—first identify the structure, then justify the chosen transformation or theorem.

A decision route

Step 1. Fix the object and domain. Is the input a sequence index or a real variable? For a function, record the actual domain and whether the approach is two-sided, one-sided, or at infinity.

Step 2. Try the structurally simplest operation. Use known finite-limit laws only when their hypotheses hold. If direct substitution produces an indeterminate form, regard that form as a signal to compare rates, not as an answer.

Step 3. Expose the small or dominant quantity. Factor, rationalize, normalize by the largest scale, match the argument of a standard limit, or take logarithms for a positive power.

Step 4. Choose the proof mechanism. Use a direct ε-argument for quantitative local control, squeeze for domination, monotonicity plus boundedness for recursive sequences, or a sequence criterion to detect incompatible approaches.

Step 5. Check the logical hinge. Verify nonzero denominators, eventual signs, transformed domains, punctures in compositions, and every hypothesis of a global continuity theorem.

Step 6. State exactly what has been proved. Separate existence from uniqueness, boundedness from attainment, convergence from a merely possible fixed point, and pointwise continuity from uniform continuity.

Sequence patterns

Signal	First move	What must be checked
Rational expression in n	Divide by the highest relevant power of n.	The normalized denominator has a nonzero limit; lower-order terms really vanish.
Difference of comparable roots	Factor a difference of powers; for square roots, multiply by the conjugate.	Root domains and signs, especially when extracting $\sqrt{x^2} = \|x\|$ on a tail.
Bounded oscillation times a shrinking factor	Take absolute values and squeeze by a positive majorant.	The oscillatory factor has one bound valid for every sufficiently late index.
Recursive sequence	Find an invariant interval, then prove monotonicity and boundedness.	Solve the fixed-point equation only after convergence has been established; discard roots outside the invariant interval.
Sum or average with a growing number of terms	Split it into a fixed head and a uniformly controlled tail.	A finite-sum limit law cannot be applied directly when the number of summands depends on n.
Several positive growth scales	Normalize by the largest plausible scale and use $\ln n \ll n^\alpha \ll a^n \ll n! \ll n^n$.	The restrictions $\alpha > 0$, $a > 1$, and any cancellation between leading terms.

Proof and disproof templates

- **Direct limit proof.** Work backward from the desired output error to discover a sufficient input restriction, then write the proof forward with the quantifiers in their prescribed order.
- **Negating convergence.** Fix one tolerance $\varepsilon_0 > 0$ and produce bad terms or inputs arbitrarily far out or arbitrarily close to the limiting point. The fixed tolerance is the essential feature.
- **Sequential disproof.** For a sequence, find two subsequences with different limits. For a function, find two admissible input sequences approaching the same point whose output sequences have incompatible limits.
- **Uniform-continuity disproof.** Choose pairs x_n, y_n whose input distance tends to 0 while the output distance remains at least one fixed positive number.
- **Closed-interval theorem.** Write the function, domain, continuity statement, and relevant endpoint or range data before naming the theorem. Then state only the conclusion supplied by that theorem.

Function-limit and continuity patterns

Signal	First move	What must be checked
Algebraic $0/0$ or cancellation	Factor, use an exact identity, or rationalize before applying limit laws.	Cancellation is valid on the punctured domain; a remaining denominator stays away from zero.
Standard small trigonometric, logarithmic, or exponential factors	Match each argument with its denominator and use a proved equivalent in products or quotients.	Equivalent infinitesimals cannot be substituted term by term inside a difference with leading cancellation.
Positive varying power $F(x)^{G(x)}$	Rewrite it as $\exp(G(x)\ln F(x))$.	The base is positive on the relevant domain and the single exponent $G \ln F$ has been evaluated.
Change of variables $t = \phi(x)$	Translate the limit into the variable t and write its exact new approach.	Side reversal, tail-to-zero direction, transformed domain, and the puncture condition for composition.
Competing infinite quantities	Establish eventual signs, normalize, compare, or transform the expression.	Forms such as $\infty - \infty$ and $0 \cdot \infty$ have no arithmetic meaning by themselves.
Suspected failure of a function limit	Compare one-sided limits or construct two admissible input sequences.	One bad sequence disproves a limit; finitely many successful sequences never prove the universal assertion.
Existence of a root or intersection	Build a continuous auxiliary difference and locate a sign change on an interval.	Continuity and the interval are explicit; uniqueness requires a separate order or injectivity argument.
Extrema or uniform control	Check the closed bounded interval hypotheses, then use the appropriate global theorem.	Attainment is stronger than boundedness; a direct estimate may yield sharper uniform information.

Stop signs. Pause before making any of the following conclusions.

- A fixed-point equation supplies possible limits of a recurrence, not convergence.
- The intermediate value theorem supplies existence, not uniqueness.
- Boundedness near a point does not supply a limit, and unboundedness does not supply a signed infinite limit.
- The value $f(a)$ does not affect a punctured limit, but it is essential for continuity at a.
- Equivalent infinitesimals are safe in the stated product and quotient setting, not across arbitrary addition or subtraction.
- A theorem name never replaces verification of its domain and hypotheses.

Before finishing a solution, check the domain once more, record every one-sided direction, identify the theorem or estimate that justifies the last passage to the limit, and make sure the conclusion answers exactly the question asked.